

CHAPTERS

[\[Immunology Home Page\]](#)

The chapter resources portion of this site was designed and created by **Duane W. Sears**, Professor of Immunology and Biochemistry at the University of California, Santa Barbara. We greatly appreciate Professor Sears' time and effort in preparing the design, content and images for these chapter resources.

[Chapter 1: Overview Of The Immune System](#)

[Chapter 2: Experimental Systems](#)

[Chapter 3: Cells And Organs Of The Immune System](#)

[Chapter 4: Antigens](#)

(Netscape Navigator v3.x or greater and Chemscape Chime plug-in required)

[Chapter 5: Immunoglobulins: Structure And Function](#)

(Netscape Navigator v3.x or greater and Chemscape Chime plug-in required)

[Chapter 6: Antigen-Antibody Interactions](#)

(Microsoft Excel helper recommended for Scatchard plot)

[Chapter 7: Organization And Expression Of Immunoglobulin Genes](#)

(Netscape Navigator v3.x or greater and Chemscape Chime plug-in required)

[Chapter 8: B-Cell Maturation, Activation, and Differentiation](#)

[Chapter 9: Major Histocompatibility Complex](#)

(Netscape Navigator v3.x or greater and Chemscape Chime plug-in required)

[Chapter 10: Antigen Processing And Presentation](#)

[Chapter 11: T-Cell Receptor](#)

[Chapter 12: T-Cell Activation, Maturation, And Differentiation](#)

(Netscape Navigator v3.x or greater and Chemscape Chime plug-in required)

[Chapter 13: Cytokines](#)

(Netscape Navigator v3.x or greater and Chemscape Chime plug-in required)

[Chapter 14: The Complement System](#)

[Chapter 15: Leukocyte Migration and Imflammation](#)

[Chapter 16: Cell-Mediated and Humoral Effector Responses](#)

[Chapter 17: Hypersensitive Reactions](#)

[Chapter 18: Vaccines](#)

Chapter 19: Immune Response To Infectious Diseases

Chapter 20: Autoimmunity

Chapter 21: Immunodeficiency Diseases

Chapter 22: The Immune System in AIDS

(Netscape Navigator v3.x or greater and Chemscape Chime plug-in required)

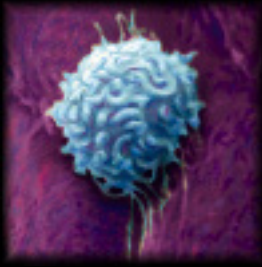
Chapter 23: Transplantation Immunology

Chapter 24: Cancer and the Immune System

[\[Immunology Home Page\]](#)

W. H. Freeman & Co. and Sumanas, Inc.
Immunology, January, 1997

Last Updated: July 17, 1997



IMMUNOLOGY

Janis Kuby

Welcome to the Kuby, *Immunology*, **Third Edition** web site. The site for the **Fourth Edition** is live, click [here](#) to access.

ABOUT THIS SITE

ABOUT THE BOOK

PREFACE

ABOUT THE AUTHOR

TABLE OF CONTENTS

EDUCATIONAL RESOURCES

CHAPTER RESOURCES

MOLECULAR VISUALIZATION

SEND US FEEDBACK

© 1997 [W. H. Freeman & Company](#)

Web design:
SUMANAS Multimedia Development Services
Sumanas, Inc.

KUBY IMMUNOLOGY



FOURTH
EDITION

RICHARD A. GOLDSBY | THOMAS J. KINDT | BARBARA A. OSBORNE

NEWS & NOTES

- ▶ For more information and updates on the field of immunology, click [here](#) to access the home page of the American Association of Immunologists.
- ▶ [For Instructors](#)

KUBY IMMUNOLOGY

RICHARD A. GOLDSBY | THOMAS J. KINDT | BARBARA A. OSBORNE

Welcome to the Web Site for *Kuby Immunology 4e* by Richard A. Goldsby, Thomas J. Kindt and Barbara A. Osborne. This Web site is designed to help students review key concepts from the textbook through interactive exercises and learning tools.

Resources are organized by chapter of the textbook and by content type. To access resources, please select a Chapter or Category below.

Users of the Third Edition, click [here](#) to access the 3e site.

Select a Chapter:

Select a Category:



Clinical Case Studies



Self-Tests



Animations



Molecular Visualizations



End-of-Chapter Useful Web Site References

System Requirements

This site requires Internet Explorer (version 4.0 or above) or Netscape Navigator (version 4.0 or above). In addition, many resources require the Macromedia Flash Player (version 3.0 or above) , Shockwave (version 7.0 or above), or Chime plug-in for Netscape browsers (Windows version 2.0.2 and Macintosh version 2.0a or greater). The latest versions of this software can be downloaded free of charge by clicking the icons below.



W.H. Freeman | [Biology](#) | [About This Book](#) | [Biology](#) | [Order a Book](#) | [Contact Us](#) | [Technical Support](#)



KUBY IMMUNOLOGY

RICHARD A. GOLDSBY | THOMAS J. KINDT | BARBARA A. OSBORNE

Welcome to the Instructor Site for *Immunology 4e*

Coming in November, the new FIFTH EDITION! For more information, start here to [contact your rep.](#)

NEWS & NOTES

- ▶ [Take our online faculty survey!](#)
- ▶ [Back to Student Site](#)

Textbook Figures

Download JPEG and/or GIF images of the figures in the textbook, in either PC (.zip files) or Mac (.sit files) to use in your own presentations.

PC (Zipped GIF .zip files)	PC (Zipped JPEG.zip files)	Mac (Stuffed GIF .sit files)	Mac (Stuffed JPEG .sit files)
• Chapter 1 GIF • Chapter 2 GIF • Chapter 3 GIF • Chapter 4 GIF • Chapter 5 GIF • Chapter 6 GIF • Chapter 7 GIF • Chapter 8 GIF • Chapter 9 GIF • Chapter 10 GIF • Chapter 11 GIF • Chapter 12 GIF • Chapter 13 GIF	• Chapter 1 JPEG • Chapter 2 JPEG • Chapter 3 JPEG • Chapter 4 JPEG • Chapter 5 JPEG • Chapter 6 JPEG • Chapter 7 JPEG • Chapter 8 JPEG • Chapter 9 JPEG • Chapter 10 JPEG • Chapter 11 JPEG • Chapter 12 JPEG • Chapter 13 JPEG	• Chapter 1 GIF • Chapter 2 GIF • Chapter 3 GIF • Chapter 4 GIF • Chapter 5 GIF • Chapter 6 GIF • Chapter 7 GIF • Chapter 8 GIF • Chapter 9 GIF • Chapter 10 GIF • Chapter 11 GIF • Chapter 12 GIF • Chapter 13 GIF	• Chapter 1 JPEG • Chapter 2 JPEG • Chapter 3 JPEG • Chapter 4 JPEG • Chapter 5 JPEG • Chapter 6 JPEG • Chapter 7 JPEG • Chapter 8 JPEG • Chapter 9 JPEG • Chapter 10 JPEG • Chapter 11 JPEG • Chapter 12 JPEG • Chapter 13 JPEG

• Chapter 14 GIF	• Chapter 14 JPEG	• Chapter 14 GIF	• Chapter 14 JPEG
• Chapter 15 GIF	• Chapter 15 JPEG	• Chapter 15 GIF	• Chapter 15 JPEG
• Chapter 16 GIF	• Chapter 16 JPEG	• Chapter 16 GIF	• Chapter 16 JPEG
• Chapter 17 GIF	• Chapter 17 JPEG	• Chapter 17 GIF	• Chapter 17 JPEG
• Chapter 18 GIF	• Chapter 18 JPEG	• Chapter 18 GIF	• Chapter 18 JPEG
• Chapter 19 GIF	• Chapter 19 JPEG	• Chapter 19 GIF	• Chapter 19 JPEG
• Chapter 20 GIF	• Chapter 20 JPEG	• Chapter 20 GIF	• Chapter 20 JPEG
• Chapter 21 GIF	• Chapter 21 JPEG	• Chapter 21 GIF	• Chapter 21 JPEG
• Chapter 22 GIF	• Chapter 22 JPEG	• Chapter 22 GIF	• Chapter 22 JPEG
• Chapter 23 GIF	• Chapter 23 JPEG	• Chapter 23 GIF	• Chapter 23 JPEG

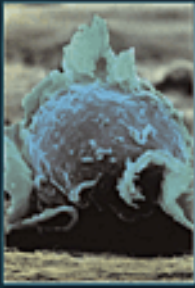
Your Syllabus Here . . .

Posting your syllabus to the Web doesn't get any easier. Just write your syllabus as you normally do. If you use a word processor, great. If you use a typewriter--or even pen and paper--that's okay, too. We'll post your syllabus to a private location on our Web site, where only you and your students will be able to access it. And we'll leave it there as long as you'd like.

Three easy steps

1. Write your syllabus as you normally do.
2. If you've created your syllabus electronically, email it to us at syllabus@bfwpub.com. Or just fax it to us at (212) 576-9497 (be sure to include your email address).
3. We will post your syllabus to a private location on our Web site within three working days and send you an email telling you where to find it.

KUBY
IMMUNOLOGY



FOURTH
EDITION

RICHARD A. GOLDSBY | THOMAS J. KINDT | BARBARA A. OSBORNE

KUBY
IMMUNOLOGY

RICHARD A. GOLDSBY | THOMAS J. KINDT | BARBARA A. OSBORNE

Welcome to the Instructor Site for *Immunology 4e*

Coming in November, the new FIFTH EDITION! For more information, start here to [contact your rep.](#)

NEWS & NOTES

- ▶ [Take our online faculty survey!](#)
- ▶ [Back to Student Site](#)

Textbook Figures

Download JPEG and/or GIF images of the figures in the textbook, in either PC (.zip files) or Mac (.sit files) to use in your own presentations.

PC (Zipped GIF .zip files)	PC (Zipped JPEG.zip files)	Mac (Stuffed GIF .sit files)	Mac (Stuffed JPEG .sit files)
• Chapter 1 GIF • Chapter 2 GIF • Chapter 3 GIF • Chapter 4 GIF • Chapter 5 GIF • Chapter 6 GIF • Chapter 7 GIF • Chapter 8 GIF • Chapter 9 GIF • Chapter 10 GIF • Chapter 11 GIF • Chapter 12 GIF • Chapter	• Chapter 1 JPEG • Chapter 2 JPEG • Chapter 3 JPEG • Chapter 4 JPEG • Chapter 5 JPEG • Chapter 6 JPEG • Chapter 7 JPEG • Chapter 8 JPEG • Chapter 9 JPEG • Chapter 10 JPEG • Chapter 11 JPEG • Chapter 12 JPEG • Chapter	• Chapter 1 GIF • Chapter 2 GIF • Chapter 3 GIF • Chapter 4 GIF • Chapter 5 GIF • Chapter 6 GIF • Chapter 7 GIF • Chapter 8 GIF • Chapter 9 GIF • Chapter 10 GIF • Chapter 11 GIF • Chapter 12 GIF • Chapter	• Chapter 1 JPEG • Chapter 2 JPEG • Chapter 3 JPEG • Chapter 4 JPEG • Chapter 5 JPEG • Chapter 6 JPEG • Chapter 7 JPEG • Chapter 8 JPEG • Chapter 9 JPEG • Chapter 10 JPEG • Chapter 11 JPEG • Chapter 12 JPEG • Chapter

	13 GIF • Chapter 14 GIF • Chapter 15 GIF • Chapter 16 GIF • Chapter 17 GIF • Chapter 18 GIF • Chapter 19 GIF • Chapter 20 GIF • Chapter 21 GIF • Chapter 22 GIF • Chapter 23 GIF	13 JPEG • Chapter 14 JPEG • Chapter 15 JPEG • Chapter 16 JPEG • Chapter 17 JPEG • Chapter 18 JPEG • Chapter 19 JPEG • Chapter 20 JPEG • Chapter 21 JPEG • Chapter 22 JPEG • Chapter 23 JPEG	13 GIF • Chapter 14 GIF • Chapter 15 GIF • Chapter 16 GIF • Chapter 17 GIF • Chapter 18 GIF • Chapter 19 GIF • Chapter 20 GIF • Chapter 21 GIF • Chapter 22 GIF • Chapter 23 GIF	13 JPEG • Chapter 14 JPEG • Chapter 15 JPEG • Chapter 16 JPEG • Chapter 17 JPEG • Chapter 18 JPEG • Chapter 19 JPEG • Chapter 20 JPEG • Chapter 21 JPEG • Chapter 22 JPEG • Chapter 23 JPEG
--	--	---	--	---

Your Syllabus Here . . .

Posting your syllabus to the Web doesn't get any easier. Just write your syllabus as you normally do. If you use a word processor, great. If you use a typewriter--or even pen and paper--that's okay, too. We'll post your syllabus to a private location on our Web site, where only you and your students will be able to access it. And we'll leave it there as long as you'd like.

Three easy steps

1. Write your syllabus as you normally do.
2. If you've created your syllabus electronically, email it to us at syllabus@bfwpub.com. Or just fax it to us at (212) 576-9497 (be sure to include your email address).
3. We will post your syllabus to a private location on our Web site within three working days and send you an email telling you where to find it.



The chapter resources portion of this site was designed and created by [Duane W. Sears](#), Professor of Immunology and Biochemistry at the University of California, Santa Barbara. We greatly appreciate Professor Sears' time and effort in preparing the design, content and images for the chapter resources.

[\[Immunology Home Page\]](#)

W. H. Freeman & Co. and Sumanas, Inc.
Immunology, January, 1997

BIOGRAPHICAL SKETCH

Professor Sears joined the faculty at the University of California Santa Barbara in 1977 following completion of his postdoctoral studies on the biochemistry of murine MHC antigens at Albert Einstein College of Medicine. Prior to that, he earned his Ph.D. in Biophysical Chemistry at Columbia University, analyzing the kinetics of immunoglobulin heavy and light chain assembly, a project which sparked his interest in immunology. His work and teaching at UCSB are both underscored by a fundamental desire to understand the structure-function relationships of proteins, as reflected in the Kuby web site and the instructional web sites created for his courses. The unifying theme of his research at UCSB has been the analysis of immunological cell-killing mechanisms.

Early studies in Dr. Sears' laboratory aimed at elucidating the structure-function relationships of mutant **MHC class I molecules** in order to probe the recognition of antigen by **cytotoxic T lymphocytes (CTLs)** and target cell lysis [\(1\)](#).

Subsequent studies aimed at elucidating the structure-function relationships of IgG antibody **Fcγ receptors (FcγRs)** expressed on **natural killer (NK) lymphocytes** in order to probe the killing mechanism of **antibody-dependent cell-mediated cytotoxicity (ADCC)** [\(2\)](#).

These studies led to the cloning and characterization of cDNA molecules encoding a family of **rat NK FcγRIII molecules** [\(3\)](#) and the murine **macrophage** high affinity **FcγRI** molecule [\(4\)](#).

Current studies focus on the creation of novel composite molecules with potential immunotherapeutic and diagnostic properties. In one composite, a portion of the high affinity **FcγRI** is fused to the **antibody Fc** itself creating an **FcγRI-Fc "receptorbody"**. These molecules provide a new type of tool for investigating the **FcγRI-antibody** interaction. At present the precise nature of this interaction can only be inferred since the three-dimensional structure of the **FcγRI** molecule has not yet been determined. However, the **antibody hinge (H)** and **C_H2** regions are known to be at the focal point of this interaction.

Contact information:

Duane W. Sears
Prof. Immunology and Biochemistry
Dept. Molecular, Cell, & Developmental Biology, and
Interdepartmental Program in Biochemistry and Molecular Biology
University of California
Santa Barbara, CA 93106

E-mail: sears@lifesci.ucsb.edu

Footnotes:

1. S. S. Burnside, P. Hunt, K. Ozato, and D. W. Sears. "A Molecular Hybrid of the $H-2D^d$ and $H-2L^d$ Genes Expressed in the *dm1* Mutant." *Proc. Natl. Acad. Sci. USA* 81:5204 (1984).
2. J. E. Christiaansen and D. W. Sears. "Unusually Efficient Tumor Cell Lysis by Human Effectors of Antibody-Dependent Cellular Cytotoxicity Mediated by Monoclonal Antibodies." *Cancer Res.* 44:3712 (1984).
3. D. L. Farber, R. Giorda, M. Y. Nettleton, M. Trucco, J. P. Kochan, and D. W. Sears. "Rat Class III Fc γ Receptor Isoforms Differ in IgG Subclass-Binding Specificity and Fail to Associate Productively with Rat CD3 ζ ." *J. Immunol.* 150:4364 (1993).
4. D. W. Sears, N. Osman, B. Tate, I. F. C. McKenzie, and P. M. Hogarth. "Molecular Cloning and Expression of the Mouse High Affinity Fc Receptor for IgG." *J. Immunol.* 144::371 (1990).

Copyright 1996, Duane W. Sears.

[\[TOP\]](#)

[\[Immunology Home Page\]](#)

W. H. Freeman & Co. and Sumanas, Inc.
Immunology, January, 1997

PREFACE

- **Introduction**
- **Organization and Content**
 - **Part I: Introduction.**
 - **Part II: Generation of B-Cell and T-Cell Responses.**
 - **Part III: Immune Effector Mechanisms.**
 - **Part IV: The Immune System in Health and Disease.**
- **Pedagogical Tools**
- **Immunology ... On The Web!**

Immunology has been used successfully in scores of introductory courses for both undergraduates and students in clinical medicine. The text's focus on the experimental process conveys much of the excitement of scientific discovery while presenting the relevant facts. Exposure to the landmark experiments that underlay the theoretical framework of immunology enables students to appreciate how current immunologic paradigms evolved. This approach also illustrates how experiments are designed and interpreted to test scientific hypotheses, thus encouraging students to develop the intellectual skills needed to read critically the current immunologic literature.

As a teacher, Jan Kuby recognized the need for a textbook that first and foremost emphasizes that immunology is based solidly upon experimental results and, second, that highlights the relationships between basic immunologic principles and clinical applications. The third edition preserves and builds upon these goals. Discussions of important clinical applications are included throughout the text, and the last seven chapters are devoted to the most important clinically related subjects. As the AIDS epidemic highlights, these topics are not matters of abstract research interest, but a question of life and death for ever increasing numbers of men, women, and children.

The rapidity and magnitude of exciting developments in immunology pose a rewarding challenge for instructors trying to keep abreast of the tremendous growth spurts that characterize the field. New data and methodology not only contribute to illuminating previously unanswered questions, but also may lead to changes in the interpretation of earlier findings. In this third edition Dr. Kuby has sought to describe the most recent advances clearly and to relate them to previous concepts. Students thus will have access to the most recent facts, and also see how interpretations may change over time to reflect new data. This approach equips them to synthesize new findings as they are generated.

[\[TOP\]](#)

Organization and Content

The first two editions of Immunology were enthusiastically received by students and instructors alike. The third edition, therefore, retains much of the original organizational motif. This edition, however, has been divided into four major parts and discussion of several topics has been shifted to maintain these cohesive conceptual units. As a result of this reorganization, the analogies between B-cell and T-cell responses have been clarified, some subjects that previously were discussed in several different chapters have been consolidated into one primary presentation, and the material on monoclonal antibodies, immune regulation, and tolerance has been incorporated into other chapters. This revised approach produces a more integrated, more logical, and more compact presentation without any loss of significant content.

[\[TOP\]](#)

Part I: Introduction.

As in the previous editions, the first three chapters provide a conceptual framework for the material presented in subsequent chapters. Chapter 1 presents a general, integrated overview of the immune system, introducing many of the key concepts and terminology. Chapter 2 again reviews some of the important experimental systems and techniques used in modern immunology including more extensive discussion about the production of gene-targeted knockout mice. Chapter 3 still includes an introduction to the cells and tissues of the immune system, but information on lymphocyte recirculation has been shifted to a new chapter in Part III.

[\[TOP\]](#)

Part II: Generation of B-Cell and T-Cell Responses.

This part focuses on how lymphocytes recognize and respond to foreign substances. It begins with a discussion of antigens, particularly the molecular properties that are recognized by B and T cells (Chapter 4). The next four chapters deal with the function of B cells: the structure and function of immunoglobulins (Chapter 5), antigen-antibody interactions (Chapter 6), the organization and expression of immunoglobulin genes (Chapter 7), and a new chapter on the maturation, activation, and differentiation of B cells (Chapter 8). The last four chapters in Part II deal with the function of T cells beginning with Chapter 9 on the major histocompatibility complex and Chapter 10 on antigen processing and presentation, subjects that are unique to the T-cell response. The extensively updated Chapter 11 on the T-cell receptor and Chapter 12 on the maturation, activation, and differentiation of T cells have been revised to highlight the similarities with analogous B-cell processes.

[\[TOP\]](#)

Part III: Immune Effector Mechanisms.

Various specific and nonspecific effector mechanisms associated with both humoral and cell-mediated responses are described in this part. Chapter 13 presents an overview of the numerous cytokines and their effects. Activation of the complement system and the mechanisms by which complement components function in the humoral immune response are covered in Chapter 14. Next comes a new chapter containing a consolidated discussion of leukocyte migration and inflammation (Chapter 15). The major cell-mediated effector mechanisms, the characteristics of primary and secondary humoral effector responses, and regulation of immune effector responses are covered in Chapter 16. Discussion of the four types of hypersensitive reactions concludes Part III (Chapter 17).

[\[TOP\]](#)

Part IV: The Immune System in Health and Disease.

As in the first and second editions, this edition concludes with a series of clinically oriented chapters in which the basic concepts discussed in previous chapters are applied to topics of clinical interest. Coverage includes vaccines (Chapter 18), immune response to infectious diseases (Chapter 19), autoimmunity (Chapter 20), immunodeficiency diseases (Chapter 21), AIDS (Chapter 22), transplantation immunology (Chapter 23), and cancer (Chapter 24).

[\[TOP\]](#)

Some of the significant additions and changes to the text of various chapters in the third edition include the following:

- Chapter 3, Cells and Organs of the Immune System, has been reorganized to include new coverage of cutaneous lymphoid tissue and intestinal lymphoid cells. Sections on hematopoiesis, programmed cell death, dendritic cells, and B- and T-cell activation within the secondary lymphoid organs have been updated and clarified.
- Chapter 5, Immunoglobulins: Structure and Function, now contains coverage of monoclonal antibodies and their uses. New topics include bioengineering and milk-borne immunity.
- Chapter 7, Organization and Expression of Immuno- globulin Genes, contains an updated, revised discussion of the molecular mechanisms of gene-segment joining and new material on generation of antibody diversity. The last section describes recent findings about regulation of immunoglobulin-gene transcription in B cells and the inhibition of immunoglobulin-gene expression in T cells.
- Chapter 8, B-Cell Maturation, Activation, and Differentiation, provides a unified view of the B cell paralleling the similar coverage of the T cell that was added in the second edition. In addition, it includes new information on the bone marrow microenvironment and the negative selection of self-reactive B cells during maturation. This new chapter also includes material on the identity of the B-cell coreceptor complex and much more extensive discussion of intracellular signal-transduction pathways triggered by antigen binding by B cells.

- Chapter 9, Major Histocompatibility Complex, presents extensive new information on class I and class II MHC molecules and how the structure of these molecules relates to the binding of peptides. The map of the MHC has also been updated.
- Chapter 10, Antigen Processing and Presentation, provides new information on the enzymes and transporters involved in the pathways for handling exogenous and endogenous antigens. Descriptions of the cytosolic proteolytic system, proteasome, and TAP proteins are presented in more detail than in the second edition.
- Chapter 11, T-Cell Receptor, is extensively updated with new information on the creation of a functional T-cell antigen receptor. It describes the structure of the $\alpha\beta$ and $\gamma\delta$ T-cell receptors and compares their domain structure with the membrane-bound immunoglobulin of B cells. Likewise, the organization of TCR gene segments and their joining into functional genes are compared with analogous processes in B cells, highlighting the similarities in the mechanisms for generating antigen receptors of incredible diversity. Finally, the function of the T-cell coreceptors (CD4 and CD8) is compared with that of the B-cell coreceptor complex, and new information is presented on the role of CD4 and CD8 in T-cell activation.
- Chapter 12, T-Cell Maturation, Activation, and Differentiation, includes new material on the pre-T cell receptor and the time course of TCR-gene rearrangement during maturation of T cells in the thymus. The section on positive and negative selection during T-cell maturation, including the many unanswered questions about thymic selection, has been extensively revamped. Coverage of T-cell activation and differentiation includes more details about the co-stimulatory signal and signal-transduction pathways; the differences in the ability of dendritic cells, macrophages, and B cells to function as antigen-presenting cells are clearly delineated. Finally, recent findings about ligand recognition by $\gamma\delta$ T cells are presented.
- Chapter 13, Cytokines, brings a rapidly changing field up to date. The section on the structure and function of cytokine receptors has been substantially expanded, with five receptor families described and illustrated, and new information presented on signal transduction mediated by these receptors. New or updated tables and figures summarize the various effects of cytokines and how they interact with cells and with one another.
- Chapter 15, Leukocyte Migration and Inflammation, expands the discussion on leukocyte circulation and cell-adhesion molecules and relates it to inflammation. This new chapter presents a more general view of leukocyte circulation, how leukocytes are directed to different tissues, and a detailed picture of neutrophil and lymphocyte extravasation. It also contains a new section on the role of chemokines and other mediators in the inflammatory response.
- Chapter 16, Cell-Mediated and Humoral Effector Responses, includes new descriptions of the properties of effector T cells as well as soluble and membrane-bound effector molecules. This reorganized chapter also presents new findings concerning the role of apoptosis in target-cell destruction and receptors on natural killer cells that allow them to distinguish normal cells from infected or cancerous cells.
- Chapter 17, Hypersensitive Reactions, includes updated coverage of intracellular signals involved in mast cell degranulation and expanded discussion of asthma and other manifestations of localized type 1 hypersensitivity.
- Chapter 18, Vaccines, includes a new section on DNA vaccines, a novel approach that offers several advantages. The discussion of other vaccine types has been updated and clarified.

- Chapter 21, Immunodeficiency Diseases, describes recent findings about the genetic defects causing several types of X-linked immunodeficiencies and type I bare-lymphocyte syndrome. Updated information also is presented on the progress of gene therapy for severe combined immunodeficiency caused by ADA deficiency.
- Chapter 22, The Immune System in AIDS, has been reorganized, and discussion of some earlier findings of limited importance has been deleted. The chapter describes the recent discovery of fusin, a G-protein-coupled receptor that is the likely cofactor required for HIV infection of CD4 T cells. In addition, new information is presented on a possible intriguing link between chemokine levels and susceptibility to infection, as well as the characteristics of long-term survivors.
- Chapter 23, Transplantation Immunology, includes a new section on xenotransplants, which may become more important in the future as the supply of donor organs continues to fall short of the need.

[\[TOP\]](#)

Pedagogical Tools

Words tell only part of the story for today's students. The other part is presented in pictures. Each figure in the previous edition has been carefully reevaluated, and many have been redesigned to improve their graphical appearance and pedagogical effectiveness. In some cases, previous figures were combined to allow a clear comparison of relationships; other figures were simplified so that they portrayed a single important concept.

A significant number of new full-color illustrations have been added to the third edition. These are intended to engage students' attention and help them to better understand complex topics. The use of color not only serves aesthetic purposes but also functions as an additional teaching tool. Color photographs appear throughout this edition, rather than in a single color insert, and thus are located in close proximity to the text points they illustrate.

[\[TOP\]](#)

A number of concepts are crucial for students to develop a firm understanding of immunology. These concepts are illustrated in new key diagrams called "Visualizing Concepts." These figures dramatize important ideas and processes in a way that written text alone cannot convey.

Finally, recurring elements, such as various immune-system cells and important membrane molecules, are depicted consistently, thereby facilitating recognition of these elements in different contexts. These recurring elements look more realistic than simple stylized shapes and provide students with an additional intuitive tool to help them to construct a more complete picture of the field of immunology.

Each of the four parts opens with a brief overview, orienting students to the main topics covered in each chapter. Outlines are placed at the beginning of each chapter in this edition. These provide students with a chapter preview and a guideline to follow as they plan their reading.

[\[TOP\]](#)

Key terms now appear in boldface type within the text. Definitions of many of these terms appear in the revised and updated glossary at the end of the book.

Chapter summaries not only present a concise review of the material but also include references to key illustrations that provide a visual overview of the most important concepts within each chapter.

Updated references are provided at the end of each chapter to enable students to sample the primary literature relating to topics of interest.

The study questions were a popular feature of earlier editions for both instructors and students. They continue to constitute a useful teaching tool that provides a challenging and thoughtful review for students. Additional questions related to new topics are included in the third edition, and many questions from the previous editions have been rewritten to enhance their clarity. As before, the answers to the study questions are given at the back of the book.

Immunology is replete with acronyms and abbreviations that may be daunting for students early in their studies, but soon constitute a familiar language. To assist students in mastering the use of this terminology, a list of acronyms and abbreviations is presented on the inside cover of the book for easy reference throughout the course of study.

[\[TOP\]](#)

Immunology on the Web!

Duane Sears, University of California at Santa Barbara, has created a Web page based upon Immunology that brings the subject to life for students and keeps the content refreshed with the latest breaking advances in the field. Many figures from the text, as well as additional images embedded in the Web, are available on the Web site and can be captured for classroom projection or for other use by the instructor.

The Web page also provides links to images, movies, and animations of cells undergoing an immune response, or dynamic imaging of molecules such as antibodies and MHC antigens that mediate immune functions. Students can explore the Web page to discover new areas of immunology that extend beyond the course material, enjoy the visualization of three-dimensional representations of molecules that they can rotate, and access movies and snapshots of immune functions as they appear in our daily news.

[\[TOP\]](#)

[\[Immunology Home Page\]](#)

W. H. Freeman & Co. and Sumanas, Inc.
Immunology, January, 1997

TABLE OF CONTENTS

PART I: INTRODUCTION

CHAPTER 1 OVERVIEW OF THE IMMUNE SYSTEM

Historical Perspective

- Discovery of Humoral and Cellular Immunity
- Early Theories of Immunity
- Components of Immunity

Innate (Nonspecific) Immunity

- Anatomic Barriers
- Physiologic Barriers
- Endocytic and Phagocytic Barriers
- Barriers Created by the Inflammatory Response

Acquired (Specific) Immunity

- The Cells of the Immune System
 - B Lymphocytes
 - T Lymphocytes
 - Antigen-Presenting Cells
- Functions of Humoral and Cell-Mediated Immune Responses
- Recognition of Antigen by B and T Lymphocytes
- Generation of Lymphocyte Specificity and Diversity
- Role of the Major Histocompatibility Complex
- Processing and Presentation of Antigens
- Clonal Selection of Lymphocytes
- Cellular Interactions Required for Generation of Immune Responses
 - Activation and Proliferation of T Helper Cells
 - Generation of the Humoral Response
 - Generation of the Cell-Mediated Response

Summary

References

Study Questions

[\[TOP\]](#)

CHAPTER 2 EXPERIMENTAL SYSTEMS

Experimental Animal Models

- Inbred Strains
- Adoptive-Transfer Systems
- Scid* Mice and *Scid*-Human Mice

Cell-Culture Systems

- Primary Lymphoid Cultures

Cloned Lymphoid Cell Lines

Hybrid Lymphoid Cell Lines

Recombinant DNA Technology

Restriction-Endonuclease Cleavage of DNA

Cloning of DNA Sequences

Cloning Vectors

Cloning of cDNA and Genomic DNA

Selection of DNA Clones

Southern Blotting

Northern Blotting

Polymerase Chain Reaction

Analysis of DNA Regulatory Sequences

DNA Footprinting

Gel-Shift Analysis

CAT Assay

Gene Transfer into Mammalian Cells

Transfer of Cloned Genes into Cultured Cells

Transfer of Cloned Genes into Mouse Embryos

Transgenic Mice

Gene-Targeted Knock-Out Mice

Summary

References

Study Questions

[\[TOP\]](#)

Chapter 3 CELLS AND ORGANS OF THE IMMUNE SYSTEM

Hematopoiesis

Hematopoietic Growth Factors

Regulation of Hematopoiesis

Programmed Cell Death

Regulatory Abnormalities and Leukemia

Enrichment of Hematopoietic Stem Cells

Clinical Uses of Pluripotent Stem Cells

Transplantation of Bone Marrow

Gene Therapy with Engineered Stem Cells

Immune-System Cells

Lymphoid Cells

B lymphocytes

T lymphocytes

Null cells

Mononuclear Cells

Phagocytosis

Antimicrobial and Cytotoxic Activities

- Oxygen-Dependent Killing Mechanisms
- Oxygen-Independent Killing Mechanisms
- Resistant Pathogens

- Antigen Processing and Presentation

- Secretion of Factors

- Granulocytic Cells

- Neutrophils

- Eosinophils

- Basophils

- Mast Cells

- Dendritic Cells

Organs of the Immune System

- Primary Lymphoid Organs

- Thymus

- Maturation and Selection of T Cells

- Relation between Thymic and Immune Function

- Bone Marrow

- Lymphatic System

- Secondary Lymphoid Organs

- Lymph Nodes

- Spleen

- Mucosal-Associated Lymphoid Tissue

- Cutaneous-Associated Lymphoid Tissue

Summary

References

Study Questions

[\[TOP\]](#)

PART II: GENERATION OF B-CELL AND T-CELL RESPONSES

Chapter 4 ANTIGENS

Immunogenicity Versus Antigenicity

Factors That Influence Immunogenicity

- Contribution of the Immunogen to Immunogenicity

- Foreignness

- Molecular Size

- Chemical Composition and Heterogeneity

- Susceptibility to Antigen Processing and Presentation

- Contribution of the Biological System to Immunogenicity

- Genotype of the Recipient Animal

- Immunogen Dosage and Route of Administration

- Adjuvants

Epitopes

Properties of B-Cell Epitopes

Properties of T-Cell Epitopes

Haptens and the Study of Antigenicity

Mitogens

Summary

References

Study Questions

[\[TOP\]](#)

Chapter 5 IMMUNOGLOBULINS: STRUCTURE AND FUNCTION

Basic Structure of Immunoglobulins

Immunoglobulin Sequencing Studies

Role of Multiple Myeloma

Light-Chain Sequencing

Heavy-Chain Sequencing

Immunoglobulin Fine Structure

Immunoglobulin Domains

Variable-Region Domains

CDRs and Antigen Binding

Conformational Changes Induced by Antigen Binding

Constant-Region Domains

C_H1 And C_L Domains

Hinge Region

Other Constant-Region Domains

B-Cell Receptor

Antigenic Determinants on Immunoglobulins

Isotypic Determinants

Allotypic Determinants

Idiotypic Determinants

Immunoglobulin Classes

Immunoglobulin G (IgG)

Immunoglobulin M (IgM)

Immunoglobulin A (IgA)

Immunoglobulin E (IgE)

Immunoglobulin D (IgD)

The Immunoglobulin Superfamily

Monoclonal Antibodies

Formation and Selection of Hybrid Cells

Production of Monoclonal Antibodies

Clinical Uses for Monoclonal Antibodies

Engineered Monoclonal Antibodies

Chimeric and Hybrid Monoclonal Antibodies

Monoclonal Antibodies Constructed from Ig-Gene Libraries

Catalytic Monoclonal Antibodies (Abzymes)

Summary

References

Study Questions

[\[TOP\]](#)

Chapter 6 ANTIGEN-ANTIBODY INTERACTIONS

Strength of Antigen-Antibody Interactions

Antibody Affinity

Antibody Avidity

Cross-Reactivity

Precipitation Reactions

Precipitation Reactions in Fluids

Precipitation Reactions in Gels

Radial Immunodiffusion (Mancini Method)

Double Immunodiffusion (Ouchterlony Method)

Immunoelectrophoresis

Agglutination Reactions

Hemagglutination

Bacterial Agglutination

Passive Agglutination

Agglutination Inhibition

Radioimmunoassay

Enzyme-Linked Immunosorbent Assay

Indirect ELISA

Sandwich ELISA

Competitive ELISA

Western Blotting

Immunofluorescence

Immunoelectron Microscopy

Summary

References

Study Questions

[\[TOP\]](#)

Chapter 7 ORGANIZATION AND EXPRESSION OF IMMUNOGLOBULIN GENES

Genetic Model Compatible with Ig Structure

Germ-Line and Somatic-Variation Models

Dryer and Bennett Two-Gene Model

Early Verification of the Dryer and Bennett Hypothesis

Multigene Organization of Ig Genes

λ -Chain Multigene Family

κ -Chain Multigene Family

Heavy-Chain Multigene Family

Variable-Region Gene Rearrangements

V-J Rearrangements in Light-Chain DNA

V-D-J Rearrangements in Heavy-Chain DNA

Mechanism of Variable-Region DNA Rearrangements

Recombination Signal Sequences

Enzymatic Joining of Gene Segments

Experimental Identification of *Rag-1* and *Rag-2* Genes

Defects in Ig-Gene Rearrangement

Productive and Nonproductive Rearrangements

Allelic Exclusion

Generation of Antibody Diversity

Multiple Germ-Line V, D, and J Gene Segments

Combinatorial V-J and V-D-J Joining

Junctional Flexibility

P-Nucleotide Addition

N-Region Nucleotide Addition

Somatic Hypermutation

Association of Heavy and Light Chains

Class Switching among Constant-Region Genes

Expression of Ig Genes

Differential RNA Processing of Heavy-Chain Primary Transcripts

Expression of Membrane-Bound or Secreted Immunoglobulin

Simultaneous Expression of IgM and IgD

Synthesis, Assembly, and Secretion of Immunoglobulins

Regulation of Ig-Gene Transcription

Effect of DNA Rearrangement on Transcription

Inhibition of Ig-Gene Expression in T Cells

Summary

References

Study Questions

[\[TOP\]](#)

Chapter 8 B-CELL MATURATION, ACTIVATION, AND DIFFERENTIATION

B-Cell Maturation

Bone Marrow Environment

Ig-Gene Rearrangements

Pre-B Cell Receptor

Selection of Immature Self-Reactive B Cells

Negative Selection

Rescue by Editing of Light-Chain Genes

B-Cell Activation and Proliferation

Thymus-Dependent and Thymus-Independent Antigens

Origin of Activating Signals

Transduction of Activating Signals

B-Cell Coreceptor Complex

Role of T_H Cells in Humoral Response

Formation of T-B Conjugate

Contact-Dependent Help Mediated by CD40/CD40L Interaction

Progression Signals Induced by T_H-Cell Cytokines

Negative Selection of Mature, Self-Reactive B Cells

In Vivo Sites for Induction of Humoral Response

B-Cell Differentiation

Affinity Maturation

Role of Somatic Hypermutation

Antigen Selection of High-Affinity Centrocytes

Class Switching

Generation of Plasma Cells and Memory B Cells

Regulation of B-Cell Development

Summary

References

Study Questions

[\[TOP\]](#)

Chapter 9 MAJOR HISTOCOMPATIBILITY COMPLEX

General Organization and Inheritance of the MHC

Location and Function of MHC Regions

MHC Haplotypes

Congenic MHC Mouse Strains

MHC Molecules and Genes

Structure of Class I Molecules

Structure of Class II Molecules

Organization of Class I and II Genes

Peptide Binding by MHC Molecules

Class I MHC∅Peptide Interaction

Class II MHC∅Peptide Interaction

Polymorphism of Class I and II Molecules

Class III Molecules

Detailed Genomic Map of MHC Genes

Map of Class I MHC

Map of Class II MHC

Map of Class III MHC

Cellular Distribution of MHC Molecules

Regulation of MHC Expression

MHC and Immune Responsiveness

Determinant-Selection Model

Holes-in-the-Repertoire Model

MHC and Susceptibility to Infectious Disease

Summary

References

Study Questions

[\[TOP\]](#)

Chapter 10 ANTIGEN PROCESSING AND PRESENTATION

Self-MHC Restriction of T Cells

Role of Antigen-Presenting Cells

Early Evidence for Necessity of Antigen Processing

Cells That Function in Antigen Presentation

Evidence for Two Processing and Presentation Pathways

Endogenous Antigens: The Cytosolic Pathway

Peptide Generation by Cytosolic Proteasomes

Peptide Transport from the Cytosol to the RER

Assembly of Peptides with Class I MHC Molecules

Overview of Class I Endogenous Pathway

Exogenous Antigens: The Endocytic Pathway

Peptide Generation in Endocytic Vesicles

Transport of Class II MHC Molecules to Endocytic Vesicles

Assembly of Peptides with Class II MHC Molecules

Overview of Class II Exogenous Pathway

Clinical Applications

Summary

References

Study Questions

[\[TOP\]](#)

Chapter 11 T-CELL RECEPTOR

Early Studies of the T-Cell Receptor

Self-MHC Restriction of the T-cell Receptor

Isolation of T-Cell Receptors

Structure of T-Cell Receptors

Organization and Rearrangement of TCR Genes

Identifying and Cloning the TCR Genes

TCR Multigene Families

Variable-Region Gene Rearrangements

Mechanism of TCR DNA Rearrangements

Allelic Exclusion of TCR Genes

Structure of Rearranged TCR Genes

Generation of TCR Diversity

T-Cell Receptor Complex: TCR-CD3

T-Cell Accessory Membrane Molecules

CD4 and CD8 Coreceptors

Other Accessory Membrane Molecules

Ternary TCR-Peptide-MHC Complex

Evidence That TCR Alone Recognizes Peptide-MHC Complex

Interactions Involved in Forming Ternary Complex

Affinity of TCR for Peptide-MHC Complexes

Influence of Peptide on Topology of Ternary Complex

Alloreactivity of T cells

Summary

References

Study Questions

[\[TOP\]](#)

Chapter 12 T-CELL MATURATION, ACTIVATION, AND DIFFERENTIATION

T-Cell Maturation

Pre-T Cell Receptor

T-Cell Developmental Pathways

Thymic Selection of the T-Cell Repertoire

Positive Selection

Negative Selection

Experimental Evidence for Positive Selection

Experimental Evidence for Negative Selection

Unsolved Questions Regarding Thymic Selection

T_H-Cell Activation

TCR-Coupled Signaling Pathways

The Co-Stimulatory Signal

Altered Peptide Ligands as Antagonists or Partial Agonists

Clonal Expansion Versus Clonal Anergy

Superantigen-Induced T-Cell Activation

T-Cell Differentiation

Generation of Effector and Memory Cells

Co-Stimulatory Differences among Antigen-Presenting Cells

Peripheral $\gamma\delta$ T Cells

Distribution of $\gamma\delta$ T Cells

Ligands Recognized by $\gamma\delta$ T Cells

Functions of $\gamma\delta$ T Cells

Summary

References

Study Questions

[\[TOP\]](#)

PART III: IMMUNE EFFECTOR MECHANISMS

Chapter 13 CYTOKINES

Properties of Cytokines

General Structure of Cytokines

Function of Cytokines

Cytokine Receptors

General Structure of Cytokine Receptors

Subfamilies of Class I Cytokine Receptors

IL-2 Receptor

Signal Transduction Mediated by Cytokine Receptors

Cytokine Antagonists

Cytokine Secretion by T_H1 and T_H2 Subsets

Development of T_H1 and

T_H2 Subsets

Cytokine Cross-Regulation

Role of T_H1/T_H2

Balance in Determining Disease Outcomes

Cytokine-Related Diseases

Bacterial Septic Shock

Bacterial Toxic Shock and Similar Diseases

Lymphoid and Myeloid Cancers

Chagas' Disease

Therapeutic Uses of Cytokines and Their Receptors

Summary

References

Study Questions

[\[TOP\]](#)

Chapter 14 THE COMPLEMENT SYSTEM

The Complement Components

Complement Activation

Classical Pathway

Alternative Pathway

Terminal Sequence: Formation of Membrane-Attack Complex

Regulation of the Complement System

Biological Consequences of Complement Activation

Cell Lysis

Inflammatory Response

Opsonization of Antigen

Viral Neutralization

Solubilization of Immune Complexes

Complement Deficiencies

Summary

References

Study Questions

[\[TOP\]](#)

Chapter 15 LEUKOCYTE MIGRATION AND INFLAMMATION

Lymphocyte Recirculation

Cell-Adhesion Molecules

- Selectin Family

- Mucin-Like Family

- Integrin Family

- Immunoglobulin Superfamily

Neutrophil Extravasation

Lymphocyte Extravasation

- High-Endothelial Venules (HEVs)

- Homing of Lymphocytes

 - Naive Lymphocyte

 - Effector and Memory Lymphocytes

- Adhesion-Molecule Interactions

Mediators of Inflammation

- Chemokines

- Plasma Enzyme Mediators

 - Kinin System

 - Clotting System

 - Fibrinolytic System

 - Complement System

- Lipid Inflammatory Mediators

- Cytokine Inflammatory Mediators

The Inflammatory Process

- Central Role of Neutrophils in Inflammation

- Acute Inflammatory Response

 - Localized Response

 - Systemic Acute-Phase Response

- Chronic Inflammatory Response

 - Role of IFN- γ and TNF- α

 - Chronic Inflammatory Diseases

Anti-Inflammatory Agents

- Agents That Reduce Leukocyte Extravasation

 - Corticosteroids

 - Nonsteroidal Anti-Inflammatory Drugs

Summary

References

Study Questions

[\[TOP\]](#)

Chapter 16 CELL-MEDIATED AND HUMORAL EFFECTOR RESPONSES

Effector Responses in Cell-Mediated Branch

- General Properties of Effector T Cells

 - Activation Requirements

 - Cell-Adhesion Molecules

 - Effector Molecules

- Direct Cytotoxic Responses

 - CTL-Mediated Cytotoxicity

 - Phase 1: Generation of CTLs

 - Phase 2: Destruction of Target Cells

- NK Cell-Mediated Cytotoxicity

 - NK-Cell Lineage

 - Mechanism of NK-Cell Killing

 - NK-Cell Receptors

- Antibody-Dependent Cell-Mediated Cytotoxicity

- Experimental Assessment of Cell-Mediated Cytotoxicity

 - Mixed-Lymphocyte Reaction (MLR)

 - Cell-Mediated Lympholysis (CML)

 - Graft-Versus-Host Reaction

- Delayed-Type Hypersensitivity

 - Phases of the DTH Response

 - Cytokines Involved in the DTH Reaction

 - Protective Role of the DTH Response

 - Detection of the DTH Response

 - Pathologic DTH Responses

Effector Responses in Humoral Branch

- Primary and Secondary Responses

- Experimental Assessment of Humoral Immunity

 - Hemolytic Plaque Assays

 - Elispot Assay

- Humoral Response to Hapten-Carrier Conjugates

Regulation of Immune Effector Responses

- Antigen-Mediated Regulation

- Antibody-Mediated Suppression

- Immune Complexes as Regulators

- Idiotypic Regulation: The Network Theory

- Neuroendocrine Regulation (Neuroimmunomodulation)

Summary

References

Study Questions

[\[TOP\]](#)

Chapter 17 HYPERSENSITIVE REACTIONS

Gell and Coombs Classification

IgE-Mediated (Type I) Hypersensitivity

Components of Type I Reactions

- Allergens

- Regained Antibody (IgE)

- Mast Cells and Basophils

- IgE-Binding Fc Receptors

 - High-Affinity Receptor (Fc ϵ RI)

 - Low-Affinity Receptor (Fc ϵ RII)

Mechanism of IgE-Mediated Degranulation

- Receptor Cross-Linkage

- Intracellular Events Leading to Mast Cell Degranulation

Mediators of Type I Reactions

- Histamine

- Leukotrienes and Prostaglandins

- Cytokines

Consequences of Type I Reactions

- Systemic Anaphylaxis

- Localized Anaphylaxis (Atopy)

 - Allergic Rhinitis

 - Asthma

 - Food Allergies

 - Atopic Dermatitis

- Late-Phase Reaction

Regulation of the Type I Response

Detection of Type I Hypersensitivity

Therapy for Type I Hypersensitivities

Antibody-Mediated Cytotoxic (Type II) Hypersensitivity

- Transfusion Reactions

- Hemolytic Disease of the Newborn

- Drug-Induced Hemolytic Anemia

Immune Complex-Mediated (Type III) Hypersensitivity

- Localized Type III Reactions

- Generalized Type III Reactions

T_{DTH}-Mediated (Type IV) Hypersensitivity

Summary

References

Study Questions

[\[TOP\]](#)

PART IV: THE IMMUNE SYSTEM IN HEALTH AND DISEASE

Chapter 18 VACCINES

Active and Passive immunization

Passive Immunization

Active Immunization

Designing Vaccines for Active Immunization

Whole-Organism Vaccines

Attenuated Viral or Bacterial Vaccines

Inactivated Viral or Bacterial Vaccines

Purified Macromolecules As Vaccines

Polysaccharide Vaccines

Toxoid Vaccines

Recombinant Antigen Vaccines

Recombinant Vector Vaccines

DNA Vaccines

Synthetic Peptide Vaccines

Multivalent Subunit Vaccines

Anti-Idiotypic Vaccines

Summary

References

Study Questions

[\[TOP\]](#)

Chapter 19 IMMUNE RESPONSE TO INFECTIOUS DISEASES

Viral Infections

Viral Neutralization by Humoral Antibody

Cell-Mediated Antiviral Mechanisms

Viral Evasion of Host-Defense Mechanisms

Influenza

Properties of Influenza Virus

Host Response to Influenza Infection

Bacterial Infections

Immune Response to Extracellular and Intracellular Bacteria

Bacterial Evasion of Host-Defense Mechanisms

Contribution of the Immune Response to Bacterial Pathogenesis

Diphtheria (*Corynebacterium diphtheriae*)

Tuberculosis (*Mycobacterium tuberculosis*)

Lyme Disease (*Borrelia burgdorferi*)

Protozoan Diseases

Malaria (*Plasmodium* Species)

Plasmodium Life Cycle and Pathogenesis of Malaria

Host Response to *Plasmodium* Infection

Design of Malaria Vaccines

African Sleeping Sickness (*Trypanosoma* Species)

Diseases Caused by Parasitic Worms (Helminths)

Summary

References

Study Questions

[\[TOP\]](#)

Chapter 20 AUTOIMMUNITY

Organ-Specific Autoimmune Diseases

Diseases Mediated by Direct Cellular Damage

Hashimoto's Thyroiditis

Autoimmune Anemias

Goodpasture's Syndrome

Insulin-Dependent Diabetes Mellitus

Diseases Mediated by Stimulating or Blocking Antibodies

Graves' Disease

Myasthenia Gravis

Systemic Autoimmune Diseases

Systemic Lupus Erythematosus

Multiple Sclerosis

Rheumatoid Arthritis

Animal Models for Autoimmune Disease

Spontaneous Autoimmunity in Animals

Experimentally Induced Autoimmunity in Animals

Evidence Implicating the CD4⁺ T Cell, MHC, and TCR in Autoimmunity

Role of CD4⁺ T-Cells and T_H1/T_H2 Balance

Association with the MHC

Association with the T-Cell Receptor

Proposed Mechanisms for Induction of Autoimmunity

Release of Sequestered Antigens

Molecular Mimicry

Mimicry between MBP and Viral Peptides

Mimicry Involving Heat-Shock Proteins

Inappropriate Expression of Class II MHC Molecules

Polyclonal B-Cell Activation

Treatment of Autoimmune Diseases

Current Therapies

Experimental Therapeutic Approaches

T-Cell Vaccination

Peptide Blockade of MHC Molecules

Monoclonal-Antibody Treatment

Tolerance Induction by Oral Antigens

Summary

References

Study Questions

[\[TOP\]](#)

Chapter 21 IMMUNODEFICIENCY DISEASES

Phagocytic Deficiencies

- Reduction in Neutrophil Count
- Defective Phagocytic Function
 - Adherence Defects
 - Chemotactic Defects
 - Killing Defects

Humoral Deficiencies

- X-Linked Agammaglobulinemia
- X-Linked Hyper-IgM Syndrome
- Common Variable Hypogammaglobulinemia
- Selective Immunoglobulin Deficiencies

Cell-Mediated Deficiencies

- Digeorge Syndrome (Congenital Thymic Aplasia)
- Nude Mice

Combined Immunodeficiencies

- Reticular Dysgenesis
- Bare-Lymphocyte Syndrome
- Severe Combined Immunodeficiency Disease (SCID)
 - ADA Deficiency and PNP Deficiency
 - Scid* Mice and *Scid*-Human Mice
- Wiskott-Aldrich Syndrome

Complement Deficiencies

Summary

References

Study Questions

[\[TOP\]](#)

Chapter 22 THE IMMUNE SYSTEM IN AIDS

Discovery of AIDS and Its Causative Agent

Human Immunodeficiency Virus (HIV)

- Relation between HIV and Other Retroviruses
- Structure of HIV
- HIV Infection of Target Cells
 - Entry of HIV into Cells
 - Integration of HIV into Host Genome
 - Activation of HIV Provirus
- Transmission of HIV

HIV Genome

Factors That Promote Activation of HIV Provirus

Expression of HIV Proviral DNA

Genetic Variation in HIV

Role of Immune Response in Emergence of HIV Variants

Variation in the Envelope Glycoproteins

Diagnosis of HIV Infection and AIDS

Serologic Profile of HIV Infection

Screening Tests for HIV Infection

Clinical Diagnosis of AIDS

Destruction of CD4⁺ T Cells

Depletion of HIV-Infected CD4⁺ T Cells

Depletion of Uninfected CD4⁺ T Cells

Destruction Mediated by Anti-gp120 Antibody

Gp120-Induced Disruption of T-Cell Activation

Gp120-Stimulated Syncytia Formation

Interference with T-Cell Maturation by gp120

Immunologic Abnormalities in AIDS

Pathologic Changes in Lymph Nodes

Reduced Antigen-Specific Responses by T_H Cells

Ineffective Antibody Response

Cytokine Imbalance

Decreased DTH Response

Impaired CTL Activity

Disturbances in Immune Regulation

Development of an AIDS Vaccine

Obstacles to Development of an AIDS Vaccine

Experimental AIDS Vaccines

Inactivated Whole Viruses

Cloned Envelope Glycoproteins

Attenuated Viruses

Recombinant Viruses Carrying HIV Genes

Synthetic Crown-Sequence Peptides

Summary

References

Study Questions

[\[TOP\]](#)

Chapter 23 TRANSPLANTATION IMMUNOLOGY

Immunologic Basis of Graft Rejection

Specificity and Memory of Rejection Response

Role of Cell-Mediated Responses

Transplantation Antigens

Mechanisms Involved in Graft Rejection

Sensitization Stage

Effector Stage

Clinical Manifestations of Graft Rejection

Hyperacute Rejection

Acute Rejection

Chronic Rejection

Tissue Typing

General Immunosuppressive Therapy

Mitotic Inhibitors

Corticosteroids

Cyclosporin A, FK506, and Rapamycin

Total Lymphoid Irradiation

Specific Immunosuppressive Therapy

Monoclonal Antibodies to T-Cell Components or Cytokines

Agents That Block the Co-Stimulatory Signal

Donor-Cell Microchimerism

Clinical Transplantation

Bone Marrow Transplants

Organ Transplants

Xenotransplants

Transplants to Immunologically Privileged Site

Summary

References

Study Questions

[\[TOP\]](#)

Chapter 24 CANCER AND THE IMMUNE SYSTEM

Cancer: Origin and Terminology

Malignant Transformation of Cells

Oncogenes and Cancer Induction

Function of Oncogenes

Induction of Cellular Proliferation

Inhibition of Cellular Proliferation

Regulation of Programmed Cell Death

Conversion of Proto-Oncogenes to Oncogenes

Induction of Cancer: A Multistep Process

Tumors of the Immune System

Tumor Antigens

Tumor-Specific Antigens

Chemically or Physically Induced Tumor Antigens

Virally Induced Tumor Antigens

Tumor-Associated Antigens

- Oncofetal Tumor Antigens
- Oncogene Proteins as Tumor Antigens
- TATAs on Human Melanomas

Immune Response to Tumors

- Role of NK Cells and Macrophages
- Immune Surveillance Theory

Tumor Evasion of the Immune System

- Immunologic Enhancement of Tumor Growth
- Modulation of Tumor Antigens
- Reduction in Class I MHC Molecules
- Lack of Co-Stimulatory Signal

Cancer Immunotherapy

- Manipulation of Co-Stimulatory Signal
- Enhancement of APC Activity
- Cytokine Therapy
 - Interferons
 - Tumor Necrosis Factors
 - In Vitro Activated LAK And TIL Cells
- Monoclonal Antibodies
- Tumor-Cell Vaccines

Summary

References

Study Questions

GLOSSARY

ANSWERS

CREDITS

INDEX

[\[TOP\]](#)

[\[Immunology Home Page\]](#)

W. H. Freeman & Co. and Sumanas, Inc.
Immunology, January, 1997

Chapter 1: Overview of the Immune System

"Innate" or "nonspecific" immunity refers to various physical and cellular attributes that collectively represent the first lines of defense against infectious disease. Obviously, the skin and the extensive epithelial lining of the mucosal tissues act as nonspecific barriers impeding, if not preventing, infectious agents from entering the body. However, once these barriers are penetrated, in a [puncture wound](#) for example, infectious agents usually encounter the activities of "defensive" cells, either resident to the site or [recruited](#) there from the blood by [chemotactic](#) signals which promote [diapedesis](#), or [extravasation](#) ([Figure 1-5](#)). Most notable among the early defensive cells are [phagocytes](#) -- mainly [macrophages](#), neutrophils, and dendritic cells -- which destroy an invading organism by [phagocytosis](#), a multi-step process beginning with the complete engulfment of the organism and ending with modification and chemical breakdown of its structural components ([Figure 1-4](#)). A host of toxic chemical reactions may occur in phagocytes including biomolecular breakdown by digestive enzymes and chemical modification by highly reactive oxygen (and nitrogen) intermediates generated by the "[oxidative burst](#)" reaction.

One consequence of phagocytosis is that macrophages and dendritic cells "present" antigens on their cell surfaces corresponding to the chemical breakdown products of the organism. Antigens presented in this way activate specific immune lymphocytes against the organism and its associated components ([Figure 1-14](#)). This in turn leads to acquired immunity as defined by the antigen-specific activities of immune lymphocytes and their effects on other cells ([Figure 1-8](#)).

© 1997, Duane W. Sears

[\[TOP\]](#)

[\[Chapter Resources\]](#)

[\[Immunology Home Page\]](#)

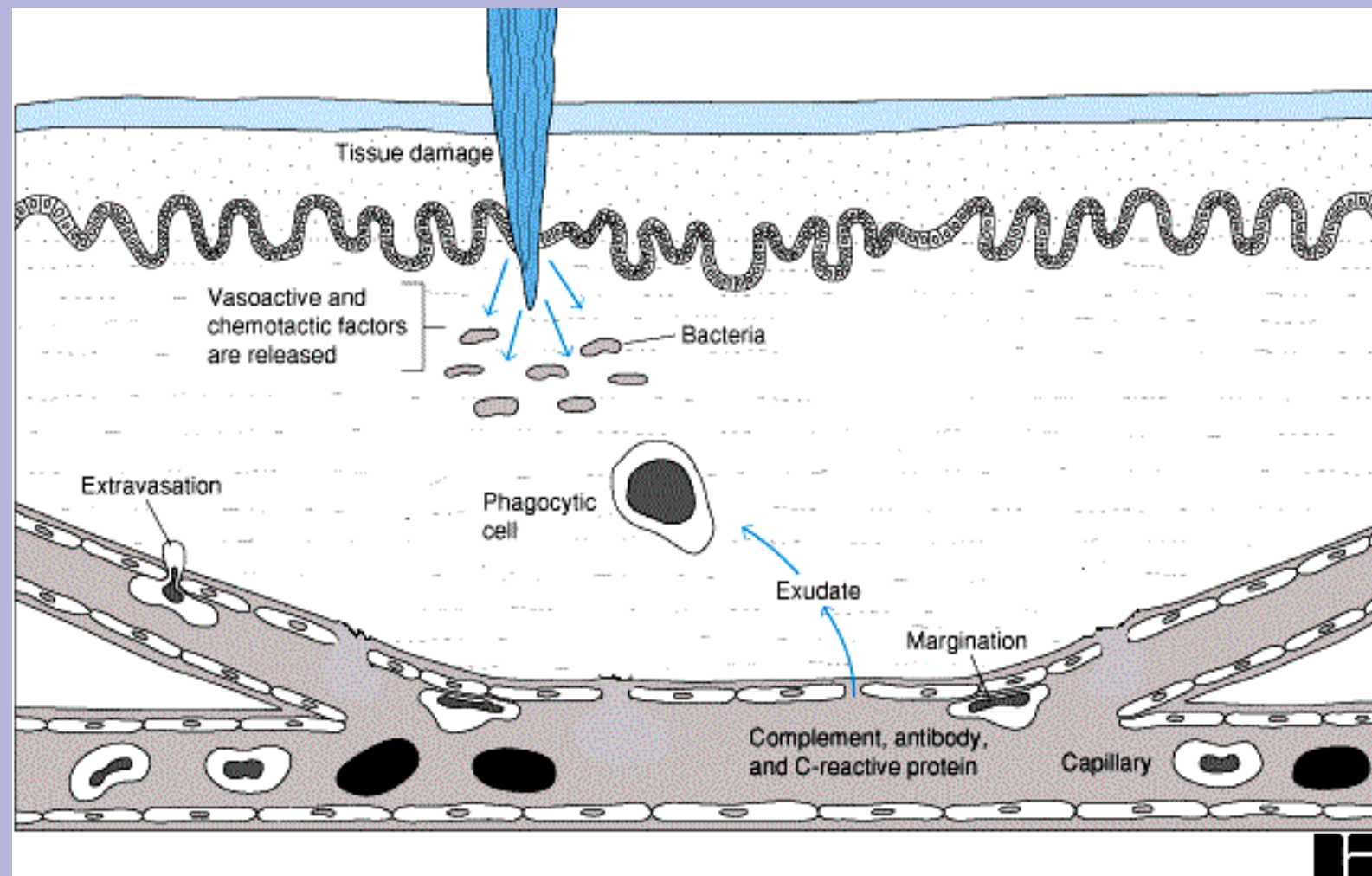
W. H. Freeman & Co. and Sumanas, Inc.
Immunology, January, 1997

Last Updated: June 24, 1997

Figure 1-5: Diapedesis (Extravasation) & Chemotaxis

Leukocytes cross the endothelial lining of the capillary and migrate interstitially by the process called **diapedesis** or **extravasation** as illustrated in the image below. Chemotactic factors drive this process, particularly during an inflammatory response when immune-related cells migrate to and become concentrated at the site of inflammation.

Major events in the inflammatory response. A bacterial infection causes tissue damage with release of various vasoactive and chemotactic factors. These factors induce increased blood flow to the area, increased capillary permeability, and an influx of white blood cells, including phagocytes and lymphocytes, from the blood into the tissues. The serum proteins contained in the exudate have antibacterial properties, and the phagocytes begin to engulf the bacteria.



© 1997, Duane W. Sears

[\[TOP\]](#)

[\[Chapter 1: Overview of the Immune System\]](#)

[\[Chapter Resources\]](#)

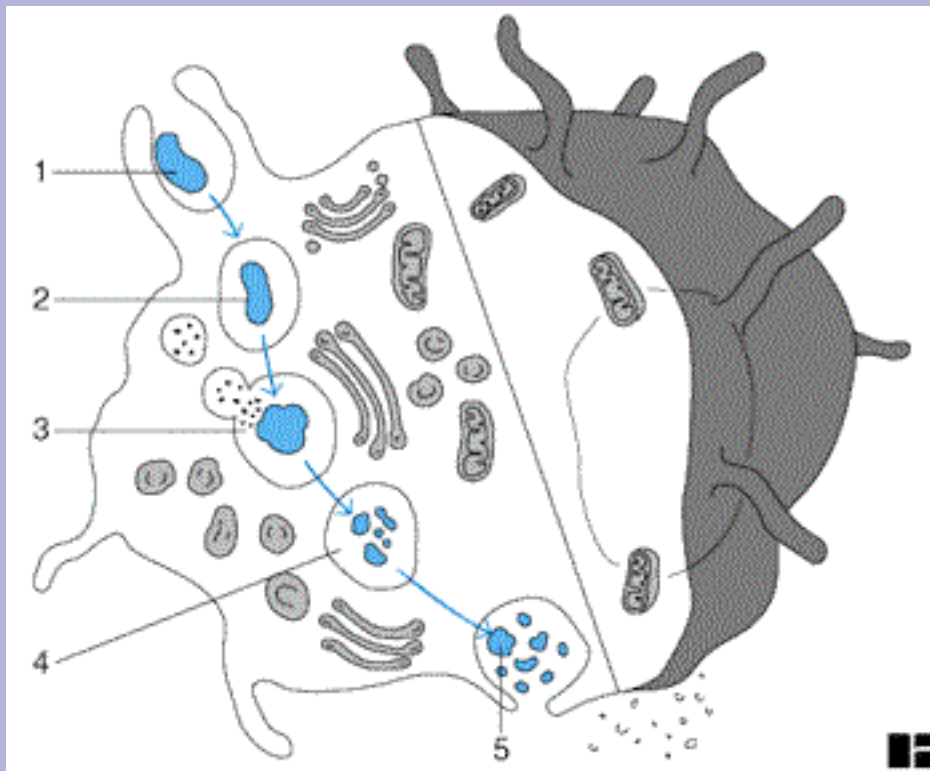
[\[Immunology Home Page\]](#)

W. H. Freeman & Co. and Sumanas, Inc.
Immunology, January, 1997

Figure 1-4: Phagocytosis

Phagocytosis is a process describing the engulfment and destruction of extracellularly-derived materials by phagocytic cells, such as macrophages and neutrophils. Five steps of phagocytosis are illustrated in the image below.

Phagocytosis of bacteria



Schematic diagram of the steps in phagocytosis:

1. Attachment of the bacterium to the long membrane evaginations, called pseudopodia.
2. Ingestion of the bacterium forming a "phagosome," which moves toward the lysosome.
3. Fusion of the lysosome and phagosome, releasing lysosomal enzymes into the phagosome.
4. Digestion of the ingested material.
5. Release of digestion products from the cell.

© 1997, Duane W. Sears

[\[TOP\]](#)

[\[Chapter 1: Overview of the Immune System\]](#)

[\[Chapter Resources\]](#)

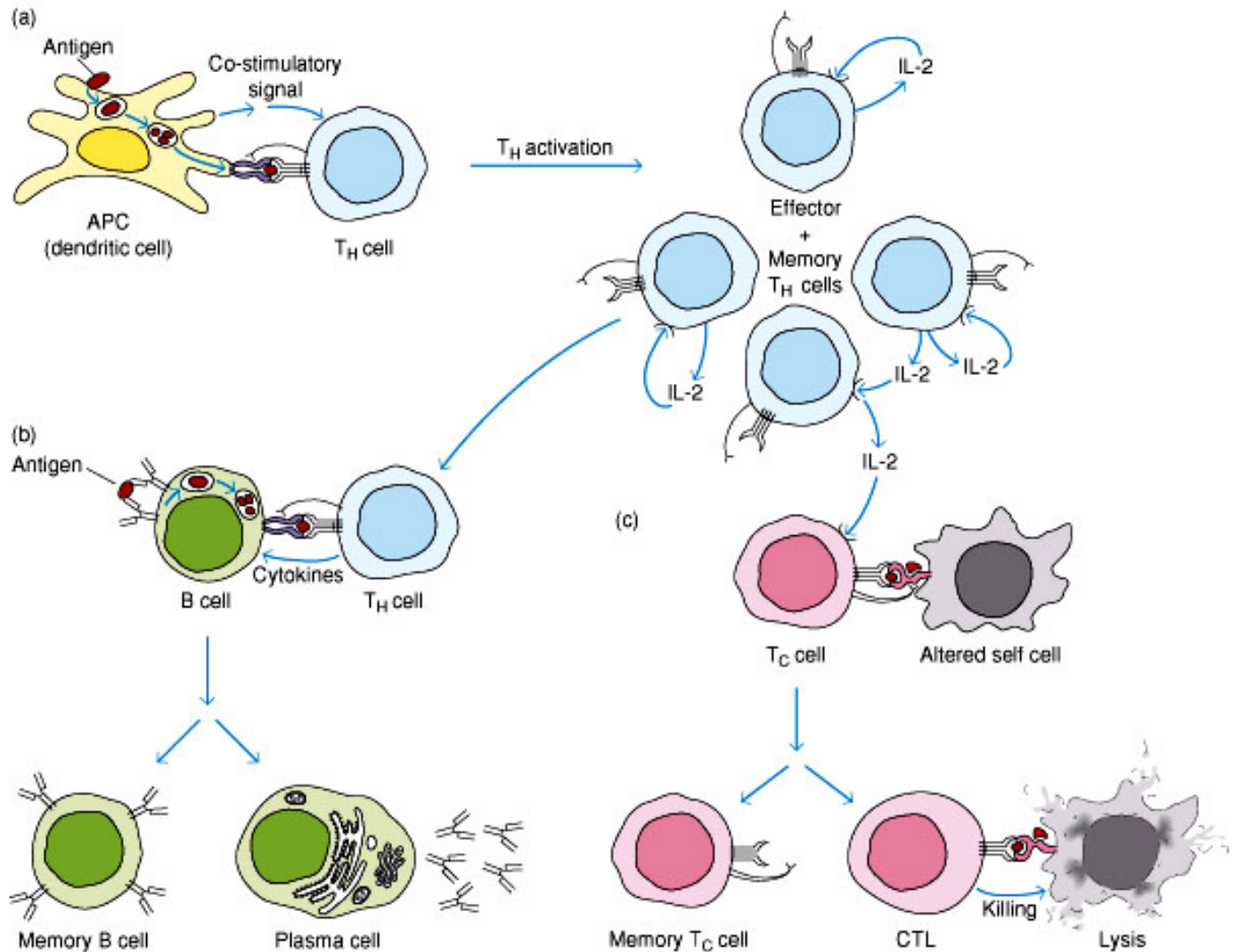
[\[Immunology Home Page\]](#)

W. H. Freeman & Co. and Sumanas, Inc.

Immunology, January, 1997

Figure 1-14: Cellular Interactions

Cellular interactions involved in induction of immune responses. Activation and proliferation of T_H cells. (a) is required for generation of humoral response (b) and cell-mediated response to altered self-cells (c). APC = antigen-presenting cell; Ag = antigen.



W. H. Freeman & Co. and Sumanas, Inc.
Immunology, January, 1997

Figure 1-8: Cellular and Humoral Immunity

Overview of the humoral and cell-mediated branches of the immune system. The humoral response involves interaction of B cells with antigen (Ag) and their differentiation into antibody-secreting plasma cells. The secreted antibody (Ab) binds to the antigen and facilitates its clearance from the body. The cell-mediated responses involve various subpopulations of T cells that recognize antigen presented on self-cells. T_H cells respond to antigen by producing cytokines. T_C cells respond to antigen by developing into cytotoxic T lymphocytes (CTLs), which mediate killing of altered self-cells (e.g., virus-infected cells).

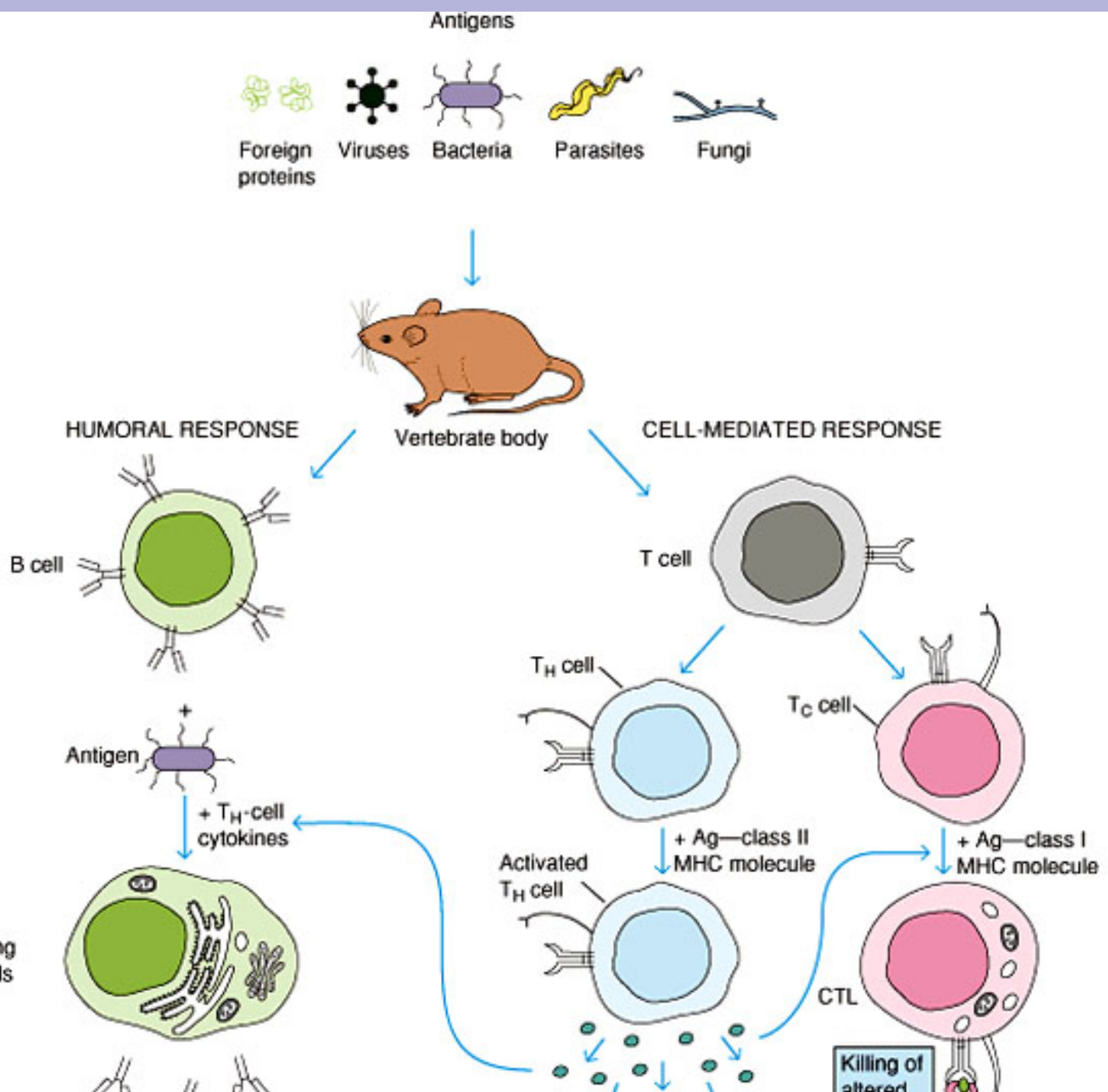
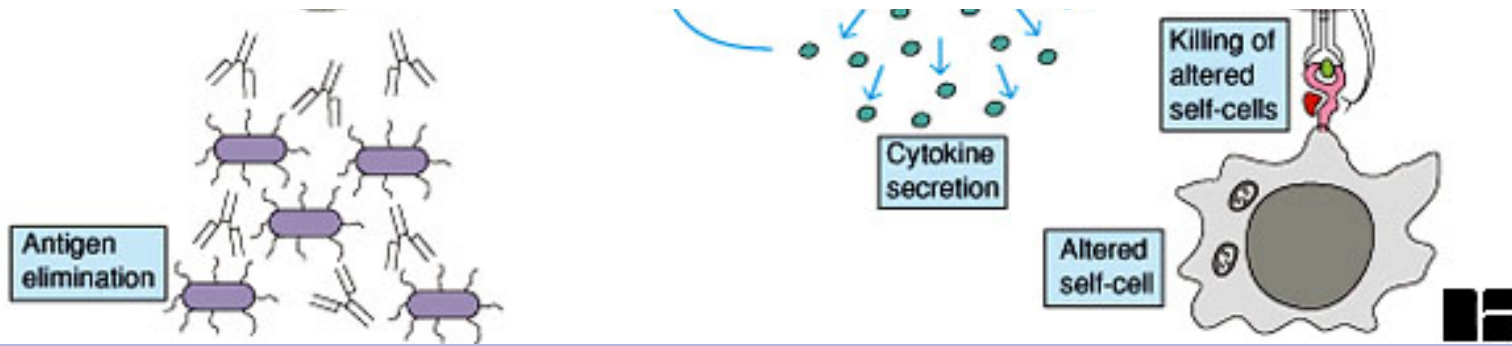


Figure 1-8



© 1997, Duane W. Sears

[\[TOP\]](#)

[\[Chapter 1: Overview of the Immune System\]](#)

[\[Chapter Resources\]](#)

[\[Immunology Home Page\]](#)

W. H. Freeman & Co. and Sumanas, Inc.
Immunology, January, 1997

Chapter 2: Experimental Systems

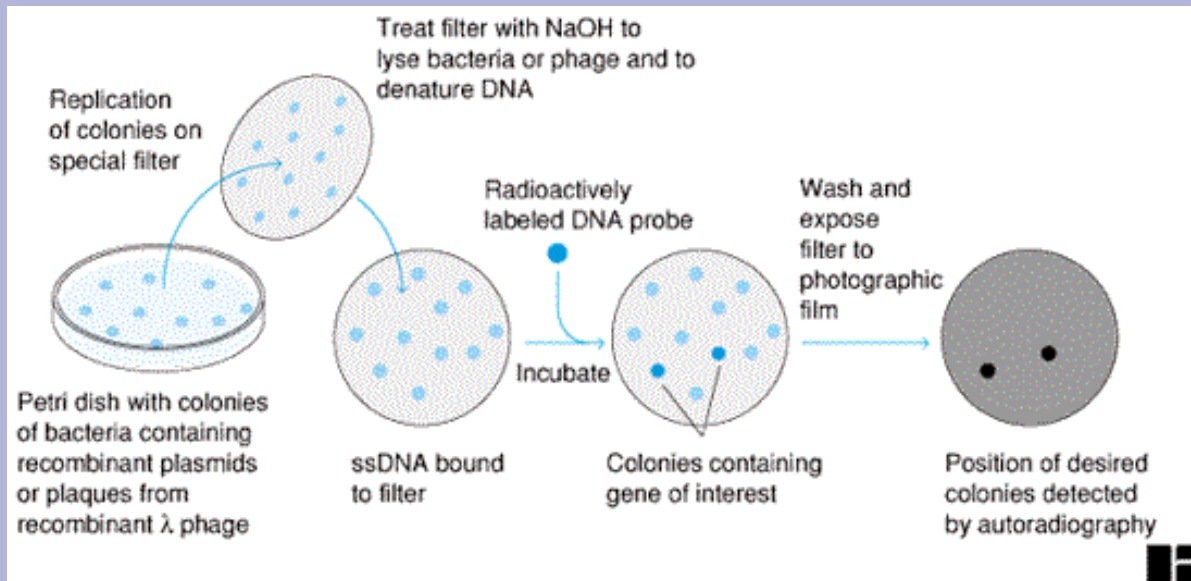
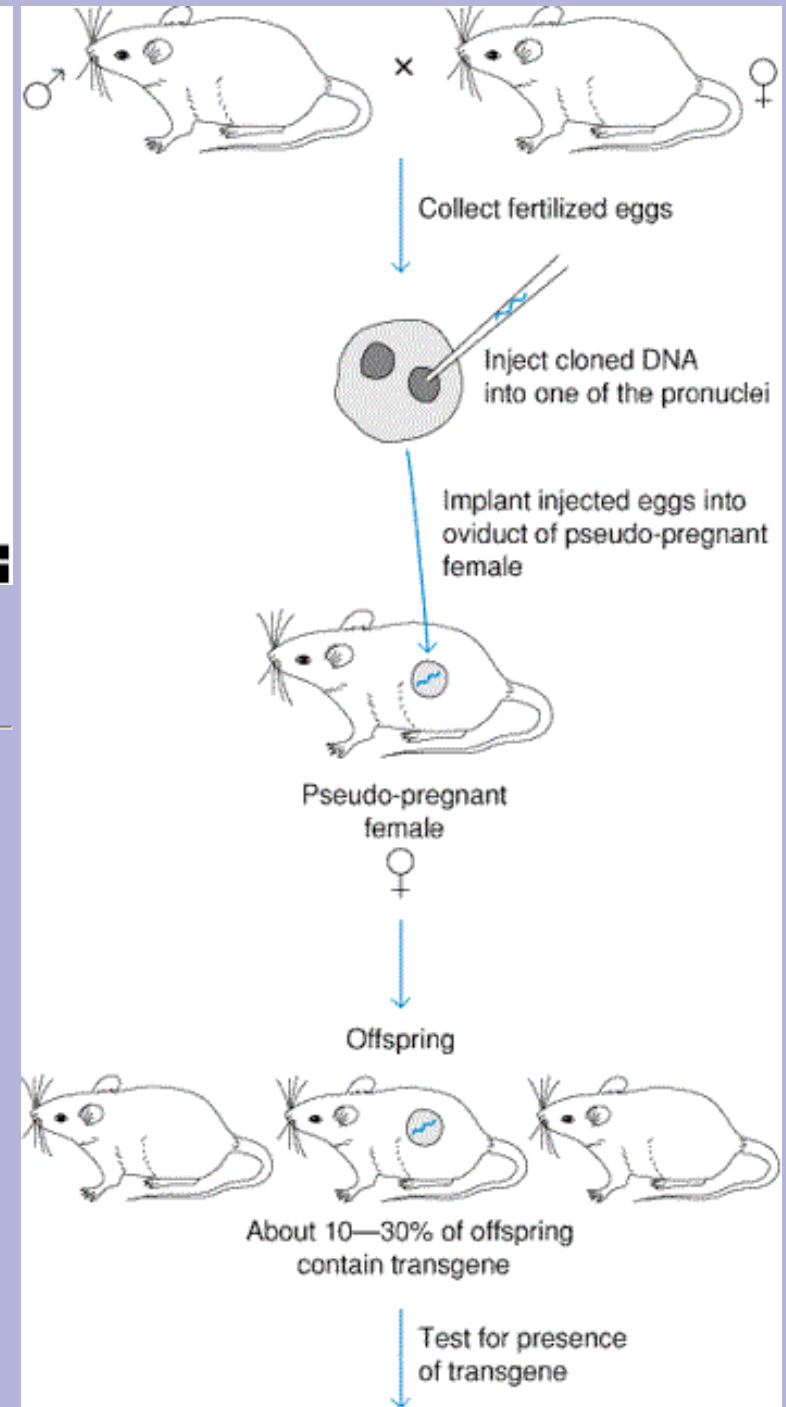


Figure 2-5. Selection of specific clones from a cDNA or genomic DNA library by in situ hybridization.

Chapter 2:
Experimental Systems

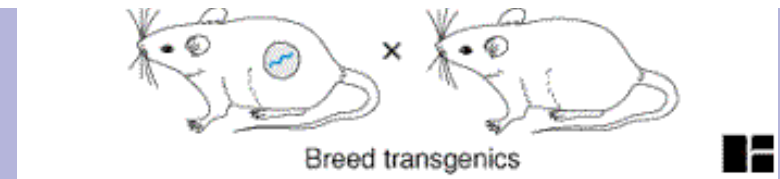


Figure 2-11. General procedure for producing transgenic mice.

Our basic understanding of immunology, like other areas of modern biology, has undergone an astounding transformation in the past thirty years. No doubt, much of this transformation can be traced to the extraordinary wealth of powerful experimental tools and methodologies that have emerged for analyzing biological systems in general. Many of these methodologies are unified under the common theme of "cloning." For example, countless genes encoding numerous immune molecules have now been cloned. Also, many of the protein products encoded by these genes have been isolated to near homogeneity from cloned cell lines. For example, a vast number of monoclonal antibodies have been isolated from cloned, antibody-producing **hybridoma cells** ([Figure 2-2](#)) many of which are available from the [American Type Culture Collection](#) along with many other types of cloned, immunologically-related cell lines. Finally, "cloned" (*i.e.*, genetically inbred) mice are also available for immunological study from places like the [The Jackson Laboratory](#) whose animal stocks include mice with immunodeficiency disorders and also a rapidly growing number of transgenic and targeted gene-knockout mice. With their inheritable DNA permanently altered by experimental design, transgenic and targeted gene-knockout mice will no doubt play a central role in revolutionizing our understanding of the immune system in years to come as immunologists shift their focus from the individual components defining the immune system to the complex physiological and functional interplay between these components in the whole individual.

The [American Type Culture Collection](#) (ATCC) serves as a repository for biological samples made available for biomedical research at nominal commercial fees.

Included among their stocks are:

- hundreds of hybridoma cell lines producing a variety of monoclonal antibodies
- a variety of cell-type-specific tumor cell lines derived from humans and other species (such as those listed in Kuby's Table 2-2)
- recombinant DNA molecules cloned from several species
- microorganisms

[The Jackson Laboratory](#) houses and breeds a phenomenal number of genetically defined mice that are commercially available for biomedical research.

Included among their stocks are:

- most standard inbred mouse strains (*e.g.*, see Kuby's [Table 2-1](#))
- transgenic mice
- targeted gene-knockout mice

© 1997, Duane W. Sears

[\[TOP\]](#)

[\[Chapter Resources\]](#)

[\[Immunology Home Page\]](#)

W. H. Freeman & Co. and Sumanas, Inc.
Immunology, January, 1997

Last Updated: June 24, 1997

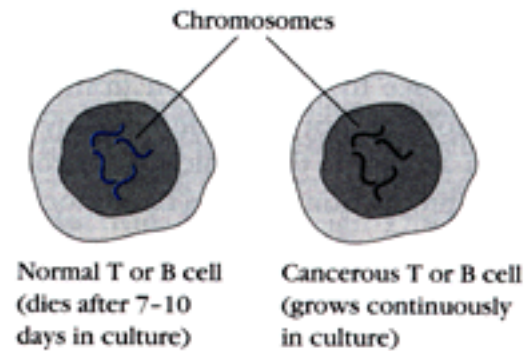
Figure 2-2: Hybridomas

Figure 2-2: Hybridomas

Production of B-cell and T-cell hybridomas by somatic-cell hybridization.

The resulting hybridomas express some of the genes of the original normal B or T cell but also exhibit the immortal-growth properties of the tumor cell.

This procedure is used to produce B-cell hybridomas secreting **monoclonal antibody** and T-cell hybridomas secreting various growth factors (*e.g.*, **IL-2**).



© 1997, Duane W. Sears

[\[TOP\]](#)

[\[Chapter 2: Experimental Systems\]](#)

[\[Chapter Resources\]](#)

[\[Immunology Home Page\]](#)

W. H. Freeman & Co. and Sumanas, Inc.
Immunology, January, 1997

Chapter 3: Cells & Organs of the Immune System

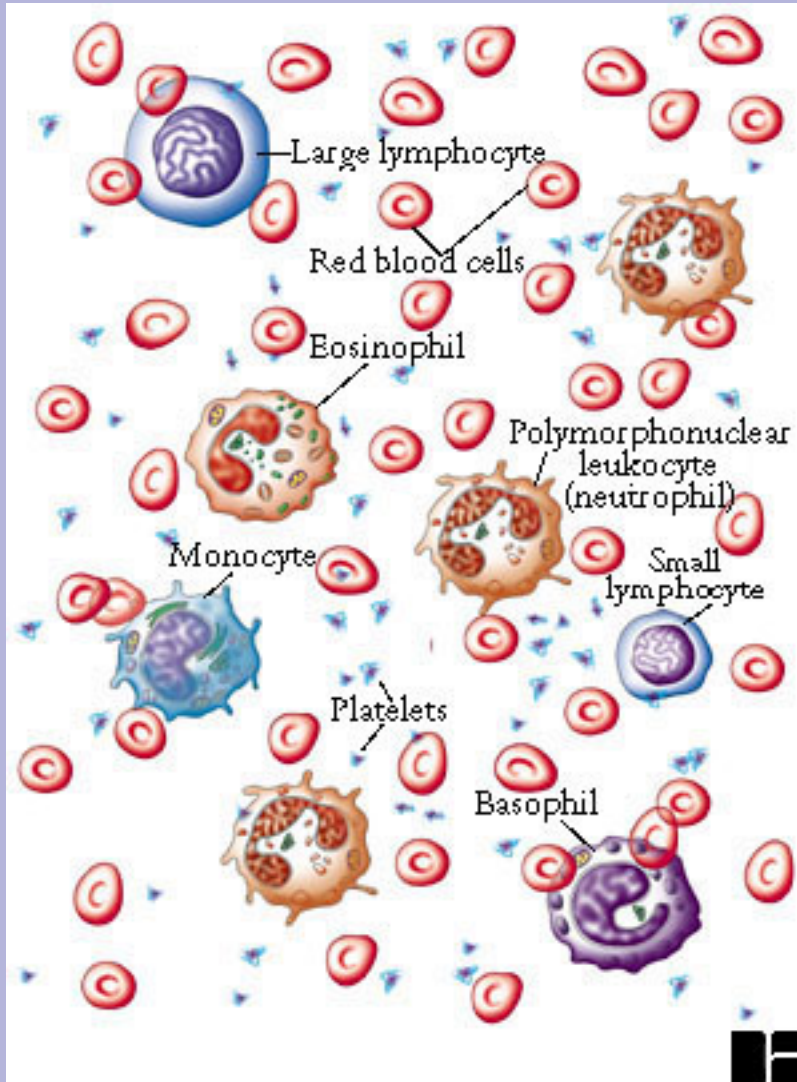


Figure 3-1. Morphology and staining characteristics of various types of blood cells. Red blood cells and platelets, which both lack nuclei, are the most numerous. Most numerous of the leukocyte populations are the neutrophils. Lymphocytes are the predominant cell type responsible for immune responses.

The immune system is constituted by a surprising variety of different cell types disseminated throughout the body and collectively defining one's capacity to mount an immune response. All blood cells and certain other cells located throughout the body - particularly in the **reticuloendothelial system (RES)** - are continuously regenerated throughout life by the process called **hematopoiesis**. Most hematopoietic cells are short lived, some surviving for only a day or two, and thus hematopoiesis serves to maintain a steady renewal of these cells on physiological demand. Hematopoiesis is believed to be the function of a single precursor cell called the pluripotent **stem cell (SC)**. In addition to being self-renewing, SCs undergo multi-lineage differentiation driven by stage-specific cytokine and cell-cell interactions ([Figure 3-2](#)).

In general, hematopoietic cells are highly mobile, moving with the flow of blood in the

cardiovascular system or with the flow of lymph in the lymphatic system. Many of these cells also cross-migrate between these two circulatory systems, and many migrate directly into the tissues, particularly under the influence of inflammatory cytokines with chemoattractive properties. High concentrations of hematopoietic cells, particularly lymphocytes, are also localized in the [primary lymphoid organs](#), *i.e.*, [thymus](#) and [bone marrow](#), and [secondary lymphoid organs](#), *e.g.*, [spleen](#), [lymph nodes](#), *etc.* ([Figure 3-16](#)).

Obviously, the task of identifying different hematopoietic cells is a special challenge considering the sheer diversity of these cells. Most of the basic cell types can be distinguished on a gross morphological level using various [cytochemical staining techniques](#). However, some of the basic cell types are comprised of heterogeneous mixtures of functionally distinct subpopulations that are morphologically indistinguishable. In this case, immunologists resort to monoclonal antibodies directed against specific cell-surface molecules called [CD antigens](#). CD antigens are differentially expressed on leukocytes and distinct cell subpopulations can be identified and even isolated according to their patterns of CD antigen expression.

© 1997, Duane W. Sears

[\[TOP\]](#)

[\[Chapter 3: A. Blood Cell Morphology & Staining\]](#)

[\[Chapter 3: B. Cluster of Differentiation \(CD\) Antigens\]](#)

[\[Chapter Resources\]](#)

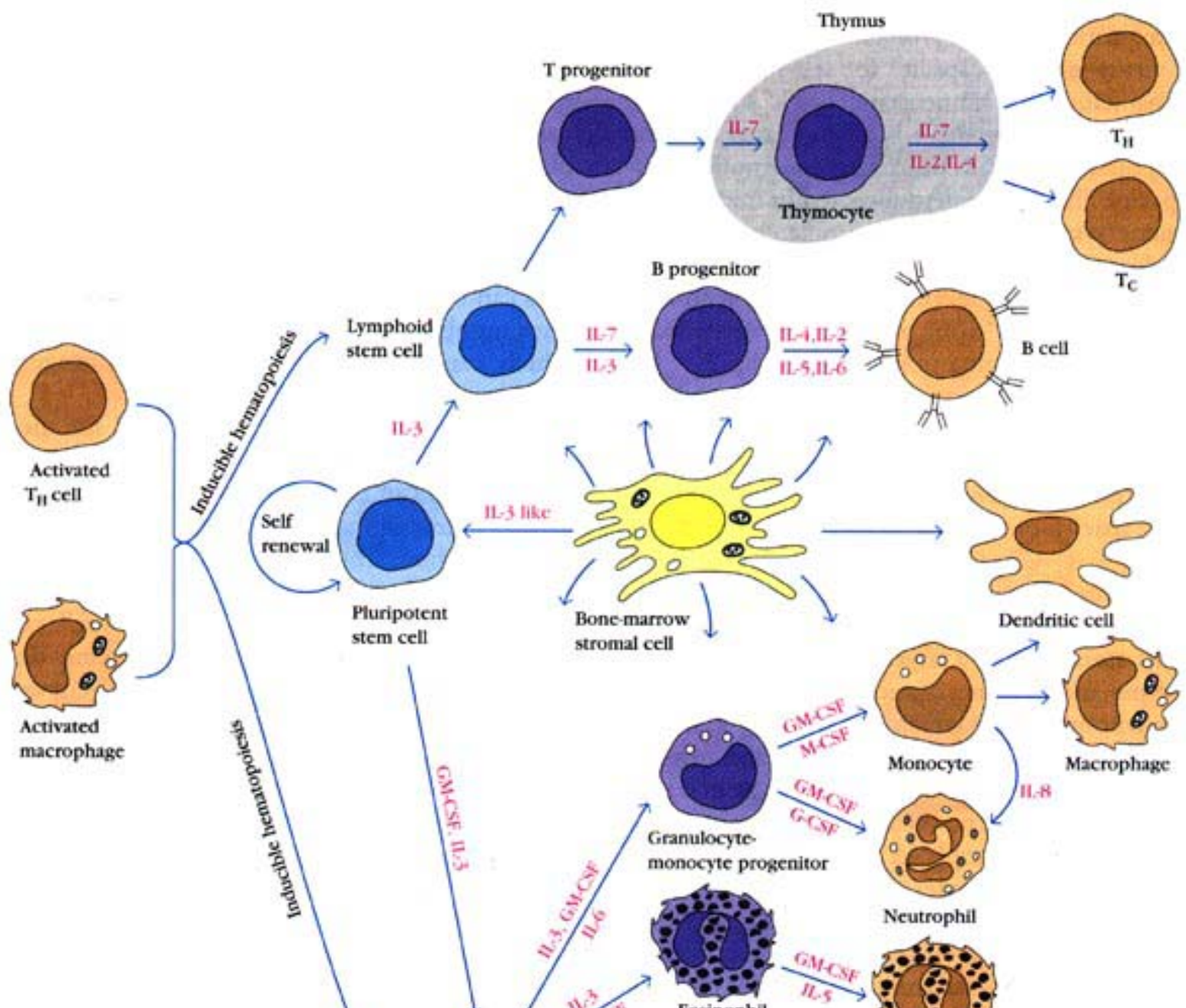
[\[Immunology Home Page\]](#)

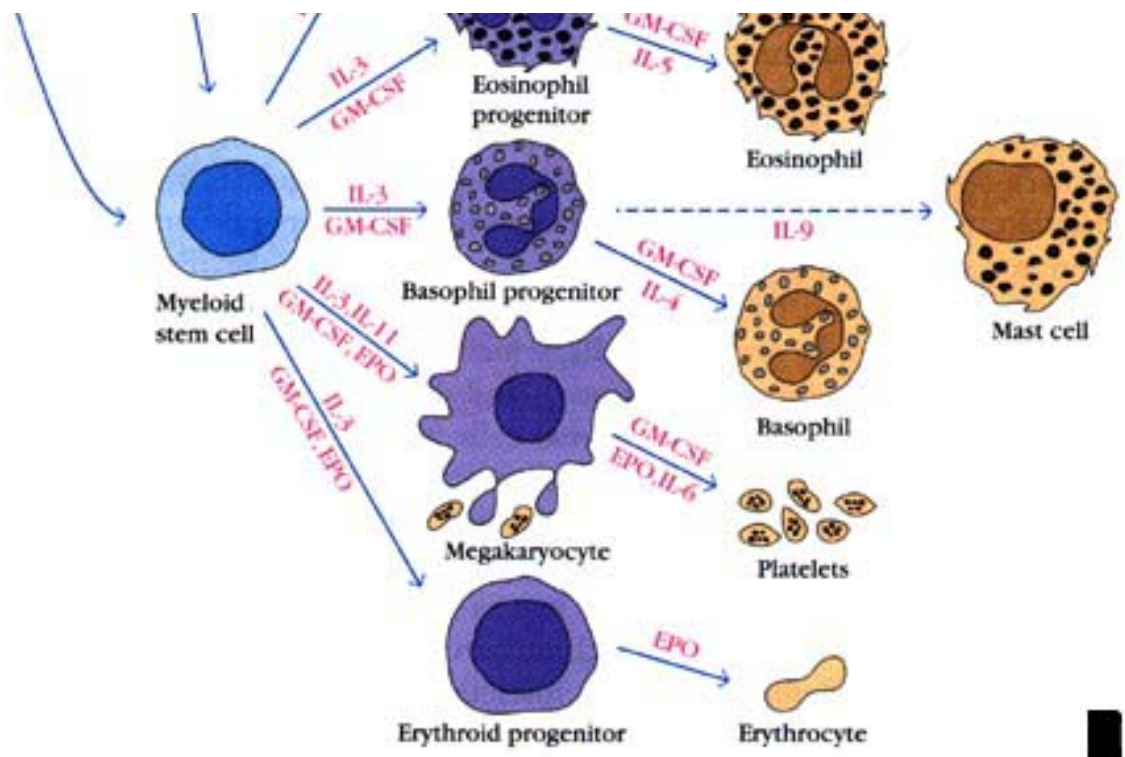
W. H. Freeman & Co. and Sumanas, Inc.
Immunology, January, 1997

Last Updated: June 24, 1997

Figure 3-2: Hematopoiesis

Regulation of hematopoiesis by cytokines that stimulate the proliferation and/or differentiation of various hematopoietic cells. In the absence of infection, bone-marrow stromal cells are the major source of hematopoietic cytokines. In the presence of infection, cytokines produced by activated macrophages and T-helper cells induce additional hematopoietic activity, resulting in rapid expansion of white blood cells that participate in fighting infection. IL = interleukin; CSF = colony-stimulating factor; EPO = erythropoietin. Stem cells are shown in blue; progenitor and immune cells in purple; stromal cell in yellow; and mature differentiated cells in tan.





© 1997, Duane W. Sears

[\[TOP\]](#)

[\[Chapter 3: Cells & Organs of the Immune System\]](#)

[\[Chapter 3: A. Blood Cell Morphology & Staining\]](#)

[\[Chapter 3: B. Cluster of Differentiation \(CD\) Antigens\]](#)

[\[Chapter Resources\]](#)

[\[Immunology Home Page\]](#)

W. H. Freeman & Co. and Sumanas, Inc.
Immunology, January, 1997

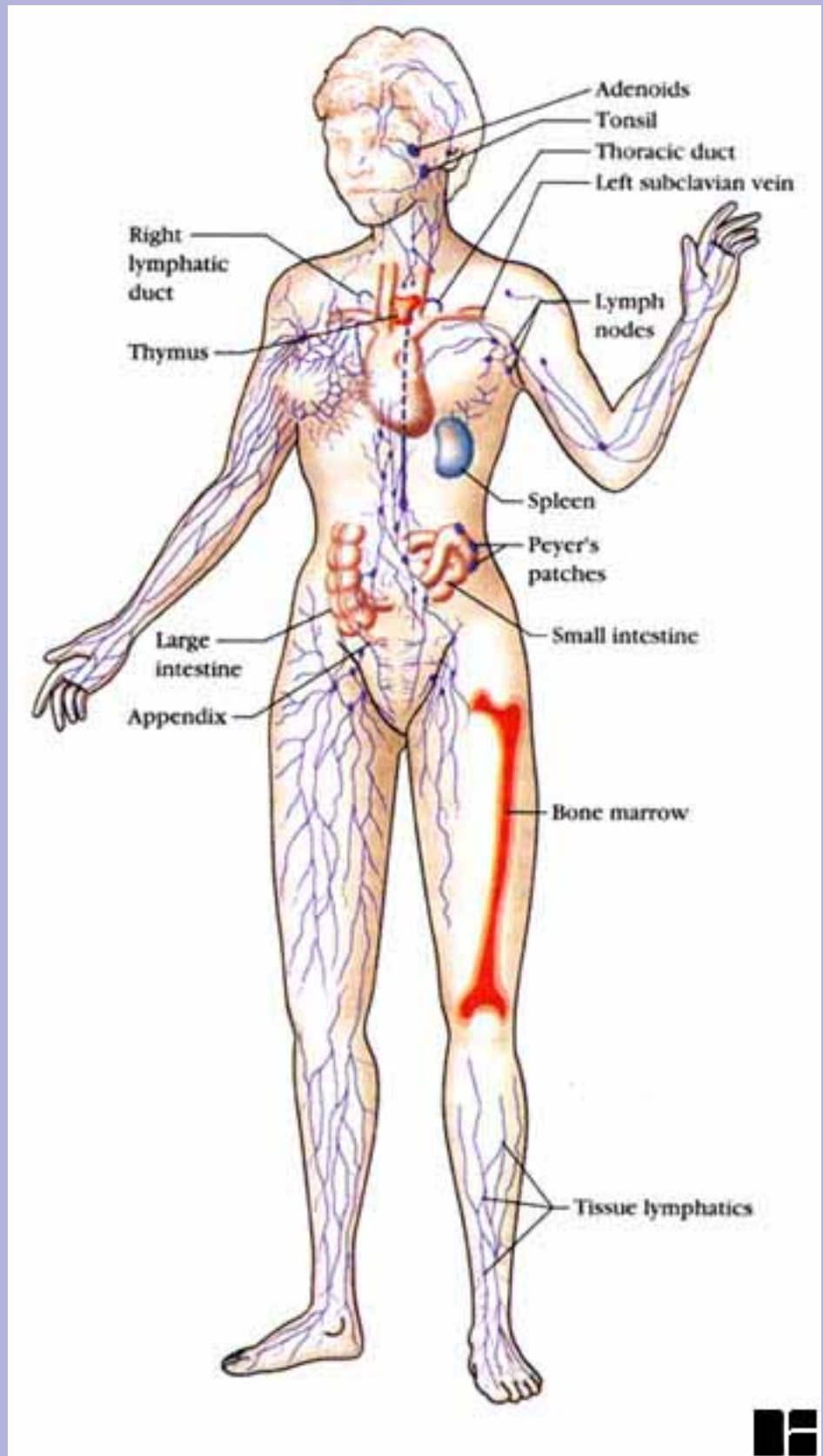
Last Updated: June 24, 1997

Figure 3-16: Human lymphoid system

The primary organs (bone marrow and thymus) are shown in red; secondary organs and tissues, in blue.

These structurally and functionally diverse lymphoid organs and tissues are interconnected by the blood vessels (not shown) and lymphatic vessels (purple) through which lymphocytes circulate. Only one bone is shown, but all major bones contain marrow and thus are part of the lymphoid system.

[Adapted from Harvey Lodish *et al.* 1995, *Molecular Cell Biology*, 3rd ed. Scientific American Books]



[\[TOP\]](#)

[\[Chapter 3: Cells & Organs of the Immune System\]](#)

[\[Chapter 3: A. Blood Cell Morphology & Staining\]](#)

[\[Chapter 3: B. Cluster of Differentiation \(CD\) Antigens\]](#)

[\[Chapter Resources\]](#)

[\[Immunology Home Page\]](#)

W. H. Freeman & Co. and Sumanas, Inc.
Immunology, January, 1997

Last Updated: June 24, 1997

Chapter 3: A. Blood Cell Identification by Staining and Morphology

[White blood cells](#) comprise a diverse collection of leukocytes mediating a variety of immunologically related functions. Individual cell types can be microscopically distinguished by gross morphology and by [staining](#) with cytochemical dyes. For example, Wright-Giemsa stain, with its combination of acidic and basic dyes, will differentially stain the **granules**, **cytoplasm**, and **nuclei** of various blood cell types as illustrated by some of the images linked to this site ([1](#)).

Differential staining morphology of different blood cell types

 red blood cells (RBCs):	 monocytes :
cytoplasm = orange-pink to rose	cytoplasm = pale gray-blue nucleus = deep bluish-purple
 lymphocytes :	 neutrophils :
cytoplasm = light blue nucleus = deep blue-violet	granules = purple-to-lilac cytoplasm = pale pink nucleus = deep blue-violet
 eosinophils :	 basophils :
granules = orange to pink	granules = deep blue to violet
 platelets :	
central granules = red-purple surrounded by light blue	

[Buttons of identical color point to the same web link.]

Morphologically, white blood cells are classified into two broad categories -- **granulocytes** and **mononuclear cells**.

Granulocytes ([Figure 3-13](#)), which typically have [multi-lobed nuclei](#) (often shaped like sausages on a string) and a granular cytoplasm, are comprised of three basic cell types ([2](#)):

- **neutrophils**, also referred to as polymorphonuclear cells, or more simply PMNs
- **eosinophils**; and
- **basophils**, the blood-borne precursors of [mast cells](#).

Mononuclear cells, which typically have [rounded or kidney-shaped nuclei](#) and often little cytoplasm, are comprised of two basic cell types:

- **monocytes** ([Figure 3-11](#)), the blood-borne precursors of macrophages ([Figure 3-12b](#)); and
- **lymphocytes**, morphologically classified as **small** and **large** ([Figure 3-10](#)).

Lymphocytes themselves ([Figure 3-9](#), 2nd ed.) comprise a heterogeneous mixture of functionally distinct cell types, including **B lymphocytes**, the precursors of antibody-producing **plasma cells**, several types of **T lymphocytes**, and **natural killer (NK) lymphocytes**.

In addition to their distinctive cytochemical staining characteristics, blood cells can be distinguished on a gross level by their average size and granularity as measured by [flow cytometry](#). With a [flow cytometer](#), the optical effects of passing a single cell through a laser light beam can be measured in terms of light scattered by the cell in two directions -- parallel to the beam ("forward scattering" or **FSC**) and perpendicular to the beam ("side scattering" or **SSC**). Greater FSC correlates with larger cell size while greater SSC correlates with more granularity in the cytoplasm and nucleus of a cell. A two-dimensional plot of [FSC versus SSC for human blood cells](#), reveals that different cell types exhibit distinct average ranges of size and granularity. Thus, flow cytometry can be used to analyze and even physically isolate different blood cell populations.

With a modified flow cytometer designed to detect fluorescent light stimulated by the laser beam, *i.e.*, a [fluorescence-activated cell sorter \(FACS\)](#), even finer distinctions between different cell populations can be made if they have been treated with fluorescently tagged monoclonal antibodies directed against specific cell surface molecules, generically referred to as [cluster of differentiation \(CD\) antigens](#).

References:

(1) J. D. Bauer, *Clinical Laboratory Methods* (9th ed.) Mosby, St. Louis, p. 111 (1982)

(2) Other primary resources on the web:

- [Department of Cell Biology, Vanderbilt University Medical Center](#)
Prepared by: Michael A. Aleman and John Heusinkveld
Edited by: L. H. Hoffman, Ph.D.
- [University of Leicester, Department of Microbiology & Immunology](#)
- [University of Washington Health Sciences Center](#)
- [Atlas of Hematology](#)
Nagoya University School of Medicine Department of Medicine, The Branch Hospital
Takuji Ichihashi, Tomoki Naoe
Nagasaki University School of Medicine Department of Hematology
Kazutaka Kuriyama
Kyoto University College of Medical Technology
Masataka Sasada
Hamamatsu University School of Medicine Department of Medicine Three
Ryuzo Ohno
- [Inflammation and Fever](#), Academic Electronic Press, 1995
- [Jerry Chin's Immunology Web Page](#)
- [Elements of the Immune Defense System](#): Troels Wind, Svend Kjaer, and Morten Kjeldgaard
- [Plasma cells and Myeloma cells](#): by Michio M. Kawano

© 1997, Duane W. Sears

[\[TOP\]](#)

[\[Chapter 3: Cells & Organs of the Immune System\]](#)

[\[Chapter 3: B. Cluster of Differentiation \(CD\) Antigens\]](#)

[\[Chapter Resources\]](#)

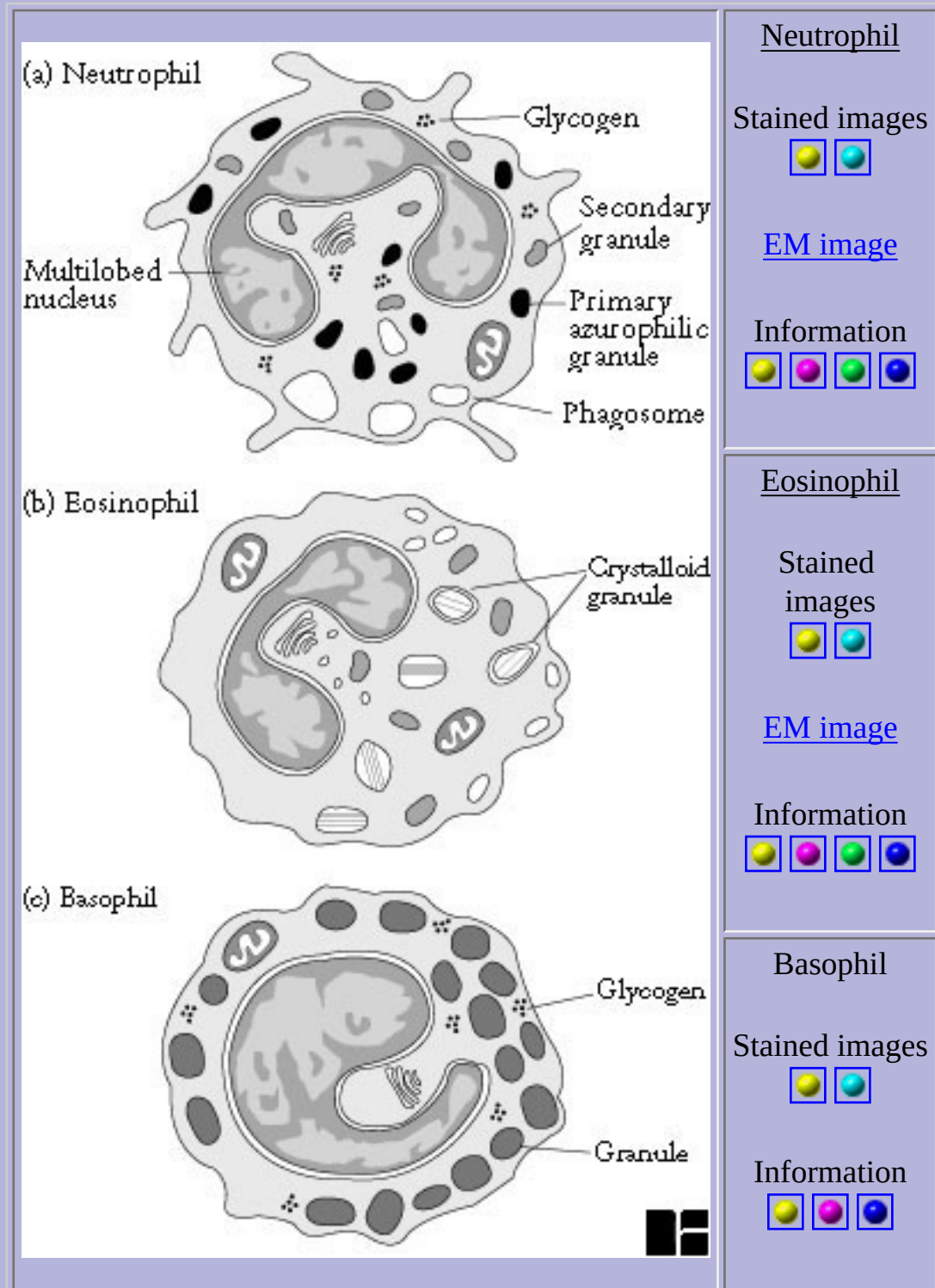
[\[Immunology Home Page\]](#)

W. H. Freeman & Co. and Sumanas, Inc.
Immunology, January, 1997

Last Updated: June 24, 1997

Figure 3-13: Granulocytes

Drawings showing typical morphology of granulocytes. Note differences in the shape of the nucleus and in the number and shape of cytoplasmic granules.



[Buttons of identical color point to different links on the same web site.]

© 1997, Duane W. Sears

[\[TOP\]](#)

[\[Chapter 3: Cells & Organs of the Immune System\]](#)

[\[Chapter 3: A. Blood Cell Morphology & Staining\]](#)

[\[Chapter 3: B. Cluster of Differentiation \(CD\) Antigens\]](#)

[\[Chapter Resources\]](#)

[\[Immunology Home Page\]](#)

W. H. Freeman & Co. and Sumanas, Inc.
Immunology, January, 1997

Last Updated: June 24, 1997

Chapter 3: B. Cluster of Differentiation (CD) Antigens

Leukocytes express distinct assortments of molecules on their cell surfaces, many of which reflect either different stages of their lineage-specific differentiation or different states of activation or inactivation. Leukocyte cell surface molecules are routinely detected with anti-leukocyte [monoclonal antibodies \(mAbs\)](#). Using different combinations of mAbs, it is possible to chart the [cell surface immunophenotypes](#) of different [leukocyte subpopulations](#), including the functionally distinct mature lymphocyte subpopulations of [B-cells](#), [helper T-cells \(\$T_H\$ \)](#), [cytotoxic T-cells \(\$T_C\$ \)](#), and [natural killer \(NK\) cells](#) ([Figure 3-9](#), 2nd ed.).

The abbreviations used for identifying different leukocyte cell surface molecules are somewhat complex because historically three conventions have been used:

Conventions for Naming Leukocyte Surface Molecules

1. By one convention, cell surface molecules are named according to a particular function affected by an anti-leukocyte mAb. For example, the [lymphocyte function-associated antigen 1](#), or [LFA-1](#), was so named because antibodies recognizing this structure interfere with lymphocyte cell adhesion events and optimal lymphocyte function.
2. The second convention is effectively no convention at all. Molecules are named arbitrarily according to individual laboratory preferences. For example, no obvious logic follows in the designations [B7](#) and [B220](#), except that the leading "B" reminds us that these antigens are typically expressed on [B lymphocytes](#).
3. By a third convention, leukocyte cell surface molecules are named systematically by assigning them a [cluster of differentiation \(CD\) antigen](#) number that includes any antibody having an identical and unique reactivity pattern with different leukocyte populations.

Although less intuitive, the [CD antigen](#) convention has become the accepted norm for formal designation of leukocyte surface molecules. This is a logical consequence of the fact that most leukocyte cell surface molecules were discovered only by their reactivity patterns with anti-leukocyte antibodies when little, if anything, was known about their

functions. Also, because most leukocyte cell surface molecules are recognized by more than one antibody, it is easy to group antibodies according to identical reactivity patterns even though they may recognize different portions of the same molecule. An obvious problem, though, is that some antibodies recognize more than one distinct leukocyte surface structure as is the case, for example, with the structurally heterogeneous **CD45** splice variants. Different variants have identical protein domains, and anti-CD45 antibodies recognizing these shared domains will cross-react with more than one CD45 variant.

A useful [Table of CD Antigens](#) has been prepared by **Paul Travers** for web viewing with well over 100 human leukocyte cell surface molecules assigned CD designations. Note that some CD antigens have also been named by one of the other conventions. For example:

- **CD54** = **LFA-1**;
- **CD80** = **B7** (or now **B7-1**); and
- **CD45** = **B220**.

Some CD antigens are widely expressed on different leukocyte populations while others are expressed on only one type or just a few types of leukocytes. For example:

- **CD54/LFA-1** is widely expressed on a variety of hematopoietic cells.
- Conversely, **CD4** is expressed almost exclusively on **T helper (T_H) lymphocytes** and cells of the **monocyte/macrophage** lineage, an important consideration for understanding the pathology of [human immunodeficiency virus \(HIV\)](#) which gains cell entry via the **CD4** molecule.

Commercial listings of **anti-CD monoclonal antibodies** are easy to find on the web. For example, a fairly comprehensive listing is electronically published by [Immunotech](#).

© 1997, Duane W. Sears

[\[TOP\]](#)

[\[Chapter 3: Cells & Organs of the Immune System\]](#)

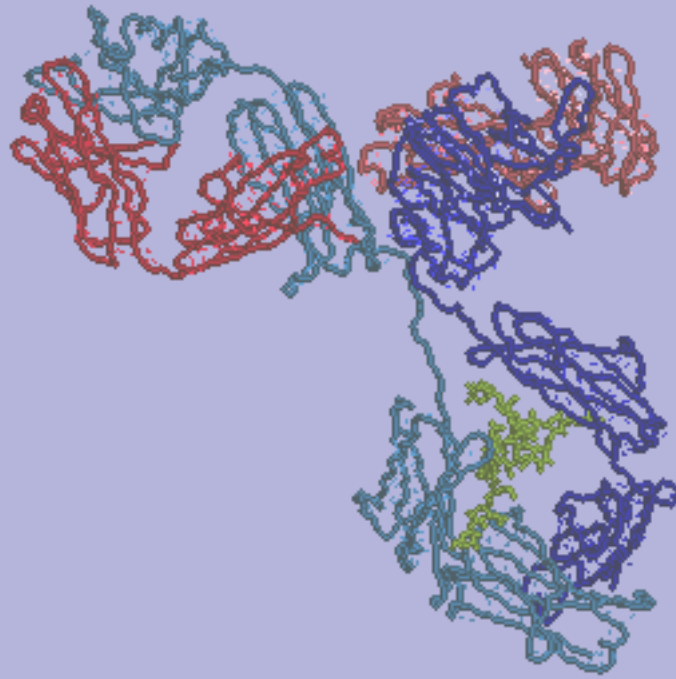
[\[Chapter 3: A. Blood Cell Identification by Staining and Morphology\]](#)

[\[Chapter Resources\]](#)

[\[Immunology Home Page\]](#)

W. H. Freeman & Co. and Sumanas, Inc.
Immunology, January, 1997

Last Updated: June 24, 1997



Monoclonal Antibodies

Monoclonal antibodies are widely used in research and medicine, and they are available to from any number of publicly supported and commercial institutions with a host of antigenic specificities that rival the imagination (see for example: [1](#), [2](#), [3](#), [4](#), & [5](#)). Many applications of monoclonal antibodies are quite novel. For example, mouse monoclonal antibodies against human antigens are can be modified or customized in ways that make them useful [therapeutic agents](#) for treatment of human diseases such as cancer. Before [therapeutic antibodies](#) are injected into patients, however, they are frequently "[humanized](#)" in order to make them more compatible with the human immune system in a process that utilizes recombinant DNA methodology to substitute the constant region sequences of mouse- or rat-derived monoclonal antibodies with the corresponding human constant region sequences, without compromising antigen specificity. Another novel spin-off of monoclonal antibody technology has been the creation of [catalytic antibodies or abzymes](#) produced by immunizing mice with enzyme transition state analogs. Abzymes represent a whole new approach to the study of enzyme mechanisms and the creation of useful biological catalysts.

© 1997, Duane W. Sears

[\[TOP\]](#)

[\[Chapter 5: Overview of the three-dimensional structure of human IgG1 antibodies\]](#)

[\[Chapter 5: Overview of the three-dimensional structure of a catalytic monoclonal antibody\]](#)

[\[Chapter Resources\]](#)

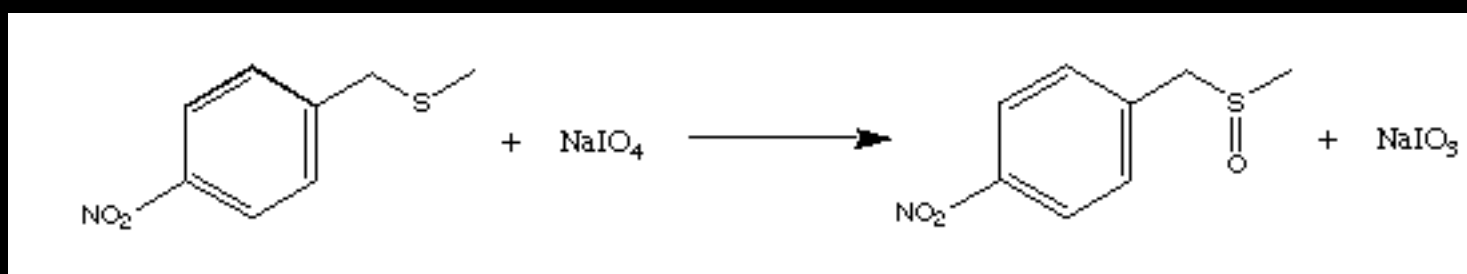
[\[Immunology Home Page\]](#)

W. H. Freeman & Co. and Sumanas, Inc.
Immunology, January, 1997

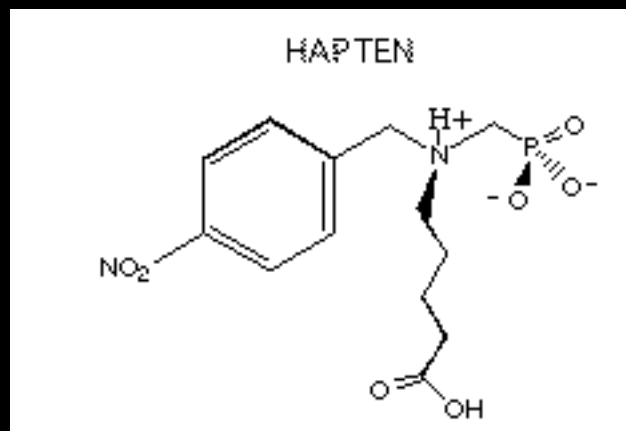
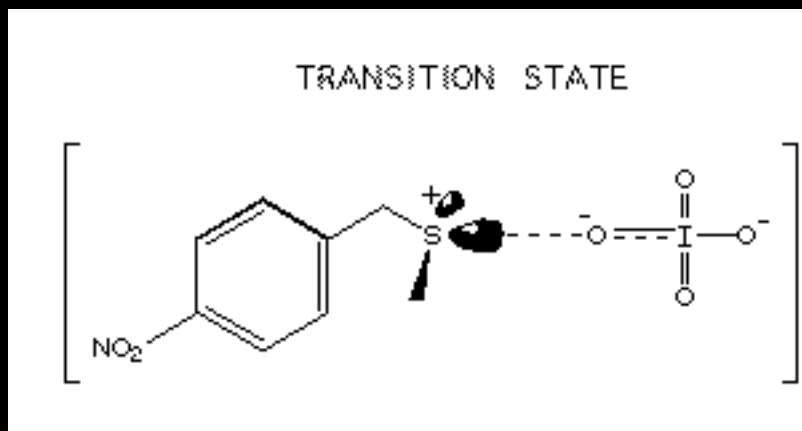
Last Updated: June 25, 1997

Abzymes: Catalytic Antibodies

By exploiting the highly specific antigen binding properties of antibodies, experimental strategies have been devised to produce antibodies that catalyze chemical reactions. These catalytic antibodies, or *abzymes*, are selected from monoclonal antibodies generated by immunizing mice with haptens that mimic the transition states of enzyme-catalyzed reactions. For example, the 28B4 abzyme catalyzes periodate oxidation of *p*-nitrotoluene-methyl sulfide to sulfoxide, as shown below, where electrons from the sulfur atom are transferred to the more electronegative oxygen atom.



The rate of this reaction is promoted by enzyme catalysts that stabilize the transition state of this reaction, thereby decreasing the activation energy and allowing for more rapid conversion of substrate to product. In this case, the transition state is thought to involve a transient positive charge on the sulfur atom and a double-negative charge on the periodate ion as shown below on the left.



In order to generate abzymes complementary in structure to this transition state, mice were immunized with an aminophosphonic acid hapten, as shown above at the right. Obviously, its structure mirrors the structure and electrostatic properties of the sulfoxide transition state. Of the hapten-binding monoclonal antibodies produced with this hapten, many were found to catalyze sulfide oxidation but with a wide range of binding affinities and catalytic efficiencies. In

particular, abzyme 28B4 binds hapten with high affinity ($K_d = 52 \text{ nM}$) and exhibits a correspondingly high degree of catalytic efficiency ($k_3/K_M = 190,000 \text{ M}^{-1} \text{ s}^{-1}$).

Elucidation of the molecular structure of abzyme 28B4 bound to the hapten reveals much about the nature of its catalytic action. Highly specific structural and electrostatic interactions create a remarkable degree of structural complementarity between the antigen-binding site and the sulfoxide transition state analog as illustrated in the following series of three-dimensional views of the antibody-hapten complex.

Reset	Zoom out	Zoom in	Spacefill hapten	Spacefill active site	Hapten only	Active site only
-------	----------	---------	------------------	-----------------------	-------------	------------------

[1kel.pdb](#)



For best results, push the buttons in sequence.

Rotate around active site complex.

Label charged or partially charged centers.

Labels off.

- 1. Zoom into the antigen-binding site with bound hapten.** Initially, the hapten is visible at the upper right of the 28B4 Fab fragment bound to the antigen-binding site located between the **H chain** and **L chain**.
- 2. Zoom in to view the structural complementarity between antibody and hapten.**
- 3. Highlight ionic interactions and hydrogen bonding between sidechain atoms of Arg52, Lys56, & Tyr33 (all in the H chain) and the hapten's anionic phosphate.**
- 4. Highlight hydrophobic and aromatic interactions between several antibody sidechains and the hapten's aromatic group.**
- 5. Highlight π -electron stabilization by the aromatic group of Tyr37 (L chain) and the tetrahedral, cationic nitrogen atom of the hapten.**
- 6. Highlight hydrogen bonding between the sidechain of Asp35 (H chain) and the hapten's electronegative *p*-nitro ($-\text{NO}_2$) group.**

Analysis of abzyme 28B4 without bound hapten reveals a slightly altered hapten binding site conformation, suggesting that the induced-fit model of enzyme-substrate binding is a feature of the abzyme's catalytic mechanism. Also, sequence analyses of other antibodies that bind this hapten hint at a mechanism for enzyme evolution. Of particular interest is **asparagine 35** of the 28B4 heavy chain which forms a key hydrogen bond with the *p*-nitro group of the substrate, dramatically increasing the specificity the 28B4 abzyme compared to other hapten-specific antibodies that have a different amino acid residue at this position. The germline sequences of the

heavy chain gene segments indicate that this particular asparagine residue arose as the result of somatic mutation of the heavy chain gene during affinity maturation of hapten-specific B lymphocytes. Thus, one can begin to comprehend the evolution of intricately specific enzymes from less-active or inactive protein precursors. The study of catalytic antibodies as a whole has vastly increased current understanding of the mechanisms of enzyme catalysis and represents another step forward in the attempts to create artificially engineered biological enzymes.

Based on the experimental data of L.C.Hsieh-Wilson, P.G.Schultz, and R.C.Stevens. *Proc. Natl. Acad. Sci. USA*, 93:5363-5367 (1996).

[\[TOP\]](#)

[\[Chapter 5: Overview of the three-dimensional structure of human IgG1 antibodies\]](#)

[\[Chapter 5: Overview of the three-dimensional structure of a catalytic monoclonal antibody\]](#)

[\[Chapter Resources\]](#)

[\[Immunology Home Page\]](#)

Moving Structures with the Mouse [[Macintosh](#) 

Windows Action	Mouse button + key, pointer on image
popup menu	right button pointer stationary
X-rotation	left button drag  
Y-rotation	left button drag  
Z-rotation	right button + Shift key, drag  
X-translation	right button, + Ctrl key, drag  
Y-translation	right button + Ctrl key, drag  
zoom in	left button + Shift key, drag 
shrink	left button + Shift key, drag 

[[Windows](#) 

PowerMac Action	Mouse button + key, pointer on image
popup menu	button down pointer stationary
X-rotation	button down drag  

Y-rotation	button down drag 
Z-rotation	(none)
X-translation	button down, + Option key, drag 
Y-translation	button down + Option key, drag  
zoom in	button down + Shift key, drag 
shrink	button down + Shift key, drag 

 [Go to top.](#)

 Return to previous pages with the browser's "Back" key or "Go" pulldown menu.

 [Return to the Kuby Home Page](#)

© [Duane W. Sears](#)

April 26, 1997

Overview of IgG Structure

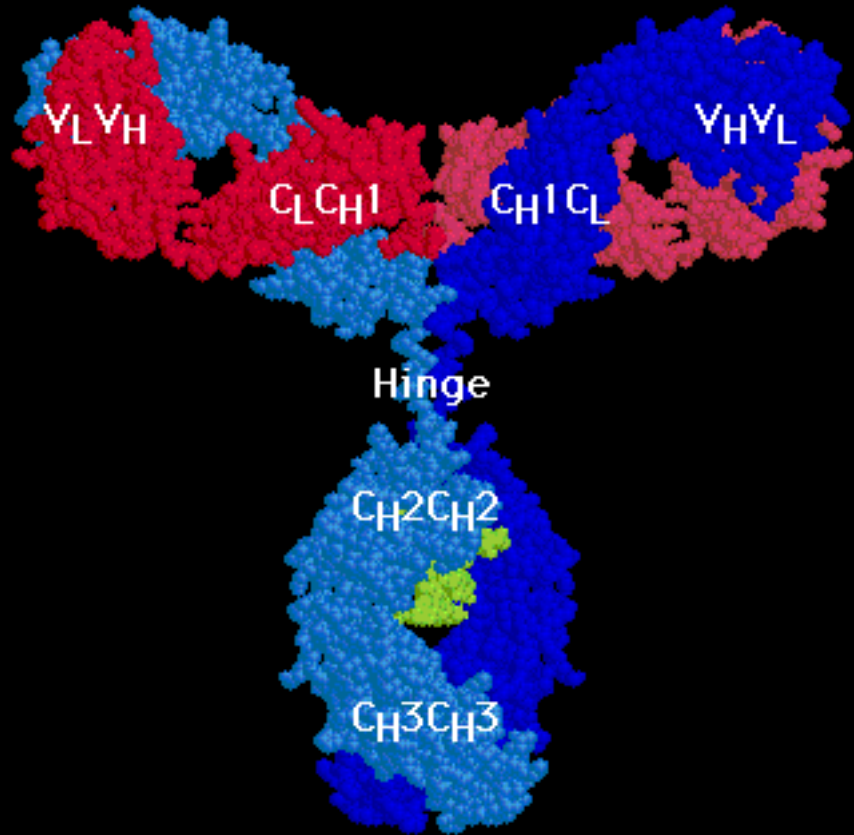
Click [here](#) for an overview of the three-dimensional structure of human IgG1 antibodies.

Reset image.

Script execution continues until "**Chime Script Completed**" is displayed.



Click [below](#) on an IgG structural domain for a close-up three-dimensional view.



This model of human IgG1 was created by [E. A. Padlan](#) -- "Anatomy of the Antibody Molecule." *Molecular Immunology* 31:169 (1994) -- who combined the Brookhaven X-ray coordinate files for the Fc ([1fc2.pdb](#)) and F(ab') ([2ig2.pdb](#)) fragments after symmetrically rotating the latter coordinates to generate the bivalent F(ab')₂ ([bio2ig2.pdb](#)).

[\[TOP\]](#)

[\[Chapter Resources\]](#)

[\[Immunology Home Page\]](#)

V_H - V_L Interface

Reset

Script execution continues until "**Chime Script Completed**" is displayed.



Note: An immunoglobulin fold is formed by a beta barrel with one intrachain disulfide bond.

Note: Three V_L CDR loops and three V_H CDR loops together form one antigen binding site on each arm of the antibody molecule.

[\[TOP\]](#)

[\[Chapter Resources\]](#)

[\[Immunology Home Page\]](#)

C_L-C_H1 Interface

Reset

Script execution continues until "**Chime Script Completed**" is displayed.



Note: An immunoglobulin fold is formed by a beta barrel with one intrachain disulfide bond.

Note: An interchain disulfide bond covalently joins one heavy to one light chain.

[\[TOP\]](#)

[\[Chapter Resources\]](#)

[\[Immunology Home Page\]](#)

Hinge Region

Reset

Script execution continues until "**Chime Script Completed**" is displayed.



Note: The extended conformation of the hinge region allows for freer movement of the three arms of the molecule.

Note: The hinge contains **two interchain disulfide bonds** promoting greater structural stability and **six proline residues** that extend this region and give the molecule greater segmental flexibility.

Click here to locate the **hinge** & **C_H2** residues implicated in high affinity FcγRI binding.

[\[TOP\]](#)

[\[Chapter 5: Overview of the three-dimensional structure of human IgG1 antibodies\]](#)

[\[Chapter 5: Overview of the three-dimensional structure of a catalytic monoclonal antibody\]](#)

[\[Chapter Resources\]](#)

[\[Immunology Home Page\]](#)

C_H2-C_H2 Carbohydrate Interface

Reset

Script execution continues until "**Chime Script Completed**" is displayed.



Note: An immunoglobulin fold is formed by a beta barrel with one intrachain disulfide bond.

Note: The two heavy-chain linked **carbohydrate moieties** associate noncovalently and are positioned between the two C_H2 domains effectively "pushing" them apart, in contrast to other domain pairs in the antibody structure.

Click here to locate the **hinge** & C_H2 residues implicated in high affinity FcγRI binding.

[\[TOP\]](#)

[\[Chapter Resources\]](#)

[\[Immunology Home Page\]](#)

C_H3-C_H3 Interface

Reset

Script execution continues until "**Chime Script Completed**" is displayed.



Note: An immunoglobulin fold is formed by a beta barrel with one intrachain disulfide bond.

Note: The heavy chain C-termini of an IgG antibody secreted from a B cell contain several **hydrophilic residues** that are replaced by hydrophobic residues in a membrane-bound IgG antibody where the heavy chain C-termini are inserted directly into the lipid bilayer of the B cell.

[\[TOP\]](#)

[\[Chapter Resources\]](#)

[\[Immunology Home Page\]](#)

Figure 3-9 (2nd ed.): Antigen Markers on Mature Lymphocyte Populations

Schematic representation of some of the membrane molecules present on **mature T** and **B lymphocytes**, and **natural killer (NK) cells**. Molecules involved in specific binding of antigen are shown in purple. **CD16** (FcγRIII) on **NK cells** and **CD32** (FcγRII) on **B cells** are receptors for the Fc region of IgG. **CD21** and **CD35** on **B cells** are receptors for complement components. Most **T_H cells** express **CD4**, whereas most **T_C cells** express **CD8** (as do some **NK cells**).

Mature Lymphocyte Populations

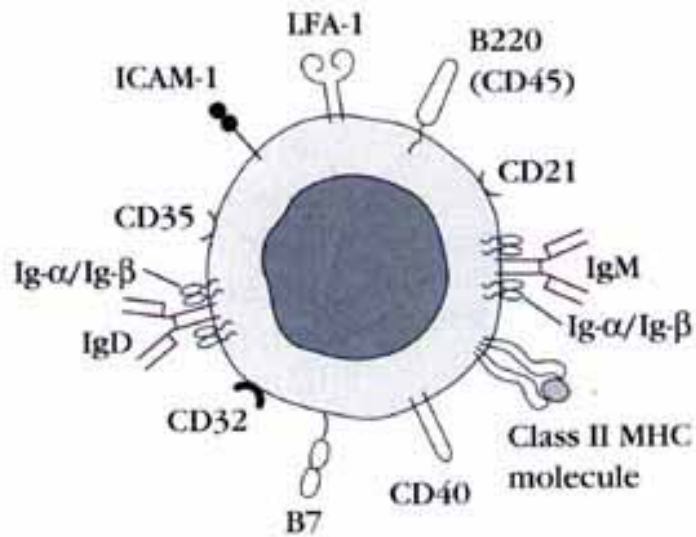
B lymphocytes

antibody-producing cells

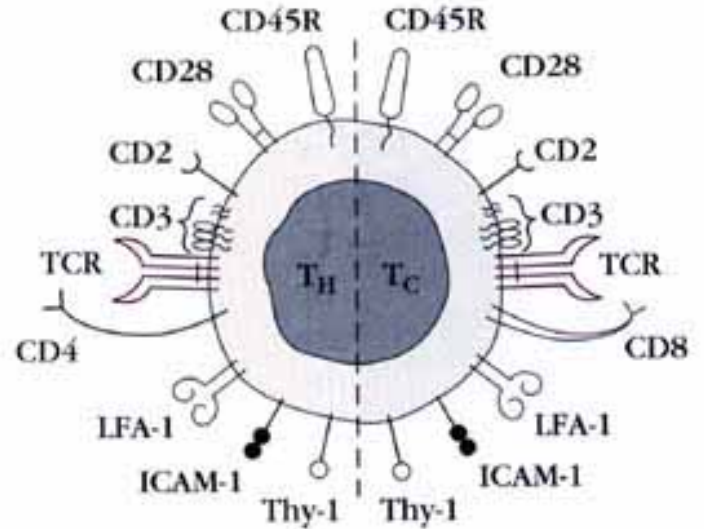
T lymphocytes

T_H cell / T_C cell

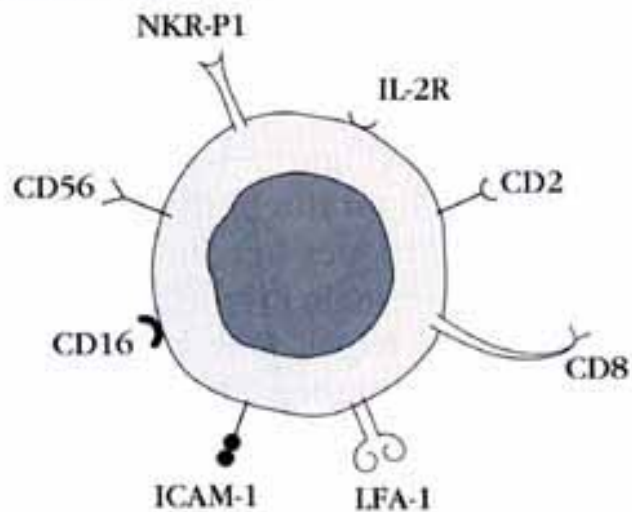
(a) B lymphocyte



(b) T lymphocyte



(c) Natural killer cell



[NK \(natural killer\) lymphocyte](#)

large granular lymphocyte (LGL)

© 1997, Duane W. Sears

[\[TOP\]](#)

[\[Chapter 3: Cells & Organs of the Immune System\]](#)

[\[Chapter 3: A. Blood Cell Morphology & Staining\]](#)

[\[Chapter 3: B. Cluster of Differentiation \(CD\) Antigens\]](#)

[\[Chapter Resources\]](#)

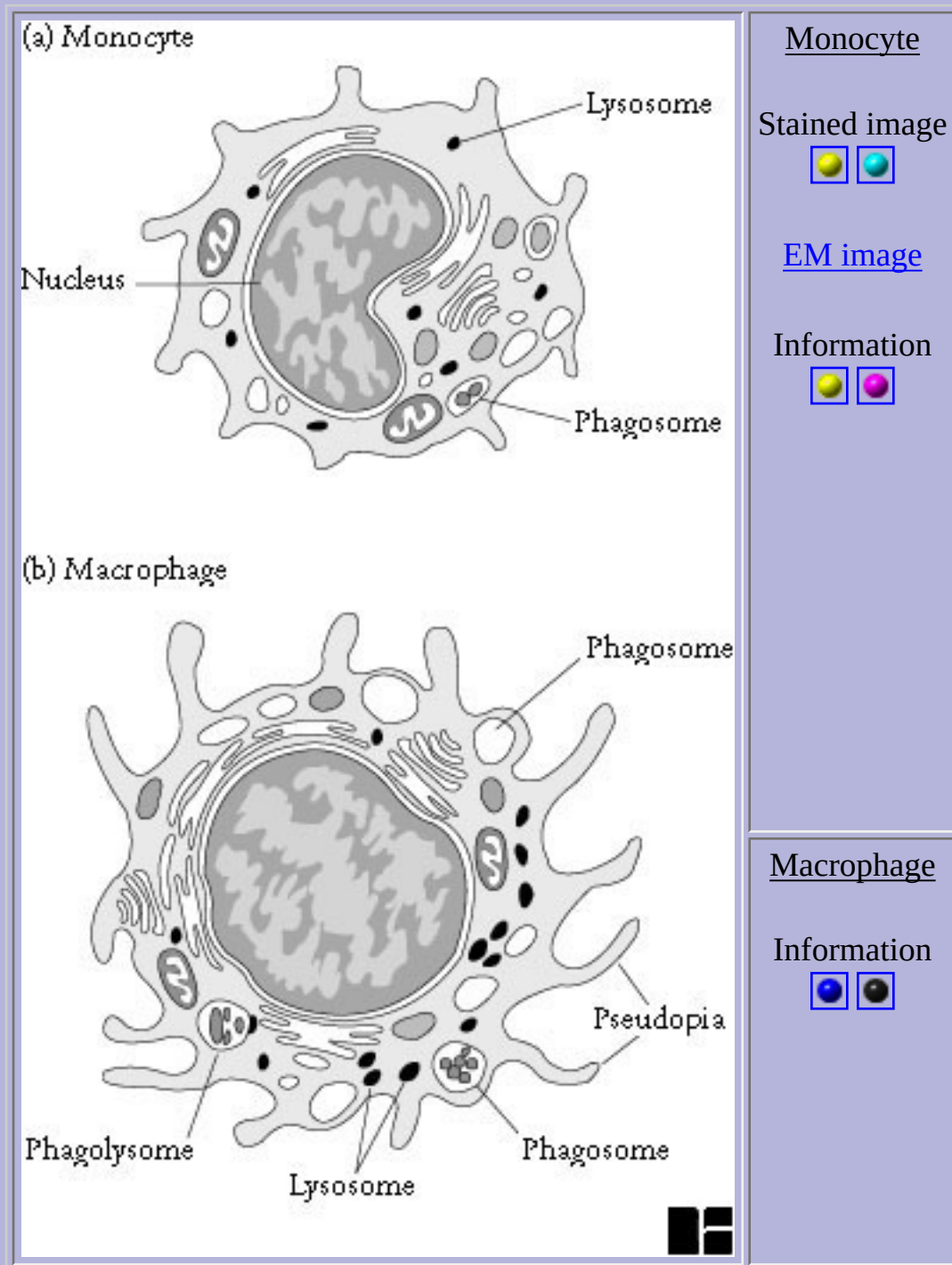
[\[Immunology Home Page\]](#)

W. H. Freeman & Co. and Sumanas, Inc.
Immunology, January, 1997

Last Updated: June 24, 1997

Figure 3-11: Monocyte/Macrophage

Drawing showing typical monocyte and macrophage morphology. Macrophages are five- to tenfold larger than monocytes and contain more organelles, especially lysosomes.



[Buttons of identical color point to different links on the same web site.]

[\[TOP\]](#)

[\[Chapter 3: Cells & Organs of the Immune System\]](#)

[\[Chapter 3: A. Blood Cell Morphology & Staining\]](#)

[\[Chapter 3: B. Cluster of Differentiation \(CD\) Antigens\]](#)

[\[Chapter Resources\]](#)

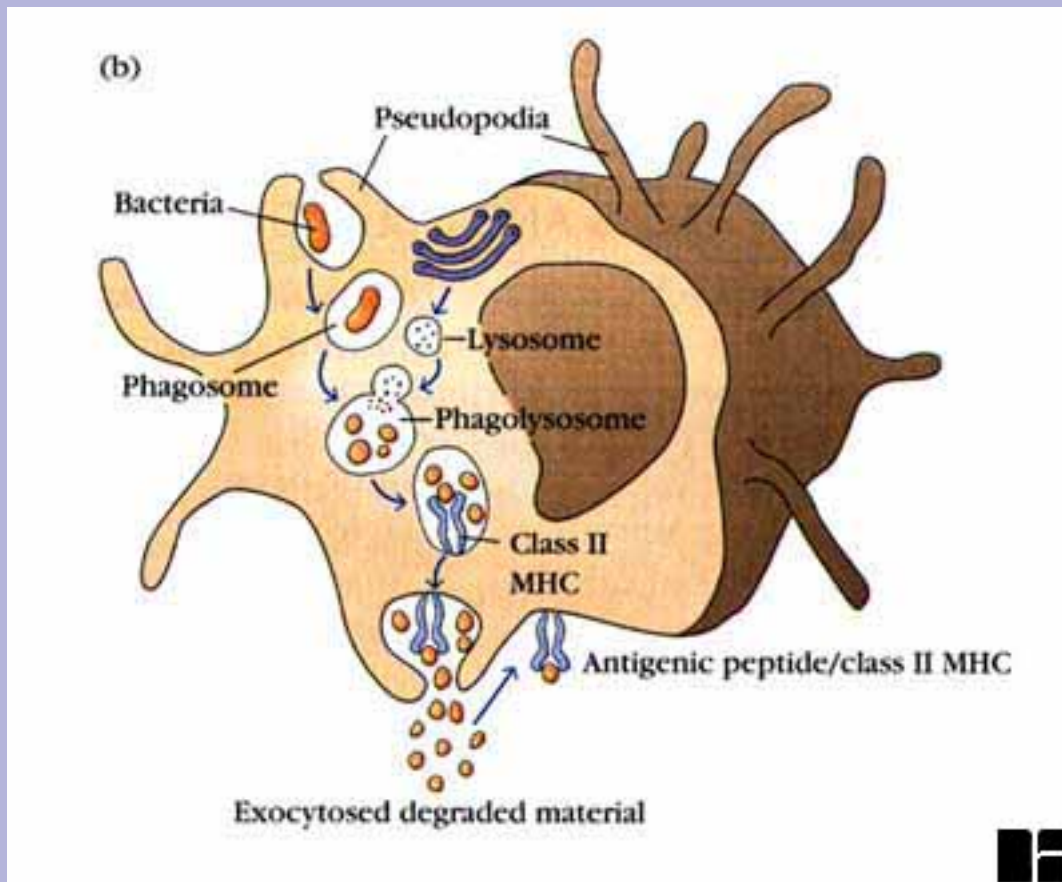
[\[Immunology Home Page\]](#)

W. H. Freeman & Co. and Sumanas, Inc.
Immunology, January, 1997

Last Updated: June 24, 1997

Figure 3-12b: Macrophage

[Phagocytosis](#) and processing of exogenous antigen by macrophages. Most of the products resulting from digestion of ingested material are exocytosed, but some peptide products interact with class II MHC molecules, forming complexes that move to the cell surface where they are presented to T_H cells.



© 1997, Duane W. Sears

[\[TOP\]](#)

[\[Chapter 3: Cells & Organs of the Immune System\]](#)

[\[Chapter 3: A. Blood Cell Morphology & Staining\]](#)

[\[Chapter 3: B. Cluster of Differentiation \(CD\) Antigens\]](#)

[\[Chapter Resources\]](#)

[\[Immunology Home Page\]](#)

W. H. Freeman & Co. and Sumanas, Inc.
Immunology, January, 1997

Last Updated: June 24, 1997

Figure 3-10: Activation and maturation of B lymphocytes

Fate of antigen-activated small lymphocytes. A small resting (unprimed or naive) lymphocyte resides in the G_0 phase of the cell cycle. At this stage, B and T lymphocytes cannot be distinguished morphologically. Following antigen activation, a B or T cell enters the cell cycle and enlarges into a lymphoblast, which undergoes several rounds of cell division and eventually generates effector cells and memory cells. Shown here are B-cell lineage cells.

B cell differentiation

Plasma cell

Small, naive B lymphocyte

KUBY'S FIGURE 3-10

Large, activated lymphoblast

© 1997, Duane W. Sears

[\[TOP\]](#)

[\[Chapter 3: Cells & Organs of the Immune System\]](#)

[\[Chapter 3: A. Blood Cell Morphology & Staining\]](#)

[\[Chapter 3: B. Cluster of Differentiation \(CD\) Antigens\]](#)

[\[Chapter Resources\]](#)

[\[Immunology Home Page\]](#)

W. H. Freeman & Co. and Sumanas, Inc.
Immunology, January, 1997

Last Updated: June 24, 1997

Chapter 6: Antibody-Antigen Interactions

NOTE: A portion of this chapter requires that Microsoft Excel be defined as a helper application (Scatchard plot).

When immunologists describe the properties of antibodies as proteins, most would include a description of the capacity of these molecules to precipitate antigens from solution, even though antibody precipitation is seldom used any more to isolate or detect antigens experimentally and even though antibodies probably rarely precipitate antigens in vivo, except in some autoimmune diseases. The instructional value of the antibody precipitation reaction, as illustrated in Figure 6-4 below, is that it neatly embodies so many of the fundamental and universal properties of antibody molecules, as first recognized many years ago by Michael Heidelberger and Elvin Kabat who advanced this technique to demonstrate, among other things, that:

- serum (IgG) antibodies are bivalent in their reactions with antigen and have the capacity to crosslink antigens;
 - antigens are often multivalent in their interactions with antibodies;
 - serum antibodies are typically polyclonal in nature; and
 - antibodies are highly specific in terms of the structures they recognize on antigenic molecules.
-

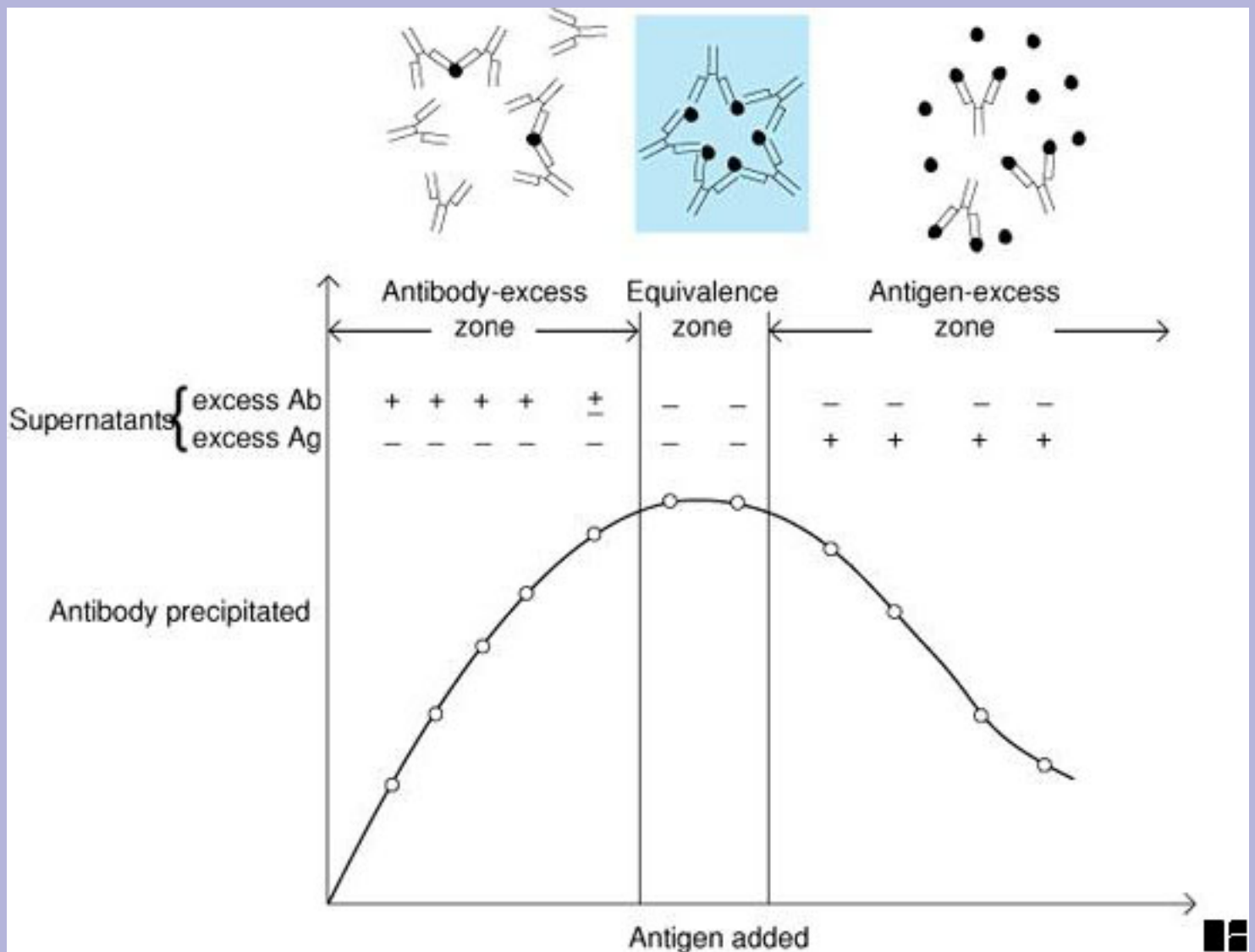


Figure 6-4, A precipitation curve for a system of one antigen and its antibodies. The plot of the amount of antibody precipitated versus increasing antigen concentration (at constant total antibody) reveals three zones: a **zone of antibody excess** in which precipitation is inhibited and excess antibody can be detected in the supernatant; an **equivalence zone** of maximal precipitation in which antibody and antigen form large insoluble complexes (shaded in blue) and neither antibody nor antigen can be detected in the supernatant; and a **zone of antigen excess** in which precipitation is inhibited and excess antigen can be detected in the supernatant.

While the precipitation reaction is particularly well suited for characterizing the interactions between polyclonal antibodies and multivalent antigens, it is generally not very useful for characterizing the interactions between antigens and monoclonal antibodies, unless the antigens exhibit multiple identical epitopes (like the repeating carbohydrate structures found in polysaccharides, for example). Only then will a monospecific monoclonal antibody be able to crosslink and precipitate antigen. However,

monoclonal antibodies are typically characterized by more refined types of measurements that provide detailed information regarding the antibody-antigen interaction, such as the equilibrium constant and the kinetic off- and on-rates. Determination of the antibody-antigen equilibrium association or dissociation constant is discussed further in reference to [Figures 6-2 and 6-3](#).

The utility of antibodies for detecting and isolating antigens can hardly be overstated given the wide spectrum of extremely powerful antibody applications developed over the years. The utility of antibodies is also greatly enhanced by their relative stability in various chemical modification reactions which alter antibody structure without destroying their capacity to bind antigens. Antibodies have been chemically tagged with fluorescent, magnetic, radioactive, and assorted other compounds as a way of facilitating antigen detection or isolation under a variety of experimental conditions. Many examples like those briefly discussed below can be found on the web.

A fluorescent microscopic web image from [Molecular Probes](#) illustrates how antibodies can be used to localize antigens in a cell. In this example, the cable-like nature of [microtubules in fibroblasts](#) is revealed by first tagging cytoplasmic structures with mouse anti- β -tubulin monoclonal antibodies and then staining bound antibodies with a goat anti-mouse IgG (H+L) antibody tagged with a fluorescent compound called Texas Red.

Fluorescent antibodies can also be used with a [fluorescence-activated cell sorter \(FACS\)](#) to identify and even physically separate different cell populations according to the antigens expressed on their surfaces. A FACS is a [flow cytometer](#) designed to detect fluorescent light stimulated by the laser beam. Cells binding fluorescently tagged monoclonal antibodies directed against specific cell surface [CD antigens](#), for example, will emit fluorescent light as they pass through the cytometer's laser beam. The relative intensity of emitted fluorescent light correlates with the amount of antibody bound to the cell surface which in turn correlates with the level of CD marker expression by the cell. With proper combinations of antibodies with different fluorescent tags, specific cell populations can be sorted and even isolated from a complex mixture of cells.

In a variation of this technique, monoclonal antibodies tagged with magnetic compounds can also be used to separate cells based on their surface antigenic determinants.

[\[TOP\]](#)

[\[Chapter 6: Measurement of equilibrium binding constants\]](#)

[\[Chapter Resources\]](#)

[\[Immunology Home Page\]](#)

W. H. Freeman & Co. and Sumanas, Inc.
Immunology, January, 1997

Last Updated: July 15, 1997

Chapter 6: Measurement of Antibody Affinity

Quantitative measurements of the affinity of an antibody for antigen can provide useful information about an antibody. For example, affinity measurements may be used to screen different isolates of an antibody in order to identify those that are most effective at binding antigen. Also, quantitative measurements of the capacity of an antibody to bind other compounds that are structurally related to the original immunizing antigen can help establish the likelihood of whether an antibody will crossreact, perhaps undesirably, with other molecules that might accompany the antigen. The **valence** of an antibody for antigen can also be found by quantitative affinity measurements, this parameter being an important distinguishing feature of different classes and subclasses of antibodies.

The simplest and most direct way of measuring antibody affinity is by the method of equilibrium dialysis, as illustrated in Figure 6-2 below and described in the legend.

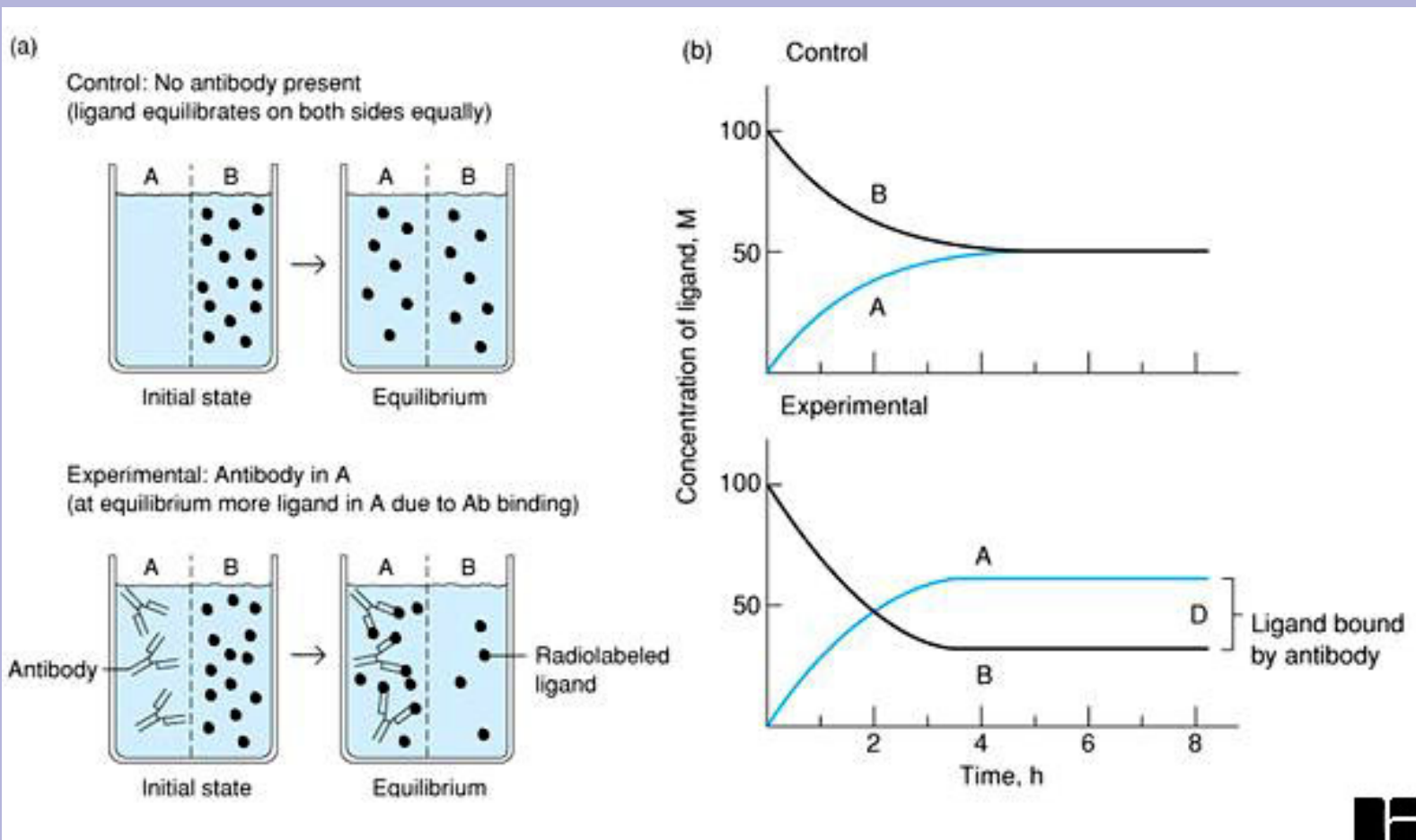


Figure 6-2. Determination of antibody affinity by equilibrium dialysis. (a) The dialysis chamber contains

two compartments (A and B) separated by a semipermeable membrane. Antibody is added to one compartment and a radiolabeled ligand to another. At equilibrium the concentration of radioactivity in both compartments is measured. (b) Plot of concentration of ligand in each compartment with time. At equilibrium the difference in the concentration of radioactive ligand in the two compartments represents the amount of ligand bound to antibody.

From measurements of the equilibrium concentrations of free and bound antigen, starting with different initial concentrations of antigen, one can apply a simple formula in order to determine the equilibrium **association constant** and valence of an antibody. The formula is the **Scatchard equation**: $r/c = K(n-r)$:

- r = moles bound ligand/mole antibody at equilibrium;
- c = free ligand concentration at equilibrium;
- K = equilibrium **association constant**; and
- n = number of antigen binding sites per antibody molecule

By graphical analysis, r/c is plotted on the Y-axis versus r on the X-axis thus producing a **Scatchard plot**, as shown in Figure 6-3.

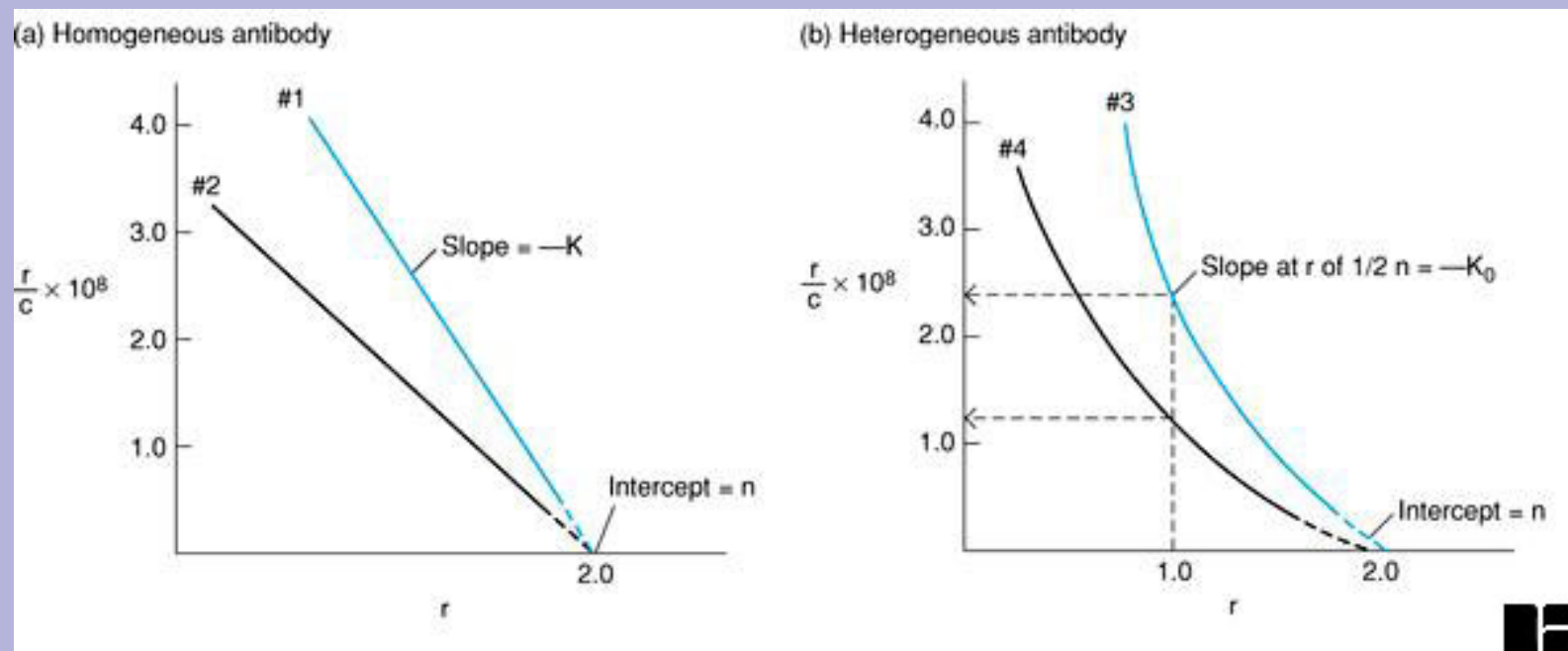


Figure 6-3. Scatchard plots are based on repeated equilibrium dialyses with a constant concentration of antibody and varying concentration of ligand. In these plots, r = moles bound ligand/mole antibody and c = free ligand. From a Scatchard plot, both the equilibrium constant (K) and the number of binding sites

per antibody molecule (n), or its valency, can be obtained. (a) If all antibodies have the same affinity, then a Scatchard plot yields a straight line with a slope of $-K$. The Y intercept is the valence of the antibody, which is 2 for IgG. In this graph antibody #1 has a higher affinity than antibody #2. (b) If the antibodies are pooled and have a range of affinities, a Scatchard plot yields a curved line, whose slope is constantly changing. The average affinity constant K_0 can be calculated by determining the value of K when one-half of the binding sites are occupied (i.e., when $r = 1$). In this graph antiserum #3 has a higher affinity ($K_0 = 2.4 \times 10^8$) than antiserum #4 ($K_0 = 1.25 \times 10^8$).

If the equilibrium reaction relates to a monoclonal antibody, the Scatchard plot will yield a straight line, as illustrated in Figure 6-3(a). In this case the equilibrium **association constant** for antigen, K , equals the negative slope of this line and the antibody's valence, n , or number of antigen binding sites, equals the r value at the point where the extrapolated graphical line intercepts the X-axis.

If the equilibrium reaction relates to a heterogeneous mixture of two or more antibodies recognizing the same antigen, the Scatchard plot will generally yield a curved line, as illustrated in Figure 6-3(b). In this case, the shape of the line depends on the molar ratio of different antibodies as well as their individual association constants. By definition, the **average affinity**, $K_0 = 1/c_{1/2}$, the reciprocal of the concentration of free antigen when one-half of the antigen binding sites are occupied. The average valence equals the r value at the point where the extrapolated graphical line intercepts the X-axis.

The relationships between the shapes of data lines observed on Scatchard plots, as illustrated in Figure 6-3 above, the association constants for antigen binding, and antibody heterogeneity are further explored in a [dynamic Scatchard plot](#), created with *Excel* and designed so that antibody equilibrium association constants and the ratio of antibodies in a heterogeneous mixture can be incrementally adjusted with interactive "dials."

© 1997, Duane W. Sears

[\[TOP\]](#)

[\[Chapter 6: Antibody-Antigen Interactions\]](#)

[\[Chapter Resources\]](#)

[\[Immunology Home Page\]](#)

W. H. Freeman & Co. and Sumanas, Inc.
Immunology, January, 1997

Last Updated: July 15, 1997

Chapter 4: Antigens

NOTE: This chapter requires Netscape Navigator v3.x or greater.

The immune system recognizes antigens in two fundamentally different ways. Antigens are recognized in their native states by [antibodies](#). Antigens are also recognized in fragmented form by [T cells](#). This duality of antigen recognition serves the immune system in two important ways.

- On the one hand, the immune system must protect against live microorganisms, especially those producing toxic substances. Antibodies are effective at neutralizing toxins by removing them from the system or blocking their actions after binding. Antibodies also trigger a host of cell-mediated clearing processes, such as phagocytosis, that can kill invading pathogens.
- On the other hand, the immune system must protect against microorganisms that live or "hide" inside cells, potentially evading the immune system and optimizing their ability to replicate. However, intracellular parasites are detected by T cells through a fairly complicated process involving T-cell receptors, Major Histocompatibility Complex (MHC) molecules, and "antigen processing." With their T-cell receptors, T cells probe the surfaces of cells "looking for" antigenic peptides complexed to MHC molecules.

The [protein-processing](#) events that normally occur inside of cells will produce antigenic peptides. Proteins synthesized internally by a cell will eventually be degraded enzymatically as a normal part of protein turnover. Peptide fragments derived from [internal protein](#) turnover are displayed on the cell surface bound to [MHC class I](#) molecules. Sometimes a cell will internalize proteins from the outside (by phagocytosis for example) and degrade these proteins in the lysosome. Peptide fragments derived from [extracellular proteins](#) are also displayed on the cell surface bound, in this case, to [MHC class II](#) molecules.

- If pathogen-derived antigenic peptides are complexed to [MHC class I](#) molecules on the cell surface, they serve to activate [cytotoxic T](#) lymphocytes which typically kill pathogen-infected cells.
- If pathogen-derived antigenic peptides are complexed to [MHC class II](#) molecules on

the cell surface, they will serve to activate **helper T** lymphocytes which in turn activate and regulate the activities of a variety of other immune cell types.

Antibody-Antigen Recognition 📌

Because immune B and T cell responses to any antigen are very heterogeneous in nature, immunologists frequently simplify matters by studying immune responses to well-characterized antigens that usually have little immunological significance in and of themselves. A case in point is illustrated by the highly refined structure of the antigen-antibody complex shown in [Figure 4-3](#). Here hen-egg white lysozyme (HEL), a concentrated protein in chicken egg whites with little immunological consequence for most individuals, is bound to one antigen-binding arm of an antibody molecule, *i.e.*, an "Fab" fragment. Since the three-dimensional structure of HEL has been thoroughly studied already, this molecule is ideal for analyzing the precise physical basis for the specific interactions between antibodies and antigens.

T-Cell Receptor Interaction with Antigen 📌

As discussed above, T cells recognize antigenic peptides via their T-cell receptors but only when complexed to an MHC molecule.

- The interaction between a **CD4-positive T_H** lymphocyte and an antigenic peptide bound to an **MHC class II** molecule is depicted in [Figure 4-9](#).
- A similar type of interaction takes place between a **CD8-positive T_C** lymphocyte and an antigenic peptide bound to an **MHC class I** molecule.

Superantigens and T Cells 📌

Some pathogens manufacture substances called "superantigens" that fool, "confuse," or hyperactivate the immune system, potentially resulting in misdirected, inefficient, or ineffectual immune responses. By their nature, superantigens induce polyclonal, antigen-independent activation responses from T cells. The inappropriate activation of T cells results from the unique binding properties of superantigens. They have affinity for both T-cell receptors and MHC class II molecules, independent of antigen, as schematically diagrammed in [Figure 4-14](#). Thus, superantigens act as a "glue" bringing together helper T cells and MHC class II-bearing, "antigen presenting" cells. Many T cells are "fooled" into activation by such interactions, potentially creating a life-threatening situation for the host.

In order to better understand their special properties, three-dimensional structures of [superantigen complexes with MHC class II molecules](#) have recently been elucidated.

© 1997, Duane W. Sears

[\[TOP\]](#)

[\[Chapter 4: Anti-HEL Antibody Fab Complexed to Hen Egg-White Lysozyme Antigen\]](#)

[\[Chapter 4: Complexes Between T-Cell Receptor, Antigenic Peptide,
MHC Class II Molecule, CD4 & Superantigen\]](#)

[\[Chapter Resources\]](#)

[\[Immunology Home Page\]](#)

W. H. Freeman & Co. and Sumanas, Inc.
Immunology, January, 1997

Last Updated: June 25, 1997

Mouse MHC Class I Molecule

MHC class I molecules are expressed on the surfaces of most cells and are recognized by CD8-positive cytotoxic T_C-cells, an essential step for initiating the elimination of virally infected cells by T_C-cell-mediated lysis.

-
- MHC class I molecules are heterodimers composed of an α and a β subunit.
 - The α subunit is a 45 kDa polypeptide, inserted in the membrane, and encoded by the MHC.
 - The first two structural domains of the α subunit – α_1 and α_2 – associate to form the peptide-binding pocket and the membrane-distal region of the molecule.
 - The third structural domain of the α subunit – α_3 – noncovalently associates with the β subunit, or β -2 microglobulin (β_2m), forming the membrane-proximal region of the molecule.
 - β_2m is a 12 kDa polypeptide not encoded by the MHC.
-

Animated images of the molecular complex between a soluble fragment of **murine H-2K^b** and a **9-amino-acid nucleoprotein peptide of Sendai Virus (SEV-9)** illustrate key structural features of MHC class I molecules and their antigen-binding pockets.



1vab.pdb

Color coding: β_2m = β -2 microglobulin subunit;

α_1 , α_2 , and α_3 = α subunit domains of **H-2K^b**;
oxygen atoms of H₂Os (H atoms are not visible); and

SEV-9 = **Sendai Virus nucleoprotein peptide**
(PheAlaProGlyAsnTyrProAlaLeu).

1. Animated overview of **H-2K^b**.
2. Examine the peptide-binding pocket formed by a network of anti-parallel β -strands and 2 α -helices, traced at one point here in the N-to-C direction, starting in α_1 and ending in α_2 .
3. View the **SEV-9 peptide-H-2K^b** complex stabilized by 9 H₂O.

4. Expose the interior of the MHC-peptide complex, initially shown as a "slab" cut vertically through the center. Note the positions of the stabilizing H_2O s as the complex rotates.
5. Reset the image.
6. View the immunoglobulin folds of the α_3 -domain and β_2m .

© 1997, Duane W. Sears & Sean R. Christensen

[\[TOP\]](#)

[\[Chapter 9: Overview of MHC class I three-dimensional structure\]](#) [Download Zip](#)

[\[Chapter 9: Animation of dual peptide binding by one MHC class I molecule\]](#) [Download Zip](#)

[\[Chapter 9: Overview of MHC class II three-dimensional structure\]](#)

[\[Chapter 9: Animation of MHC class II domain structure\]](#)

[\[Chapter Resources\]](#)

[\[Immunology Home Page\]](#)

W. H. Freeman & Co. and Sumanas, Inc.
Immunology, January, 1997

Last Updated: June 26, 1997

Figure 9-6: Human HLA-A MHC Class I Molecule

Representation based on x-ray crystallographic analysis of the three-dimensional structure of the external domains of a human class I HLA-A molecule. In the initial side view here, β -strands are depicted as thick arrows, α -helices as spiral ribbons, and **intrachain disulfide bonds** as two interconnected spheres. The α_1 domain and α_2 domain combine to form the membrane-distal peptide-binding pocket. Below the peptide-binding pocket, the α_3 domain combines with β_2 microglobulin, a separate polypeptide subunit, to form the membrane-proximal region of the MHC class I molecule. For additional views:

[Click here](#) for a top view of the peptide-binding pocket formed between the α_1 domain and α_2 domain with its base shaped by a network of antiparallel β -strands from both domains and its sides formed by two α -helices, one from each domain. With the mouse, rotate the image to view the peptide-binding pocket along its various sides. Note that the size of the pocket in class I molecules accommodates peptides containing 8-10 residues.

NOTE: This page requires Netscape Navigator v3. x or greater.

Wire	Space	Ribbon	Trace	"Picking"	In	Out	
------	-------	--------	-------	-----------	----	-----	---

Human HLA-Aw68.1 MHC Class I (chains **a** & **b**).

Enter [RasMol Commands](#) below [2hla.pdb](#)

PDB Info	Sequence	Labels off
Set Picking	None	Label
	Identify	Distance
	Monitor	Monitor off

Move the mouse pointer over the center of an atom in the Chime window and click the mouse button (while holding down the Control, Option, or Command key if using a Macintosh).

© 1997, Duane W. Sears

[\[TOP\]](#)

[\[Chapter Resources\]](#)

[\[Immunology Home Page\]](#)

W. H. Freeman & Co. and Sumanas, Inc.
Immunology, January, 1997

Last Updated: June 26, 1997

[Click here](#) to view aspartic acid residue 227 (Asp₂₂₇, rendered in [CPK colors](#)) located in the α_3 domain and in the region of MHC class I molecules where CD8 molecules bind, as expressed on T_C lymphocytes.

[Click here](#) to reset image.

© 1997, Duane W. Sears

[\[TOP\]](#)

[\[Chapter 9: Animation of MHC class I domain structure\]](#)

[\[Chapter 9: Animation of dual peptide binding by one MHC class I molecule\]](#)

[\[Chapter 9: Overview of MHC class II three-dimensional structure\]](#)

[\[Chapter 9: Animation of MHC class II domain structure\]](#)

[\[Chapter Resources\]](#)

[\[Immunology Home Page\]](#)

W. H. Freeman & Co. and Sumanas, Inc.
Immunology, January, 1997

Last Updated: June 26, 1997

NOTE: This page requires Netscape Navigator v3.x or greater.

Wire	Space	Ribbon	Trace	" Picking "	In	Out	
------	-------	--------	-------	-----------------------------	----	-----	---

Human HLA-Aw68.1 MHC Class I (chains **a** & **b**).

Enter [RasMol Commands](#) below

[2hla.pdb](#)

PDB Info	Sequence	Labels off
Set Picking	None	Label
	Identify	Distance
	Monitor	Monitor off

Move the mouse pointer over the center of an atom in the Chime window and click the mouse button (while holding down the **Control, **Option**, or **Command** key if using a Macintosh).**

© 1997, Duane W. Sears

[\[TOP\]](#)

[\[Chapter Resources\]](#)

[\[Immunology Home Page\]](#)

W. H. Freeman & Co. and Sumanas, Inc.
Immunology, January, 1997

Last Updated: June 26, 1997

©RasMol v2.6a Quick Reference Card

Copyright 1994 [Roger Sayle](#).

[Adapted for this web site by Duane W. Sears]

More comprehensive information on using RasMol can be found in the hypertext version of the [RasMol help manual](#), prepared by **Eric Martz** at the University of Massachusetts.

The most current versions of the RasMol program can be found at internet sites maintained by [Roger Sayle](#) in the UK and [Eric Martz](#) at the University of Massachusetts.

Index

Rendering Commands	Mouse Buttons	Manipulation Commands	General Commands	Set Picking Commands
Atom Expressions	Predefined Sets	Color Commands	Color Schemes	Amino Acid Colors

Rendering Commands

Screen Action	Command	Argument
Set background color	background	< color >
Depth-cueing/lighting	set ambient	[value]
Enable/disable shadows	set shadows	[boolean]
Enable atom highlights	set specular	[boolean]
Control atom 'shininess'	set specpower	[value]

Mouse Buttons

[return to [Index](#)]

Moving the mouse while holding mouse buttons and/or control keys manipulates the molecule. The default bindings are described below.

Screen Action	Windows Mouse button + key (pointer on image)	Macintosh Mouse button + key (pointer on image)
popup menu	right button pointer stationary	button down pointer stationary
X-rotation	left button drag 	button down drag 
Y-rotation	left button drag 	button down drag 
Z-rotation	right button + Shift key, drag 	(none)
X-translation	right button, + Ctrl key, drag 	button down, + Option key, drag 
Y-translation	right button + Ctrl key, drag 	button down + Option key, drag 
zoom in	left button + Shift key, drag 	button down + Shift key, drag 
shrink	left button + Shift key, drag 	button down + Shift key, drag 

Manipulation Commands

[return to [Index](#)]

Screen Action	Command	Argument
Rotate molecule	rotate	<axis> [-] <value>
Translate molecule	translate	<axis> [-] <value>
Scale molecule	zoom	[boolean]
Specify magnification	zoom	<value>
Enable/disable slabbing	slab	[boolean]
Move Z-clipping plane	slab	<value>
Set center of rotation	center	[expression]
Reset orientation to default settings	reset	blank

General Commands

[return to [Index](#)]

Screen Action	Command	Argument
Create hard spheres with van der Waals radii for of all of the selected atoms	spacefill = spacefill	(blank) selected
Create hard spheres with arbitrarily defined radii for of all of the selected atoms; number = 0-700; or turns spacefill off .	spacefill	<number> off

Select all atoms of all molecules in the pdb file	select	*
Select all atoms of chain "y"	select	*y
Display only part of a molecule as defined by the "expression"	restrict	< <u>expression</u> >
Labels selected items with single amino acid code	label	%m
Labels selected items with triple amino acid code		%n
Labels selected items with residue numbers		%r
Renders a ribbon trace of the backbone; or turns ribbons off .	ribbon(s)	<number> off
Renders a cartoon trace of the backbone with alpha helical ribbons and beta sheet arrowheads pointing in the <i>N</i> -to- <i>C</i> direction; or turns cartoons off .	cartoon(s)	<number> off
Renders a line trace of the backbone; or turns trace off .	trace	<number> off

Set Picking Commands

[return to [Index](#)]

"**Set Picking**" commands activate the mouse pointer and button for retrieving information from the image.

Note: After issuing the "**set picking <value>**" command, move the mouse pointer over an atom in the Chime window and click the mouse button (while holding down the **Control**, **Option**, or **Command key** if using a **Macintosh**).

Turns " set picking " off thereby <u>inactivating</u> the mouse pointer and button.	set picking	none
An identification label for the atom selected with the mouse is returned to the message box .		identify
An identification label for the atom selected with the mouse is placed on the image in the Chime window .		label
The distance between 2 atoms sequentially selected with the mouse is returned to the message box .		distance
" Monitors " the distance between 2 atoms sequentially selected with the mouse by placing a dashed connecting line on the image in the Chime window along labeled with the distance..		monitor(s)
Monitors off removes all distances and connecting lines shown on the image in the Chime window but <u>does not turn off the distance monitor</u> . (Use the " set picking none " command to accomplish this.)	monitor(s)	off

The angle defined by 3 atoms sequentially selected with the mouse is returned to the message box .	set picking	angle
The torsion angle defined by 4 atoms sequentially selected with the mouse is returned to the message box .		torsion
The center of image rotation is changed to the atom selected with the mouse.		center

Atom Expressions

[return to [Index](#)]

Type of Expression	Expression (example)	Comments
Wild Cards:	*	specifies every atom in the pdb file
	as?	as? = asp (aspartic acid) <u>plus</u> asn (asparagine)
Predefined Sets:	alpha	specifies all atoms found within all alpha-helical polypeptide segments
	sheet	specifies all atoms found within all beta-sheet polypeptide segments
	hydrophobic	specifies all atoms in hydrophobic residues
Residue Ranges:	3,16,12	specifies all atoms in residues numbered 3, 16, & 12 in the pdb file
	9-20	specifies all atoms in residues inclusive of those numbered 9-20
Boolean Operators:	backbone and not alpha	specifies all atoms defined as part of the backbone but not part of alpha-helical structures
	ser90 and *c	specifies all atoms defined as part of serine residue number 90 and also part of the chain "c"
	ligand or 196-199	specifies all atoms in residues defined as part of a (nonprotein) ligand and also all atoms comprising residues numbered 196-199
Primitive Expressions:	cys, glu, arg or (cys, glu, arg)	specifies all atoms in all residues named cys, glu, arg, as? where cys (cysteine), glu (glutaminic acid), and arg (arginine) are 3-letter abbreviations for the standard amino acids
	ser70a, **p, glu24:1	specifies all atoms in ser residue 90 in chain "a" <u>plus</u> all atoms in chain "p" <u>plus</u> all atoms in glu residue 24 in chain "l"

	hem*p.fe, *.sg	specifies <u>only</u> Fe atoms in all hem (heme) ligands in chain " p " <u>plus</u> all atoms in all chains labeled "sg" in the pdb file
Comparison Operators:	atomno=4,atomno=6	specifies those atoms designated 4 <u>and</u> 6 in the pdb file
	temperature>=900	specifies all atoms with a temperature factor greater than or equal to 900
Within Expressions:	within (8.0,ligand)	specifies all atoms which reside within 8.0 Å distance from any atom defined as part of a ligand

Predefined Sets

[return to [Index](#)]

at	acidic	acyclic	aliphatic	alpha	amino
aromatic	backbone	basic	bonded	buried	cg
charged	cyclic	cystine	helix	hetero	hydrogen
hydrophobic	hoh	ions	large	ligand	medium
neutral	nucleic	polar	protein	purine	pyrimidine
selected	sheet(s)	sidechain	small	solvent	surface
turn	water	User-defined sets: define <identifier> <expression>			
protein sets		nucleic acid sets			
protein or nucleic acids sets		"other molecule/atom" sets			

Color Commands

[return to [Index](#)]

Color representation: color [object] <color>									
Objects:			Predefined Colors:				Atom Color Schemes:		
atoms	bonds	backbone	blue	black	cyan	green	cpk	amino	shapely
ribbon(s)	labels	hbonds	greenblue	magenta	orange	purple	group	chain	structure
ssbonds	dots	axes	red	redorange	violet	white	temperature	charge	user
cartoon(s)			yellow						

Color Schemes

[return to [Index](#)]

Amino Acid Colors

CPK (Corey, Pauling, & Kultun) Atom Colors			ASP, GLU	bright red	[230,10,10]
carbon	gray	[200,200,200]	CYS, MET	yellow	[230,230,0]
oxygen	red	[240,0,0]	LYS, ARG	blue	[20,90,255]
nitrogen	light blue	[143,143,255]	SER, THR	orange	[250,150,0]
hydrogen	white	[255,255,255]	PHE, TYR	mid-blue	[50,50,170]
sulfur	yellow	[255,200,50]	ASN, GLN	cyan	[0,220,220]
phosphorous	orange	[255,165,0]	GLY	light gray	[235,235,235]
chlorine	green	[0,255,0]	LEU, VAL, ILE	green	[15,130,15]
calcium, metals	dark gray	[128,128,144]	ALA	gray	[200,200,200]
unknown	deep pink	[255,20,147]	TRP	pink	[180,90,180]
[R#,G#,B#] numerically specifies a color R# = 0-255, G# = 0-255, and B# = 0-255			HIS	pale blue	[130,130,210]
			PRO	flesh	[220,150,130]

[\[TOP\]](#)

W. H. Freeman & Co. and Sumanas, Inc.
Immunology, January, 1997

Last Updated: May 20, 1997

Figure 9-6: Human HLA-A MHC Class I Molecule

Representation based on x-ray crystallographic analysis of the three-dimensional structure of the external domains of a human class I HLA-A molecule. In the initial side view here, β -strands are depicted as thick arrows, α -helices as spiral ribbons, and **intrachain disulfide bonds** as two interconnected spheres. The α_1 domain and α_2 domain combine to form the membrane-distal peptide-binding pocket. Below the peptide-binding pocket, the α_3 domain combines with β_2 microglobulin, a separate polypeptide subunit, to form the membrane-proximal region of the MHC class I molecule. For additional views:

[Click here](#) for a top view of the peptide-binding pocket formed between the α_1 domain and α_2 domain with its base shaped by a network of antiparallel β -strands from both domains and its sides formed by two α -helices, one from each domain. With the mouse, rotate the image to view the peptide-binding pocket along its various sides. Note that the size of the pocket in class I molecules accommodates peptides containing 8-10 residues.

[Click here](#) to view aspartic acid residue 227 (Asp₂₂₇, rendered in [CPK colors](#)) located in the α_3 domain and in the region of **MHC class I molecules** where **CD8** molecules bind, as expressed on **T_C lymphocytes**.

[Click here](#) to reset image.

© 1997, Duane W. Sears

[\[TOP\]](#)

[\[Chapter 9: Animation of MHC class I domain structure\]](#)

[\[Chapter 9: Animation of dual peptide binding by one MHC class I molecule\]](#)

[\[Chapter 9: Overview of MHC class II three-dimensional structure\]](#)

[\[Chapter 9: Animation of MHC class II domain structure\]](#)

[\[Chapter Resources\]](#)

[\[Immunology Home Page\]](#)

W. H. Freeman & Co. and Sumanas, Inc.
Immunology, January, 1997

Last Updated: June 26, 1997

Binding of Two Different Peptides by a Mouse MHC Class I Molecule

Like antibodies and T cell receptors, MHC molecules also bind antigens in the course of initiating specific immune responses. However, unlike these mono-specific antigen receptors, MHC molecules are capable of binding many different antigens with considerable structural variation. MHC molecules exhibit no idiotypic variability, like that associated with antibodies and T cell receptors, and yet an individual's limited number of MHC molecules must be able to bind sufficient numbers of different antigens to activate T_C and T_H lymphocytes against any invading organism. The multi-antigenic binding capacity of MHC molecules was an enigma until recently, but insights regarding this unusual property of MHC proteins have emerged from three-dimensional structural analyses of MHC-peptide complexes such as those featured in the animations below. Adjacent images compare the structures of the same **murine H-2K^b** MHC class I molecule bound to different peptides. Three key features of MHC-peptide complexes emerge from this analysis:

- 1. MHC molecules bind only small antigenic peptides.** Because small peptides are usually quite flexible in solution, they can adopt numerous conformations over a period of time. Some peptides will transiently adopt conformations that conform to the contour of the MHC peptide-binding pocket. If the surface chemistry of the peptide is also compatible with that of the pocket, the peptide can bind. Press the buttons labeled **"(1)"** below to view animations of the peptide complexes formed between the **murine H-2K^b** MHC class I molecule and a **9-amino-acid, nucleoprotein peptide fragment of Sendai Virus(SEV-9)** (left) or an **8-amino-acid peptide fragment of chicken ovalbumin (OVA-8)** (right). In each case, the "frozen" conformation of the peptide in the pocket represents only one of many possible conformations the peptide can freely adopt in solution.
- 2. The MHC binding pocket exhibits some structural flexibility,** increasing the possibilities for peptide binding. Press the buttons labeled **"(2)"** below to view animations featuring some of the subtle structural differences in the antigen binding pocket for the two complexes.
- 3. A perfect structural "fit" between peptide and the MHC antigen-binding pocket is not necessarily required because water molecules act as a kind of "putty," filling in gaps between the two.** Press the buttons labeled **"(3)-(5)"** below to view animations featuring the locations of **H₂O molecules** (oxygen atoms only) that are hydrogen-bonded to the two peptides in the MHC pocket.

Color coding: α_1 , α_2 , & α_3 = α -subunit domains; β -2 microglobulin; oxygen atoms -- **polar noncharged**, **acidic negatively charged**, or **H-bonded H₂O molecules**; nitrogen atoms -- **polar noncharged** or **basic positively charged**; sidechains -- **Lys₆₆**, **Arg₆₂**, **Arg₇₉**, **Arg₁₅₅**; **Glu₁₅₂**; and peptides -- **SEV-9** and **OVA-8**.

Sendai Virus peptide (SEV-9)
 NH_3^+ -PheAlaProGlyAsnTyrProAlaLeu- COO^-
[1vab.pdb](#)

Zoom in	Zoom out	Rotate (Y-axis)	Spacefill on	Spacefill off
------------	-------------	--------------------	-----------------	------------------

<< **Murine H-2K^b** >> **Chicken ovalbumin peptide (OVA-8)**Click buttons
in sequence.
 NH_3^+ -Ser IleIleAsnPheGluLysLeu-
 COO^- [1vac.pdb](#)

Zoom in	Zoom out	Rotate (Y-axis)	Spacefill on	Spacefill off
------------	-------------	--------------------	-----------------	------------------

peptides
(1)
+ pocket
sidechains

(2)
spacefill
pocket

+H₂O
top view
(3)
side view
(4)
cross-section
(5)
spacefill

reset[\[TOP\]](#)[\[Chapter 9: Overview of MHC class I three-dimensional structure\]](#)[\[Chapter 9: Animation of MHC class I domain structure\]](#)[\[Chapter 9: Overview of MHC class II three-dimensional structure\]](#)

[\[Chapter 9: Animation of MHC class II domain structure\]](#)

[\[Chapter Resources\]](#)

[\[Immunology Home Page\]](#)

W. H. Freeman & Co. and Sumanas, Inc.
Immunology, January, 1997

Last Updated: June 26, 1997

Figure 9-8: Human HLA-DR1 MHC Class II Molecule -- Overview

Representation based on x-ray crystallographic analysis of the three-dimensional structures of the external domains of two dimerized human HLA-DR1 class II $\alpha\beta$ heterodimers with one DR1 $\alpha\beta$ pair rendered in reds and the other DR1 $\alpha\beta$ pair rendered in blues. In the initial side view here, β -strands are depicted as thick arrows, α -helices as spiral ribbons, and **intrachain disulfide bonds** as two interconnected spheres. Each class II molecule is bound to an influenza virus **antigenic peptide** derived from the viral hemagglutinin (residues 306-318). The membrane-distal, N-terminal $\alpha_1\beta_1$ or $\alpha_1\beta_1$ domains combine to form the peptide-binding pockets. Below these pockets, the immunoglobulin-like $\alpha_2\beta_2$ or $\alpha_2\beta_2$ domains combine to form the membrane-proximal regions of the class II molecules. For additional views:

NOTE: This page requires Netscape Navigator v3. x or greater.

Wire	Space	Ribbon	Trace	"Picking"	In	Out	
------	-------	--------	-------	-----------	----	-----	---

HLA-DR1: α chains = **a** & **d**; β chains = **b** & **e**.
 Flu virus hemagglutinin peptide 306-318 = chains **c** & **f**.

Enter [RasMol Commands](#) below [1dlh.pdb](#)

PDB Info	Sequence	Labels off
Set Picking	None	Label
	Identify	Distance
	Monitor	Monitor off

Move the mouse pointer over the center of an atom in the Chime window and click the mouse button (while holding down the Control, Option, or Command key if using a Macintosh).

© 1997, Duane W. Sears

[\[TOP\]](#)

[\[Chapter Resources\]](#)

[\[Immunology Home Page\]](#)

W. H. Freeman & Co. and Sumanas, Inc.
 Immunology, January, 1997

Last Updated: June 26, 1997

[Click here](#) for a top view of the two peptide-binding pockets formed between the $\alpha_1\beta_1$ or $\alpha_1\beta_1$ domains. The floor of each pocket is shaped by antiparallel β -strands from the adjacent domains and the sides are shaped by α -helices, one from each domain. With the mouse, rotate the image to view the peptide-binding pockets from the various sides. Note that the size of the pocket in class II molecules accommodates peptides containing 13-18 residues.

[Click here](#) to hide the bound peptides.

[Click here](#) to view the complete structure of the bound peptides (now rendered in [CPK colors](#)). Move the peptide-binding pocket with the mouse and note how some of the side chain atoms of the peptide protrude from the peptide-binding pocket, presenting target structures for T_H cell recognition.

[Click here](#) to view amino acid residues 134-148 (rendered in [CPK colors](#)) of the β_2 and β_2 domains identifying the region on this MHC class II complex that specifically interacts with [CD4 molecules](#), as expressed on T_H lymphocytes.

[Click here](#) to reset image.

© 1997, Duane W. Sears & Sean R. Christensen

[\[TOP\]](#)

[\[Chapter 9: Overview of MHC class I three-dimensional structure\]](#)

[\[Chapter 9: Animation of MHC class I domain structure\]](#)

[\[Chapter 9: Animation of dual peptide binding by one MHC class I molecule\]](#)

[\[Chapter 9: Animation of MHC class II domain structure\]](#)

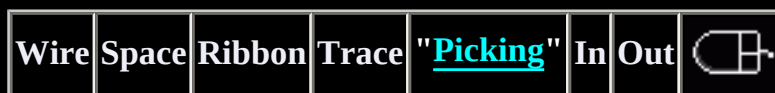
[\[Chapter Resources\]](#)

[\[Immunology Home Page\]](#)

W. H. Freeman & Co. and Sumanas, Inc.
Immunology, January, 1997

Last Updated: June 26, 1997

NOTE: This page requires Netscape Navigator v3.x or greater.



HLA-DR1: α chains = **a** & **d**; β chains = **b** & **e**.

Flu virus hemagglutinin peptide 306-318 = chains **c** & **f**.

Enter [RasMol Commands](#) below

[1dlh.pdb](#)

PDB Info	Sequence	Labels off
<u>Set Picking</u>	<u>None</u>	<u>Label</u>
	<u>Identify</u>	<u>Distance</u>
	<u>Monitor</u>	<u>Monitor off</u>

Move the mouse pointer over the center of an atom in the Chime window and click the mouse button (while holding down the **Control, **Option**, or **Command** key if using a Macintosh).**

© 1997, Duane W. Sears

[\[TOP\]](#)

[\[Chapter Resources\]](#)

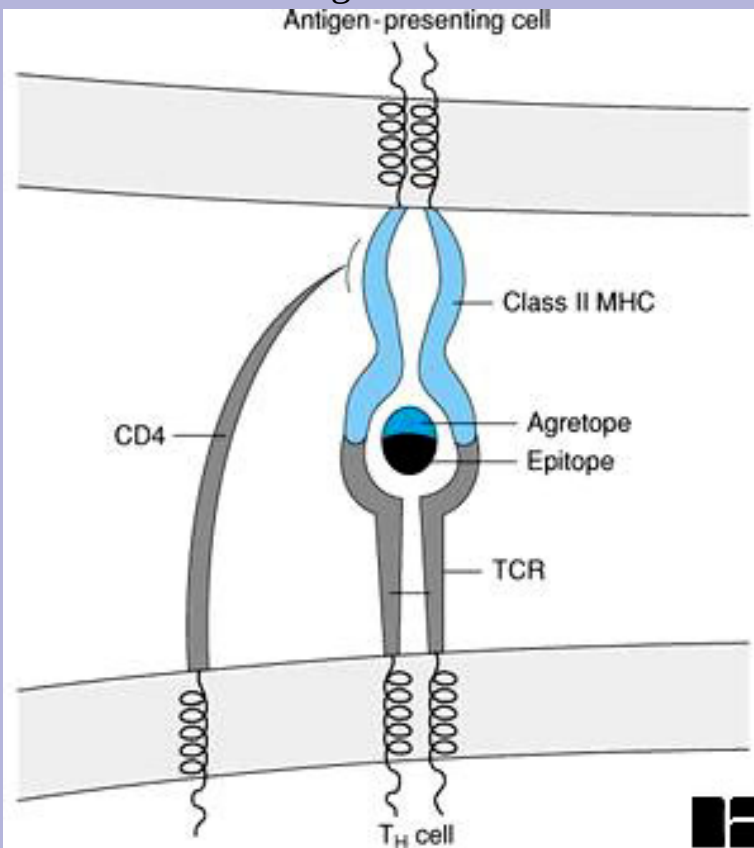
[\[Immunology Home Page\]](#)

W. H. Freeman & Co. and Sumanas, Inc.
Immunology, January, 1997

Last Updated: June 26, 1997

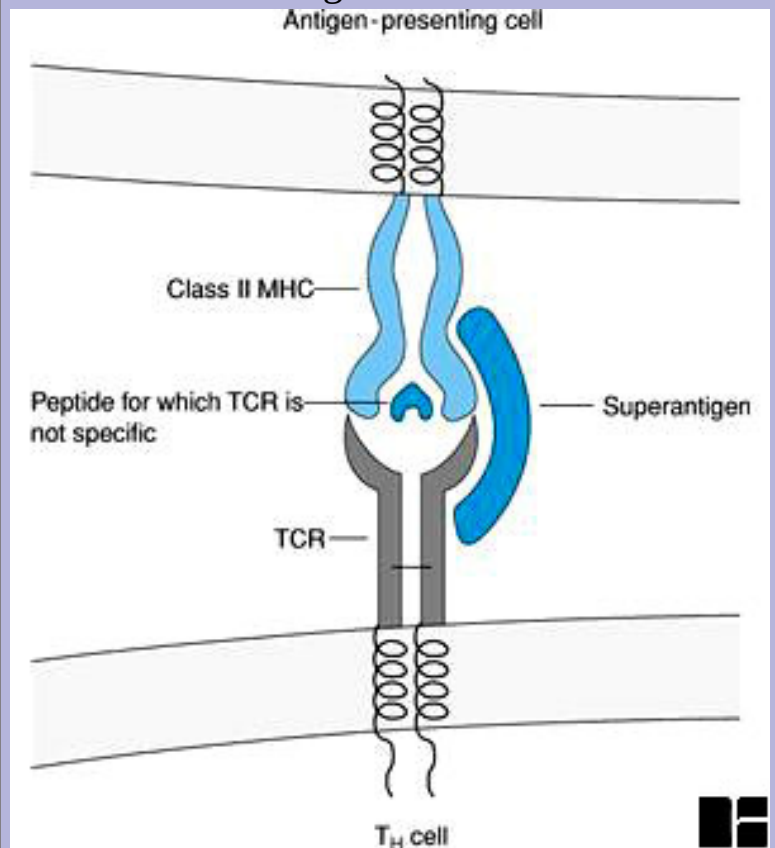
T-Cell Receptor Complex with Antigenic Peptide, MHC Class II Molecule, CD4, and Superantigen

Figure 4-9



Schematic diagram of the ternary complex formed between a T-cell receptor (TCR) on a T_H cell, an antigen, and class II MHC molecule. Antigens that are recognized by T cells have two distinct interaction sites: an agrepto, which interacts with an MHC molecule, and an epitope, which interacts with the T-cell receptor. CD4 on T_H cells also interacts with MHC molecules. T_C cells form similar ternary complexes with class I MHC molecules on target cells.

Figure 4-14:



Schematic diagram of the ternary complex formed between a T-cell receptor (TCR), superantigen, and class II MHC molecule. Superantigens bind to common sequences in class II MHC molecules and T-cell receptors that lie outside the normal antigen-binding sites. T-cell activation by superantigens is not limited by the antigenic specificity of the T cell.

[\[Immunology Home Page\]](#)

W. H. Freeman & Co. and Sumanas, Inc.
Immunology, January, 1997

Last Updated: June 25, 1997

Figure 9-8: Human HLA-DR1 MHC Class II Molecule -- Overview

Representation based on x-ray crystallographic analysis of the three-dimensional structures of the external domains of two dimerized human HLA-DR1 class II $\alpha\beta$ heterodimers with one DR1 $\alpha\beta$ pair rendered in reds and the other DR1 $\alpha\beta$ pair rendered in blues. In the initial side view here, β -strands are depicted as thick arrows, α -helices as spiral ribbons, and **intrachain disulfide bonds** as two interconnected spheres. Each class II molecule is bound to an influenza virus **antigenic peptide** derived from the viral hemagglutinin (residues 306-318). The membrane-distal, N-terminal $\alpha_1\beta_1$ or $\alpha_1\beta_1$ domains combine to form the peptide-binding pockets. Below these pockets, the immunoglobulin-like $\alpha_2\beta_2$ or $\alpha_2\beta_2$ domains combine to form the membrane-proximal regions of the class II molecules. For additional views:

[Click here](#) for a top view of the two peptide-binding pockets formed between the $\alpha_1\beta_1$ or $\alpha_1\beta_1$ domains. The floor of each pocket is shaped by antiparallel β -strands from the adjacent domains and the sides are shaped by α -helices, one from each domain. With the mouse, rotate the image to view the peptide-binding pockets from the various sides. Note that the size of the pocket in class II molecules accommodates peptides containing 13-18 residues.

[Click here](#) to hide the bound peptides.

[Click here](#) to view the complete structure of the bound peptides (now rendered in [CPK colors](#)). Move the peptide-binding pocket with the mouse and note how some of the side chain atoms of the peptide protrude from the peptide-binding pocket, presenting target structures for T_H cell recognition.

[Click here](#) to view amino acid residues 134-148 (rendered in [CPK colors](#)) of the β_2 and β_2 domains identifying the region on this **MHC class II** complex that specifically interacts with [CD4 molecules](#), as expressed on T_H lymphocytes.

[Click here](#) to reset image.

[\[Chapter 9: Overview of MHC class I three-dimensional structure\]](#)

[\[Chapter 9: Animation of MHC class I domain structure\]](#)

[\[Chapter 9: Animation of dual peptide binding by one MHC class I molecule\]](#)

[\[Chapter 9: Animation of MHC class II domain structure\]](#)

[\[Chapter Resources\]](#)

[\[Immunology Home Page\]](#)

W. H. Freeman & Co. and Sumanas, Inc.
Immunology, January, 1997

Last Updated: June 26, 1997

Human MHC Class II Molecule

MHC class II molecules are expressed exclusively on the surfaces of "antigen-presenting cells" -- including macrophages, dendritic cells, and B cells -- where they serve to stimulate CD4-positive T_H-cells, thereby initiating the release of cytokines that regulate other cellular processes associated with immune responses.

-
- MHC class II molecules are heterodimers composed of an α and a β subunit.
 - The α subunit is a 33 kDa polypeptide, the β subunit is a 28 kDa polypeptide, and both are inserted in the membrane and encoded by the MHC.
 - The N-terminal structural domains of the two subunits – the α_1 -domain and β_1 -domain – associate to form the peptide-binding pocket and the membrane-distal region of the molecule.
 - The α_2 -domain and β_2 -domain associate to form the membrane-proximal region of the molecule.
 - MHC class II $\alpha\beta$ dimers exhibit a tendency to self-associate, forming $(\alpha\beta)_2$ "dimers of dimers."
-

Animated images of the molecular complexes between 2 **human HLA-DR1** molecules and the **synthetic polyalanine peptide** -- **(Ala)₁₃** -- illustrate key structural features of MHC class II molecules and their antigen-binding pockets.

[1seb.pdb](#)



Color coding: α_1 and α_2 = α subunit domains; β_1 and β_2 = β subunit domains; and **(Ala)₁₃** = **synthetic polyalanine peptide** bound to **HLA-DR1 $\alpha\beta$** .

1. Animated overview of the **HLA-DR1** class II complex with **(Ala)₁₃** peptide.
2. Examine the peptide-binding pocket formed by a network of anti-parallel β -strands and 2 α -helices from the α_1 and β_1 domains, illustrated at one point here by a simultaneous trace in N-to-C direction through α_1 and β_1 .
3. View the immunoglobulin folds of the α_2 and β_2 domains.

4. View the $(\alpha\beta)_2$ tetrameric dimer complex with two peptide antigen binding sites.
5. Reset image.

[\[TOP\]](#)

[\[Chapter 9: Overview of MHC class I three-dimensional structure\]](#)

[\[Chapter 9: Animation of MHC class I domain structure\]](#)

[\[Chapter 9: Animation of dual peptide binding by one MHC class I molecule\]](#)

[\[Chapter 9: Overview of MHC class II three-dimensional structure\]](#)

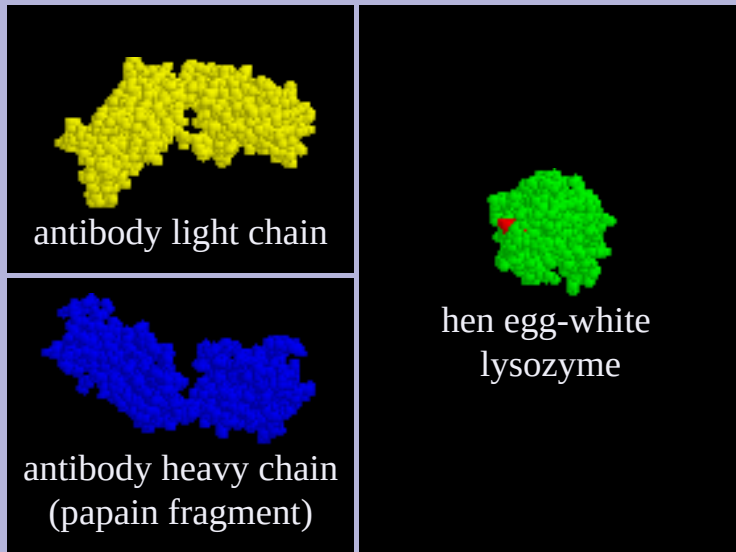
[\[Chapter Resources\]](#)

[\[Immunology Home Page\]](#)

W. H. Freeman & Co. and Sumanas, Inc.
Immunology, January, 1997

Last Updated: June 26, 1997

Figure 4-3: Complex Between HEL Antigen and Anti-HEL Antibody Fab



The structure of an antigen-antibody immune complex is shown with hen egg-white lysozyme (HEL) antigen bound to the **Fab fragment** derived by papain digestion from an **anti-HEL** monoclonal antibody. One end of the Fab fragment makes direct contact with the surface of HEL effectively burying **glutamine₁₂₁** (highlighted in red) and several neighboring surface amino acid residues of HEL. Together these amino acid residues define the "**antigenic determinant**" or "**epitope**" of HEL. Similarly, the amino acid residues of the antibody making contact with antigen define its "**antigen combining site**." Generally, these residues reside in the "**hypervariable loops**" or "**complementarity-determining regions**" (CDRs) of the antibody heavy and light chains. Overall, the specificity of an antibody for its antigen is determined by the complementary nature of the

NOTE: This page requires Netscape Navigator v3. x or greater.

Wire	Space	Ribbon	Trace	" Picking "	In	Out	
------	-------	--------	-------	-----------------------------	----	-----	--

Anti-HEL Fab fragment (chains **h** & **l**) complexed to **HEL** (chain **y**)

Enter [RasMol Commands](#) below [1fdl.pdb](#)

PDB Info	Sequence	Labels off
Set Picking	None	Label
	Identify	Distance
	Monitor	Monitor off

Move the mouse pointer over the center of an atom in the Chime window and click the mouse button (while holding down the Control, Option, or Command key if using a Macintosh).

© 1997, Duane W. Sears

[\[TOP\]](#)

[\[Chapter Resources\]](#)

[\[Immunology Home Page\]](#)

W. H. Freeman & Co. and Sumanas, Inc.
Immunology, January, 1997

Last Updated: June 25, 1997

epitope and antibody contact residues in terms of shape, charge, and chemistry. The structural basis for this antigen-antibody interaction is further explored with a series of [RasMol commands](#) entered in the command box at the right.

© 1997, Duane W. Sears

[\[TOP\]](#)

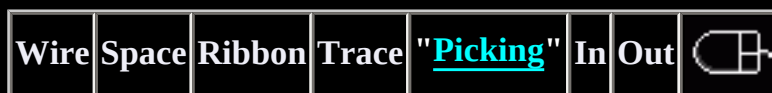
[\[Chapter Resources\]](#)

[\[Immunology Home Page\]](#)

W. H. Freeman & Co. and Sumanas, Inc.
Immunology, January, 1997

Last Updated: June 25, 1997

NOTE: This page requires Netscape Navigator v3.x or greater.



Anti-HEL Fab fragment (chains **h** & **l**) complexed to **HEL** (chain **y**)

Enter RasMol Commands below

[1fdl.pdb](#)

PDB Info	Sequence	Labels off
<u>Set Picking</u>	<u>None</u>	<u>Label</u>
	<u>Identify</u>	<u>Distance</u>
	<u>Monitor</u>	<u>Monitor off</u>

Move the mouse pointer over the center of an atom in the Chime window and click the mouse button (while holding down the **Control, **Option**, or **Command** key if using a **Macintosh**).**

© 1997, Duane W. Sears

[\[TOP\]](#)

[\[Chapter Resources\]](#)

[\[Immunology Home Page\]](#)

W. H. Freeman & Co. and Sumanas, Inc.
Immunology, January, 1997

Last Updated: June 25, 1997

Epitope/Antibody Combining Site Interaction Between HEL and Anti-HEL Fab

The interaction between HEL and the anti-HEL antibody can be explored further by using the command box at the right to issue RasMol commands that will selectively modify the image. With the first series of commands, a magnified view of the [epitope](#) or [antigenic determinant](#) of HEL is generated. With the second series of commands, a magnified view of the anti-HEL [antibody contact residues](#) is generated. Modify the image in other ways using RasMol commands as summarized in the [RasMol guide](#). The original image can always be restored by pressing the browser's "Reload" button.

Command entry can be expedited with the computer's "cut-and-paste" feature:

- Highlight the command with the mouse,
- "cut" the command,
- "paste" it into the command box, and
- press the [Enter key](#) to issue the command.

Highlighted Antigenic Epitope of HEL

Starting instructions:

Click the "Clear messages" button.

(This button may be pressed at any time without affecting commands already entered.)

Click the [mouse pointer](#) in the command box.

Press the [Enter key](#) after typing each command.

RasMol command <#>
(in black)

Comments

1. **select (18-27,117-125) and *y**

selects all residues of HEL (= *y) forming its epitope

2. **define epitope selected**

creates a custom "set" identified as "epitope" corresponding to all atoms currently selected

3. **select epitope and not gln121**

selects all epitope atoms except for those of gln121

4. **color magenta**

colors all epitope atoms magenta except for those of gln121

5. **reset**

resets orientation to default setting

6. **restrict *y**

shows only HEL = *y & hides Fab = (*l, *h)

7. **zoom 500**

"zooms-in" on the HEL epitope

8. **translate y 100**

moves HEL vertically down into the center of the Chime window

9. **center gln121**

changes image rotation point to the center of mass of gln121
#



Examine the structure anywhere along the way by moving it using the [mouse pointer and button](#).

Highlighted Contact Residues of Anti-HEL Fab

RasMol command #
(in black)

Comments

1. **select within (8.0,epitope) and not *y**
selects all atoms of Fab only ("not *y") within 8.0 Å of the atoms comprising the epitope
2. **spacefill**
selects all selected atoms
3. **define contact selected**
creates a custom "set" identified as "contact" corresponding to all atoms currently selected
4. **select contact and not *l**
selects all contact atoms in the heavy chain and not those of the light chain (not *l)
5. **color purple**
colors heavy chain contact atoms purple
6. **select contact and not *h**
selects all contact atoms in the light chain and not those of the heavy chain (not *h)
7. **color orange**
colors light chain contact atoms orange
8. **restrict epitope,contact**
restricts the image to the epitope-contact atoms
9. **spacefill off**
turns spacefilling off
10. **wireframe 75**
sets wireframe at "75"
11. **select ***
selects all atoms
12. **spacefill on**
spacefill selected atoms
13. **reset**

reorients structure to default
settings

 # Examine the structure anywhere along the way by moving it using the [mouse pointer and button](#).

© 1997, Duane W. Sears

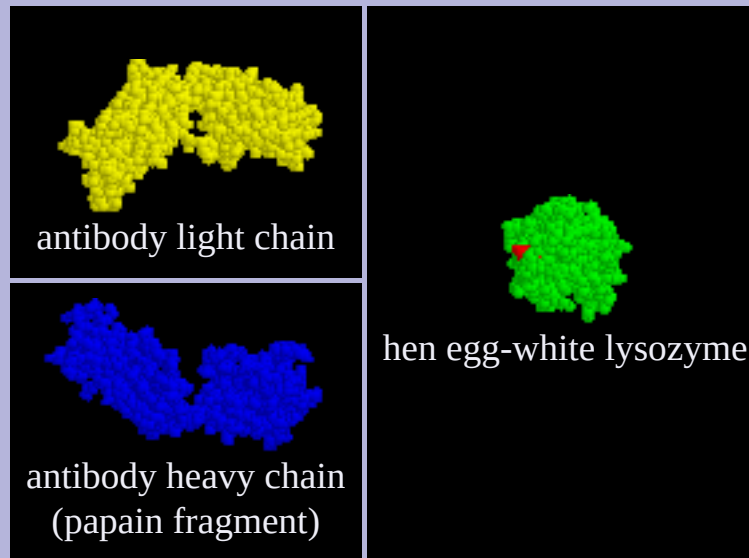
[\[TOP\]](#)

[\[Chapter Resources\]](#)

[\[Immunology Home Page\]](#)

W. H. Freeman & Co. and Sumanas, Inc.
Immunology, January, 1997

Last Updated: June 25, 1997

Figure 4-3: Complex Between HEL Antigen and Anti-HEL Antibody Fab

The structure of an antigen-antibody immune complex is shown with hen egg-white lysozyme (HEL) antigen bound to the **Fab fragment** derived by papain digestion from an **anti-HEL** monoclonal antibody. One end of the Fab fragment makes direct contact with the surface of HEL effectively burying **glutamine₁₂₁** (highlighted in red) and several neighboring surface amino acid residues of HEL. Together these amino acid residues define the "**antigenic determinant**" or "**epitope**" of HEL. Similarly, the amino acid residues of the antibody making contact with antigen define its "**antigen combining site**." Generally, these residues reside in the "**hypervariable loops**" or "**complementarity-determining regions**" (**CDRs**) of the antibody heavy and light chains. Overall, the specificity of an antibody for its antigen is determined by the complementary nature of the **epitope** and antibody **contact residues** in terms of shape, charge, and chemistry. The structural basis for this antigen-antibody interaction is further explored with a series of [RasMol commands](#) entered in the command box at the right.

© 1997, Duane W. Sears

[\[TOP\]](#)

[\[Chapter Resources\]](#)

[\[Immunology Home Page\]](#)

W. H. Freeman & Co. and Sumanas, Inc.

Immunology, January, 1997

Last Updated: June 25, 1997

Chapter 12: Enterotoxin B Superantigen Interaction with MHC Class II

Staphylococcal **enterotoxin B (SEB)** is a superantigen that binds the membrane-distal, α_1 -domain of the α subunit of the human **HLA-DR1** class II MHC molecule. Buttons **1** and **2** below animate this interaction showing that **SEB** binds near to but still outside the MHC peptide-binding pocket. In particular, **SEB** binds to a segment of the β -strands and one α -helix that partially shape the exterior of the peptide-binding pocket. Button **3** animates the complementary **hydrophobic** and **polar** surfaces of the **SEB** molecule and the α_1 -domain that define the specificity of their interaction.

Although it is not shown here, **SEB** also binds to some but not all isoforms of the human T cell receptor V_β chain, including $V_\beta 3$, $V_\beta 12$, $V_\beta 14$, $V_\beta 15$, $V_\beta 17$, and $V_\beta 20$. Button **4** highlights several **SEB** amino acid sidechains (imaged in white) that are implicated (by point-mutational analysis) in the binding reactions with T cell receptor V_β chains. **SEB** binds V_β chains near to but still outside their antigen-binding sites. Specifically, **SEB** binds to a region of the V_β chain which is structurally more conserved than the hypervariable regions that shape part of the antigen-binding site.

These data help explain how superantigen bypasses the normal routes for specific recognition of antigen by MHC class II molecules and T cell receptors resulting in antigen-nonspecific activation of the cells expressing these receptors. The consequence is an unusually vigorous polyclonal activation of T cells with an exaggerated immune-like response that can be life-threatening in the worst-case scenarios.

[1seb.pdb](#)



Reset

Zoom in	Zoom out	Rotate (Y-axis)	Spacefill on	Spacefill off
------------	-------------	--------------------	-----------------	------------------

1. Animated overview of the **HLA-DR1** ($\alpha\beta$) class II multi-peptide complex formed with 2 antigenic peptides -- (**Ala**)₁₃ -- and 2 **SEB** molecules.
2. **SEB** binds near to the antigenic peptide but still outside the peptide-binding pocket of **HLA-DR1**.

3. Complementary **hydrophobic**: **hydrophobic** and **polar**:**polar** interactions defining the binding specificity of the **SEB**: α -subunit complex. Note how the positively charged amino group of **Lys**₃₉⁺ **NH**₃⁺ of the α -subunit is positioned so as to form hydrogen bonds or salt bridges with 4 **oxygen atoms** of the **SEB** molecule, including the 2 γ -carboxylate oxygen atoms of **Glu**₆₇⁻-COO⁻ (each with 1/2 negative charge) and the phenolic oxygen atoms of **Tyr**₈₉-OH and **Tyr**₁₁₅-OH.
4. Amino acid sidechain residues of **SEB** (in white) implicated in binding to T cell receptor V _{β} chains.
Color coding: $\alpha\beta$ (**HLA-DR1**); α_1 and α_2 = α subunit domains; β_1 and β_2 = β subunit domains; (**Ala**)₁₃ = **synthetic polyalanine peptide**; **enterotoxin B (SEB)**; **polar** and **hydrophobic** regions; and amino acid sidechain residues of **SEB** implicated in V _{β} chain T cell receptor binding.

© 1997, Duane W. Sears and Sean R. Christensen

[\[TOP\]](#)

[\[Chapter 12: T-Cell Maturation, Activation, and Differentiation\]](#)

[\[Chapter Resources\]](#)

[\[Immunology Home Page\]](#)

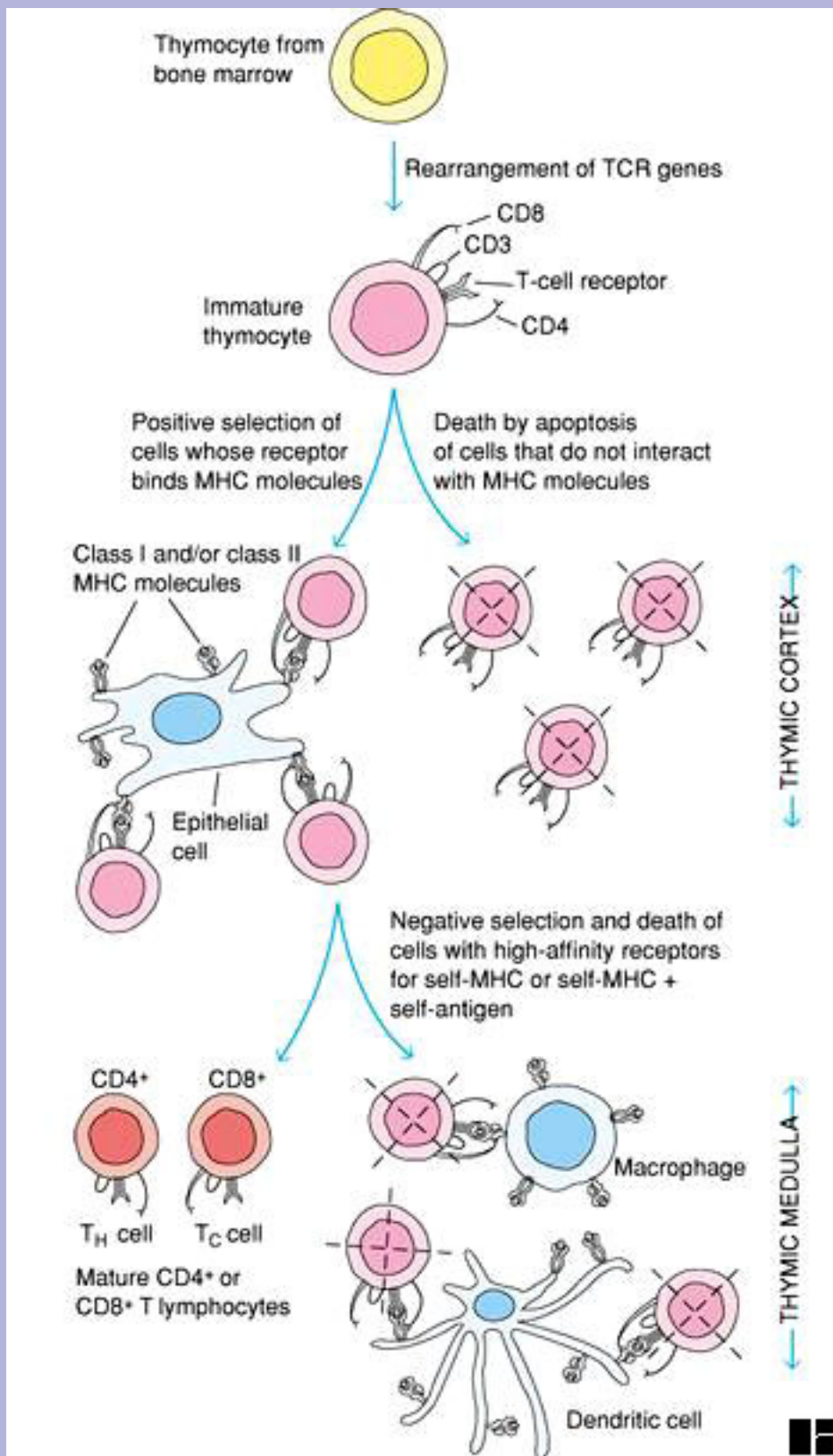
W. H. Freeman & Co. and Sumanas, Inc.
 Immunology, January, 1997

Last Updated: July 15, 1997

Chapter 12: T-Cell Maturation, Activation, and Differentiation

As cell signaling molecules, T-cell receptors are particularly intriguing because they initiate so many different cell-mediated processes that are dependent on a variety of factors, including the host cell expressing these molecules and co-signaling events triggered by various environmental stimuli or by other receptors expressed on the same cell. For example, antigen engagement of the T-cell receptors expressed on mature T cells typically results in activation of T_H helper or T_C cytotoxic effector functions along with clonal expansion of antigen-specific T-cell populations. In some situations, however, T-cell receptor engagement results in a state of clonal anergy or antigen non-responsiveness of the T-cell. In the thymus, T-cell receptors play a pivotal role in the T-lymphocyte maturation process. When the T-cell receptors on a pre-T-cell thymocyte are engaged, the thymocyte can be either positively or negatively selected depending on the balancing effects of several other factors regulating this process. As illustrated in Figure 12-5, thymocytes surviving these selection events emerge from the thymus as mature T cells.

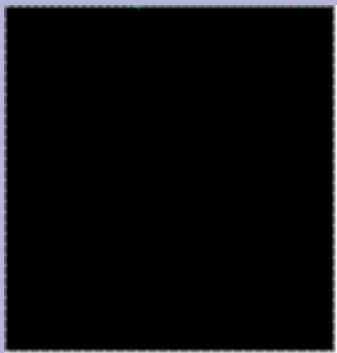
Figure 12-5. Positive and negative selection of thymocytes in the thymus. Because of thymic selection, which involves thymic stromal cells (epithelial cells, dendritic cells, and macrophages), mature T cells are both self-MHC restricted and self-tolerant.



In addition to their normal physiological functions, T-cell receptors also play a key role in some immunopathological responses such as those driven by superantigens. Superantigens are the molecular products of microbial pathogens and these effectively over-stimulate the immune system, creating an immunological "smoke screen" that diverts the activities of immune cells away from the invading microorganism. As illustrated in [Figure 12-16](#),

superantigens have the unique capacity to crosslink MHC class II molecules on antigen presenting cells simultaneously with T-cell receptors on helper T_H cells, bringing these cells together and falsely inducing their activation. By bypassing the normal steps in the antigen-specific activation of T cells, superantigens polyclonally activate sizeable populations of T_H cells, exaggerating the immune response. Hyperactive immune responses induced by superantigens can result in a number of serious diseases that may even be life-threatening owing to excessive levels of tissue-damaging cytokines and immune effector cell activity. Food poisoning, as one example, is caused by [bacterial superantigens](#), such as the *Staphylococcal* enterotoxin B (SEB) produced by bacteria in contaminated food. Toxic shock syndrome (TSS) is another superantigen-induced disease caused by yet another *Staphylococcal* toxin produced from bacteria on contaminated tampons.

Fresh insights regarding the special properties of these potent toxins have come from X-ray crystallographic analyses of superantigens complexed to MHC class II molecules, as illustrated by the image below which shows two **SEB superantigen toxins** of *Staphylococcal aureas* bound to a human MHC class II ($\alpha\beta$)₂ dimer at the center of the complex.



An exposed surface located on top of each **SEB molecule** in this image has been implicated (by mutational analysis) in T-cell receptor binding. The T-cell receptor binding sites on the superantigen molecules in the complex are also positioned near but not within the MHC peptide binding pockets. Thus, a mechanism for antigen-nonspecific, T-cell receptor crosslinking by superantigen is suggested by the fact that two **SEB molecules** are vertically oriented in similar directions in this complex.

Click here to view [animations of the enterotoxin B complex with a human MHC class II molecule](#).

© 1997, Duane W. Sears and Sean R. Christensen

[\[TOP\]](#)

[\[Chapter Resources\]](#)

[\[Immunology Home Page\]](#)

W. H. Freeman & Co. and Sumanas, Inc.

Immunology, January, 1997

Last Updated: July 15, 1997

Figure 12-16: Superantigen-mediated cross-linkage of T-cell receptor and class II MHC molecules.

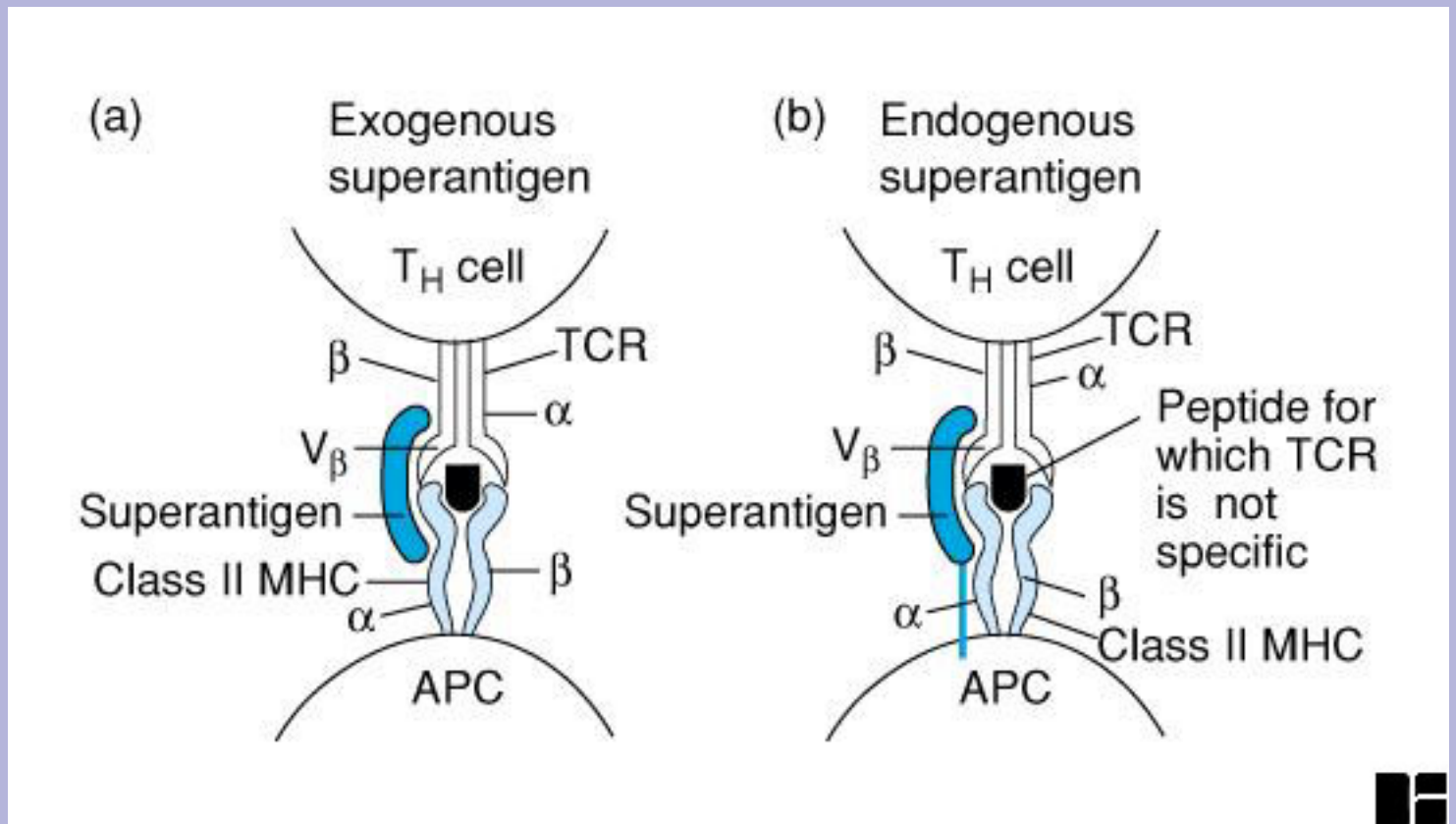


Figure 12-16. Each superantigen (dark blue) binds to all TCRs bearing a particular V_β sequence regardless of their antigenic specificity. (a) Exogenous superantigens are soluble secreted bacterial proteins, including various exotoxins. (b) Endogenous superantigens are membrane-embedded proteins produced by certain viruses; they include MIs antigens encoded by mouse mammary tumor viruses.

© 1997, Duane W. Sears

[\[TOP\]](#)

[\[Chapter 12: T-cell maturation, activation, and differentiation\]](#)

[\[Chapter Resources\]](#)

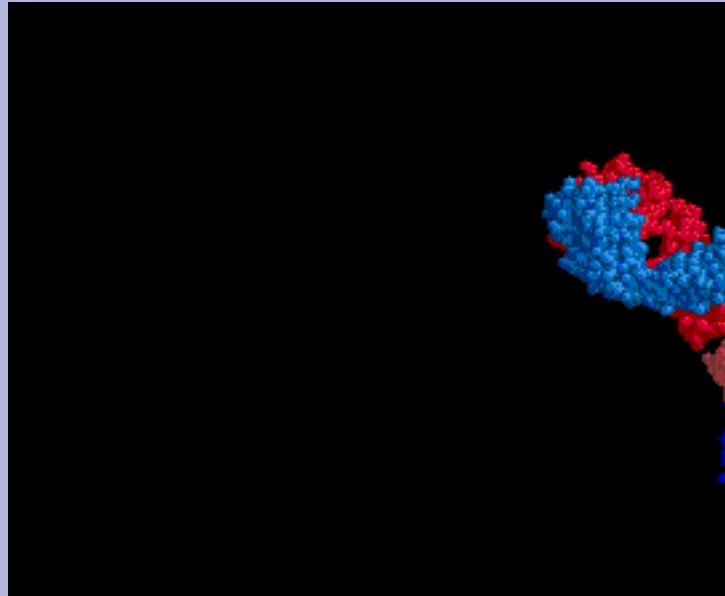
[\[Immunology Home Page\]](#)

W. H. Freeman & Co. and Sumanas, Inc.
Immunology, January, 1997

Last Updated: July 15, 1997

Chapter 5: Immunoglobulins: Structure and Function

NOTE: This chapter requires Netscape Navigator v3.x or greater.



Immunoglobulins, or antibodies, are a complex, heterogeneous mixture of proteins that exhibit two fundamental types of structural variation. Subtle structural differences in their antigen combining sites, or [variable regions](#), account for their unique antigen binding specificities (see [Figure 4-3](#)). Structural differences outside their antigen combining sites, in the so-called [constant regions](#), correlate with the different [effector functions](#) mediated by antibodies, such as complement activation or binding to one or more of the [antibody Fc receptors](#) expressed on monocytes and granulocytes. The variable and constant regions of antibodies arise from distinct structural domains as illustrated here in an [animated overview of the three-dimensional structure of IgG antibodies](#).

IgG antibodies are among one of the **five major classes of immunoglobulins** that also include **IgA**, **IgD**, **IgE**, and **IgM** antibodies (see [Figure 5-15](#)). Each antibody class is distinguished by certain effector functions and structural features including a unique heavy (H) chain isotype, designated α (IgA), δ (IgD), ϵ (IgE), γ (IgG), or μ (IgM). H chains of all isotypes associate with light (L) chains of two isotypes -- κ and λ . Thus, the basic H_2L_2 composition of an antibody can be specified in terms of its H and L isotypes; *e.g.*, $\epsilon_2\kappa_2$, $(\mu_2\lambda_2)_5$, etc.

In some species, the immunoglobulin classes are further differentiated according to **subclasses**, adding another layer of complexity to antibody structure. In humans, for example, IgG antibodies comprise **four IgG subclasses** -- IgG1, IgG2, IgG3, and IgG4 (see [Figure 5-16](#)) -- while there are **two IgA subclasses** -- IgA1 and IgA2. Each subclass corresponds to a different **heavy chain isotype**, designated $\gamma 1$ (IgG1), $\gamma 2$ (IgG2), $\gamma 3$ (IgG3), $\gamma 4$ (IgG4), $\alpha 1$ (IgA1) or $\alpha 2$ (IgA2). In all, **nine heavy chain genes** in the human haploid genome account for the subclasses and classes of human immunoglobulins.

While some of the heterogeneity of antibodies derives from the different classes and subclasses, most of the heterogeneity stems from the highly polymorphic nature of their variable regions. Estimates of the total number of distinct antibody variable regions or antigen combining sites that an individual's B cells can theoretically produce range from **10 million upward**. It is not surprising then that antibody responses to even simple antigens are generally very heterogeneous in terms of the numbers of different antibodies made, a bonus for the individual as far as immune protection goes but a hindrance to the immunologist trying to sort out and manipulate this intricate process. However, each different antibody is produced only by a single clone of B cells, the "biological key" for [creating hybridomas producing monoclonal antibodies](#). The fine-specificity of antigen recognition by monoclonal antibodies coupled with the relative ease of producing them has resulted in widespread [use of monoclonal antibodies in both research and medicine](#).

For additional overview of antibody structure, see the [Gallery of Immune Molecule Structures](#) created by Eric Martz at the University of Massachusetts, Amherst, where the structures of other immune molecules are also featured.

© 1997, Duane W. Sears

[\[TOP\]](#)

[\[Chapter 5: Overview of the three-dimensional structure of human IgG1 antibodies\]](#)

[\[Chapter 5: Overview of the three-dimensional structure of a catalytic monoclonal antibody\]](#)

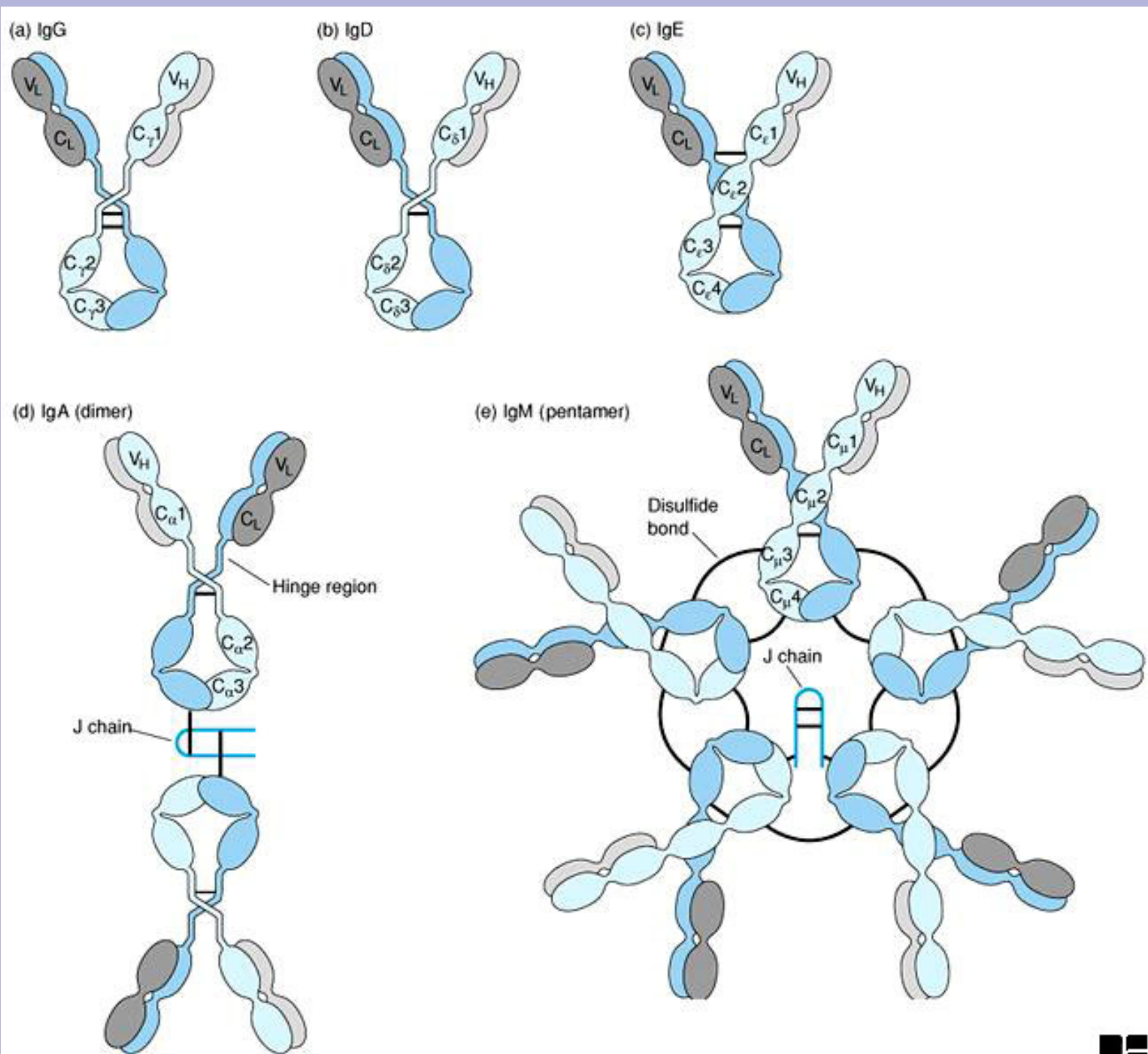
[\[Chapter Resources\]](#)

[\[Immunology Home Page\]](#)

W. H. Freeman & Co. and Sumanas, Inc.
Immunology, January, 1997

Last Updated: June 25, 1997

Figure 5-15: Major Immunoglobulin Classes



General structure of the **five major classes** of secreted antibody. Light chains are shown in shades of gray and heavy chains in shades of blue; disulfide bonds are indicated by thick black lines. Note that the **IgG**, **IgA**, and **IgD** heavy chains contain four domains but no hinge region. The polymeric forms of **IgM** and **IgA** contain a polypeptide, known as the **J chain**, that is linked by two disulfide bonds to the Fc region in two different monomers. Serum **IgM** is always a pentamer; most serum **IgA** exists as a monomer, although some dimers, trimer, and even tetramers sometimes are present. Not shown in these figures are intrachain disulfide bonds and disulfide bonds linking light and heavy chains.

© 1997, Duane W. Sears

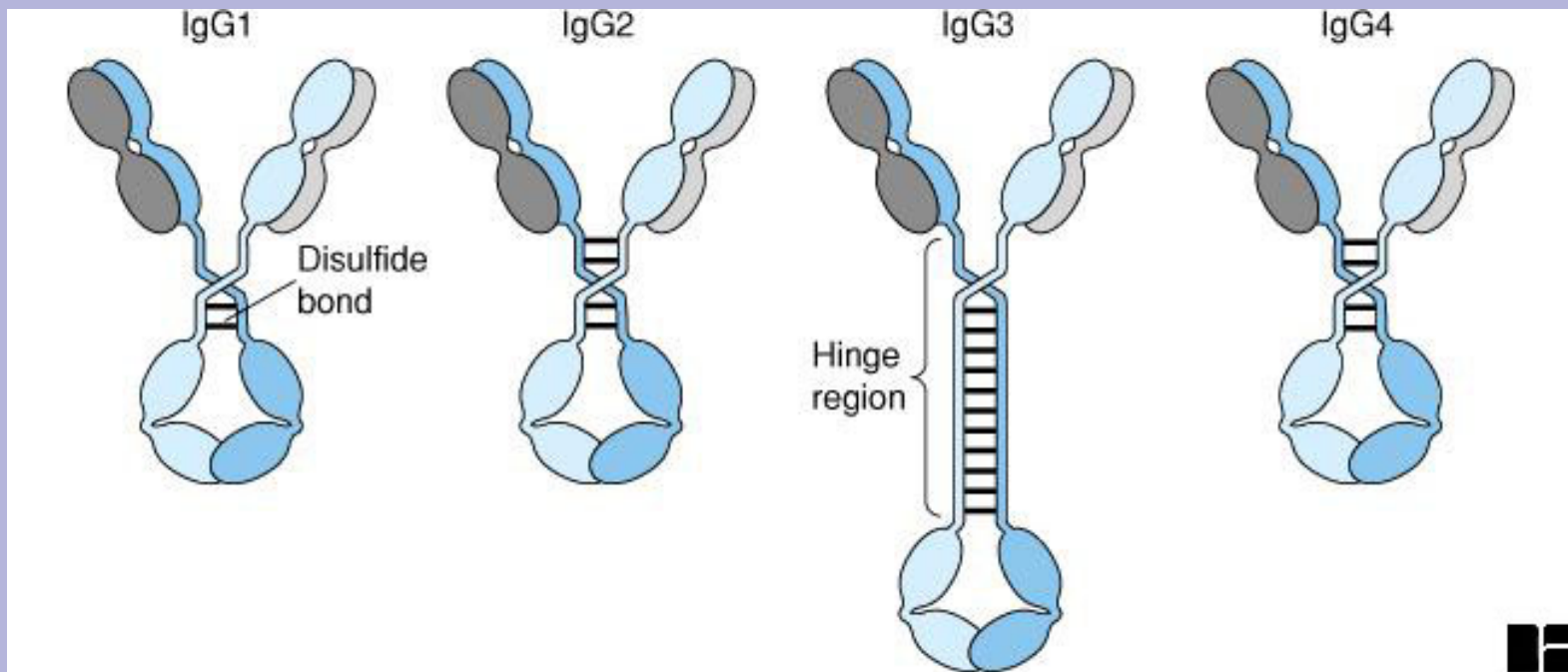
[\[TOP\]](#)

[\[Chapter Resources\]](#)

[\[Immunology Home Page\]](#)

W. H. Freeman & Co. and Sumanas, Inc.
Immunology, January, 1997

Last Updated: June 25, 1997

Figure 5-16: Major Subclasses of Human IgG

General structure of the **four subclasses of human IgG**, which differ in the number and arrangement of the interchain disulfide bonds (thick black lines) linking the heavy chains. The hinge region in **IgG3** is four times as long as the hinge region in the other **IgG** subclasses and contains 11 interchain disulfide bonds.

© 1997, Duane W. Sears

[\[TOP\]](#)

[\[Chapter Resources\]](#)

[\[Immunology Home Page\]](#)

W. H. Freeman & Co. and Sumanas, Inc.
Immunology, January, 1997

Last Updated: June 25, 1997

Chapter 7: Organization & Expression of Immunoglobulin Genes

Many of the intriguing mysteries surrounding antibody molecules, and particularly the mechanisms giving rise to their diversity, were finally resolved after the genes for these molecules were cloned and characterized. As it turns out, the structural diversity of antibodies is largely determined by various processes that modify DNA sequences in or around regions encoding portions of the complete antibody molecule. Functional antibody genes do not exist as such in the germ-line DNA but are somatically created during maturation of pre-B-cells into mature B-cells. During B-cell maturation, antibody genes are physically assembled by the translocation and joining of several gene segments to complete the coding sequences for one heavy chain gene and one light chain gene. While a single B-cell is committed to make copies of only one antibody molecule, different B cells are likely to make different antibodies since the individual antibody gene elements can be somatically rearranged in a number of different ways, resulting in highly diversified antibody coding sequences.

The somatic rearrangement processes leading to formation of the completed antibody genes in maturing B-cells require the catalytic action of DNA recombinases. These enzymes, or more likely enzyme complexes, recognize, cut, and rejoin specific DNA sequences. The DNA sequence specificity of these enzymes was initially deduced from sequence homologies found in regions neighboring the coding regions for various antibody gene elements. Two consensus sequences were identified, one being a **palindromic heptamer sequence**, always proximal to the linked coding sequence, and the other being an **AT-rich nonamer sequence**, always distal to the linked coding sequence. These heptamer and nonamer consensus sequences are always separated from one another by a nonconserved spacer sequence of either **12 base pairs -- 1 complete turn** of the double helix -- or **23 base pairs -- 2 complete turns** of the double helix. Together, the heptamer, nonamer, and spacer sequences define the "**one-turn**" and the "**two-turn**" recombination signal sequences (**RSSs**) as illustrated in Figure 7-6(a).

(a) Nucleotide sequence of RSSs

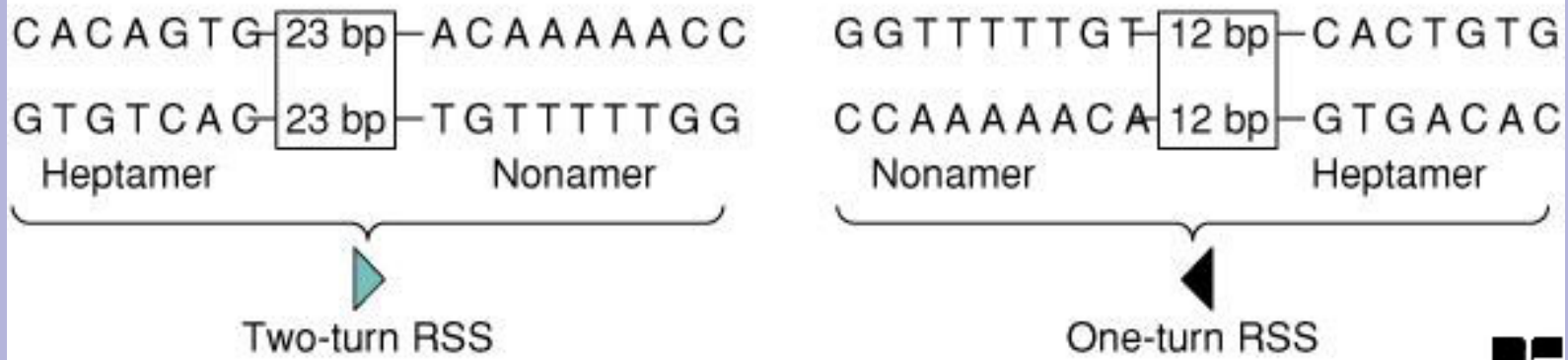


Figure 7-6(a). Two conserved sequences in light-chain and heavy-chain DNA function as recombination signal sequences (RSSs). Both signal sequences consist of a conserved palindromic heptamer and a conserved AT-rich nonamer; these are separated by nonconserved spacers of 12 or 23 base pairs.

In their germ-line DNA configuration, exons encoding certain elements of the antibody molecule are uniformly positioned adjacent to either one-turn or two-turn RSSs. One- and two-turn RSSs are both symbolized by the black triangles in Figure 7-6(b) below, illustrating the fact that every H (heavy) and L (light) chain variable (V) exon has a 3'-linked RSS; that every H or L chain joining (J) exon has a 5'-linked RSS; and that every H chain diversity (D) exon has both 5'- and 3'-linked RSSs.

(b) Location of RSSs in germ-line immunoglobulin DNA

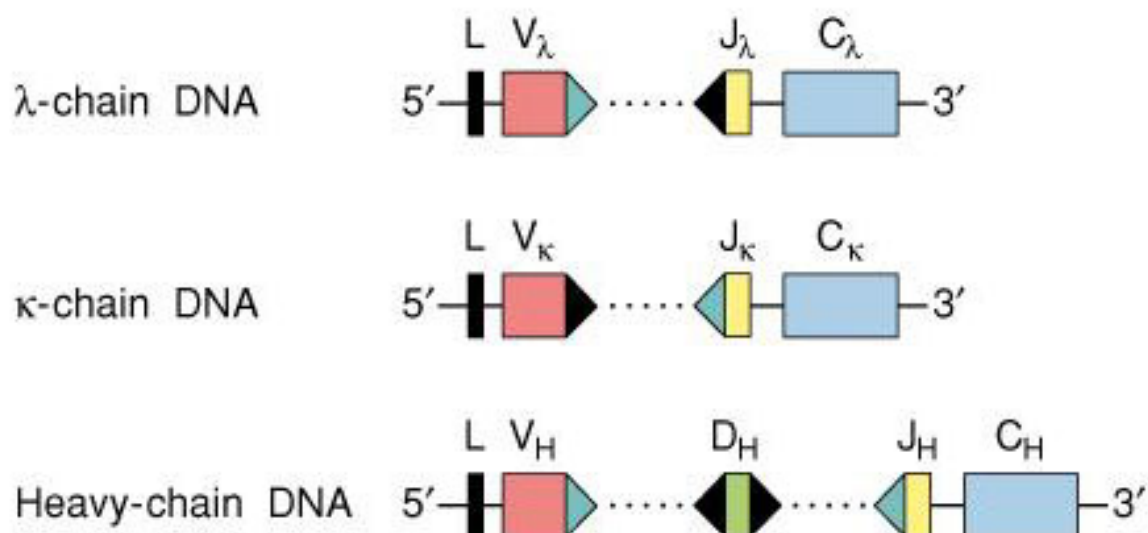


Figure 7-6(b). The two types of RSSs -- designated one-turn RSS and two-turn RSS -- have characteristic locations within λ -chain, κ -chain, and heavy-chain germ-line DNA. During DNA rearrangement, gene

segments adjacent to one-turn RSSs can join only with segments adjacent to two-turn RSSs.

At least two basic rules govern the functions of these RSSs in antibody gene rearrangements. First, all germ-line exons encoding the same antibody gene element (**V**, **D**, or **J**) are uniformly linked to RSSs with identical one- or two-turn spacer sizes. Second, antibody gene rearrangements join only exons that are linked to RSSs having non-matching spacer sizes. For example, exons linked to 3'-one-turn RSSs rearrange only with exons linked to 5'-two-turn RSSs. Conversely, exons linked to 3'-two-turn RSSs rearrange only with exons linked to 5'-one-turn RSSs. This summarizes the so-called "**one-turn/two-turn**" or "**12/23**" **joining rule** for antibody gene rearrangements.

In the first step of an immunoglobulin gene rearrangement, the recombinase creates double-stranded DNA breaks at the exon/RSS boundaries, effectively excising the intervening double-stranded DNA between the two exons. The exposed DNA ends are then ligated, creating a "**coding joint**" which corresponds to a contiguous reading frame running through the joined exons. Careful examination of the coding joint sequences in mature B-cells has revealed the interesting fact that different coding joints can be created from the same two exons. In other words, a certain degree of "**junctional flexibility**" enters into the process of antibody gene rearrangements, producing multiple coding sequences at the V_L - J_L , V_H - D_H , and D_H - J_H exon/exon boundaries as illustrated in [Figure 7-11](#). As shown in [Figure 7-13](#), antibody gene rearrangements proceed through a temporal cascade of order steps until the rearranged H and L chain genes encode a complete antibody molecule.

© 1997, Duane W. Sears

[\[TOP\]](#)

[\[Figure 7-11: V-J Junctional Flexibility\]](#)

[\[Figure 7-13: Temporal Cascade of Antibody Gene Rearrangements\]](#)

[\[Chapter Resources\]](#)

[\[Immunology Home Page\]](#)

W. H. Freeman & Co. and Sumanas, Inc.
Immunology, January, 1997

Last Updated: August 4, 1997

Figure 7-11: Junctional Flexibility in V_K to J_K Rearrangements

Junctional flexibility is but one of several mechanisms responsible for generating the vast diversity of antibody sequences. Figure 7-11 illustrates the mechanism for junctional flexibility at the V_K -exon/ J_K -exon boundary resulting from slightly different antibody gene rearrangement events. Click here for [animations of the V-J exon rearrangements](#) that result in different V-J coding joints as shown in this figure.

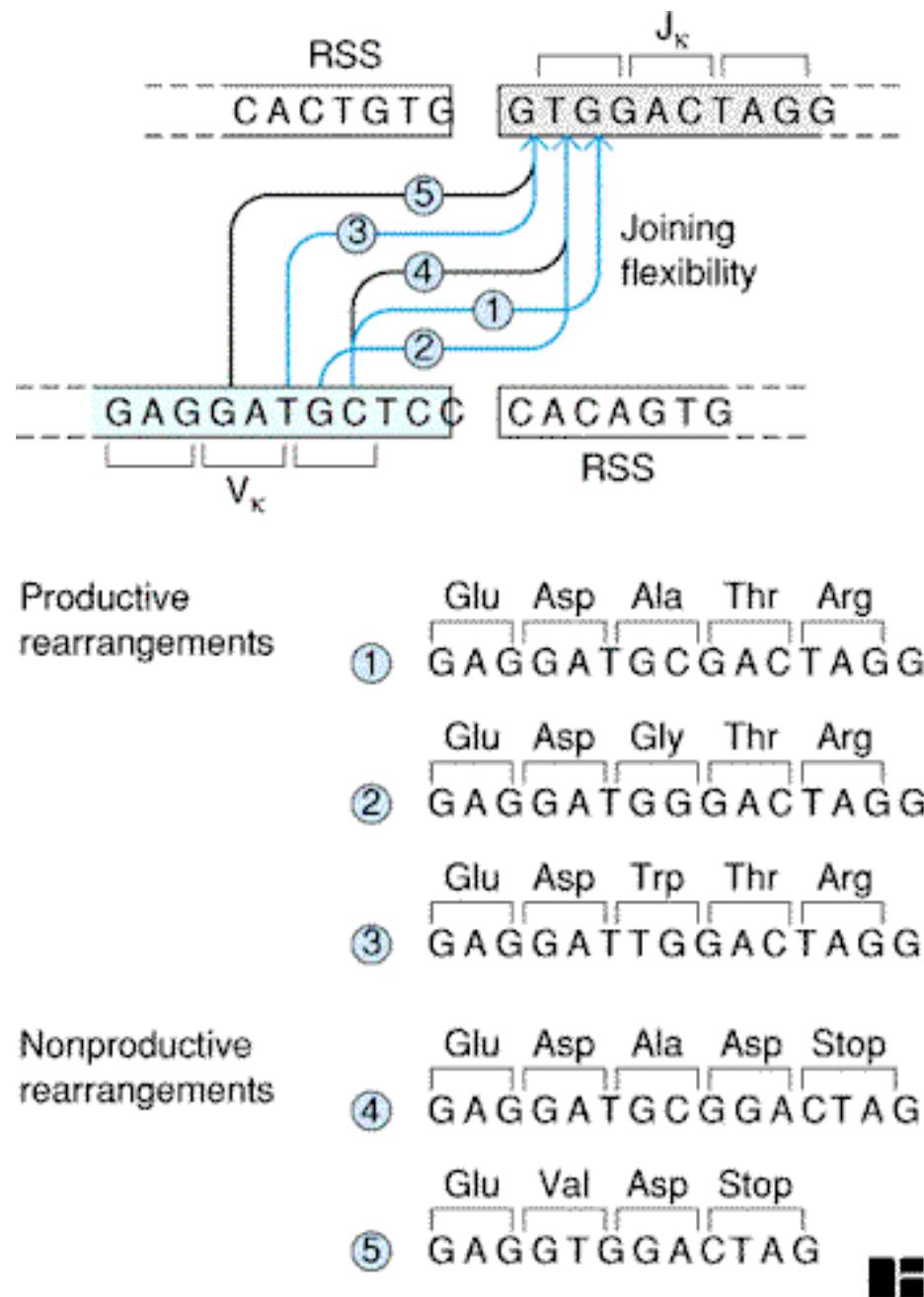


Figure 7-11. Junctional flexibility in the joining of immunoglobulin gene segments is illustrated with V_K and J_K . In-phase joining (arrows 1, 2, and 3) generates a productive rearrangement, which can be translated into protein. Out-of-phase joining (arrows 4 and 5) leads to a nonproductive rearrangement, which contains stop codons and is not translated into protein.

© 1997, Duane W. Sears

[\[TOP\]](#)

[\[Chapter 7: Organization & Expression of Immunoglobulin Genes\]](#)

[\[Figure 7-13: Temporal Cascade of Antibody Gene Rearrangements\]](#)

[\[Chapter Resources\]](#)

[\[Immunology Home Page\]](#)

W. H. Freeman & Co. and Sumanas, Inc.
Immunology, January, 1997

Last Updated: August 4, 1997

Chapter 7: V-J Joining at RSS Junctions

Mechanism of V κ -J κ Boundary Junctional Diversity

Reset Genomic Configuration	VJ-1 (Junction 1) --Glu-Asp- <u>Ala</u> -Thr-Arg--	VJ-2 (Junction 2) --Glu-Asp- <u>Gly</u> -Thr-Arg--	VJ-3 (Junction 3) --Glu-Asp- <u>Trp</u> -Thr-Arg--	VJ-4 (Junction 4) --Glu-Asp- <u>Ala</u> - <u>Asp</u> - <u>STOP</u>	VJ-5 (Junction 5) --Glu- <u>Val</u> - <u>Asp</u> - <u>STOP</u>
--	--	--	--	--	---

- Adjacent **vertical white bars** ('|') mark **one turn** of the double helix.
- These bars also mark locations where the **major DNA groove** faces outward.
- The "*" symbols mark the points of excision and ligation catalyzed by a recombinase.
- The **12** and **23 bp** "spacer" sequences (**white**) separating adjacent nonamer (**magenta**) and heptamer (**cyan**) sequences serve to orient the **major DNA groove** in the same direction in these two consensus regions, presumably bringing them into register for specific interactions with DNA-binding proteins.

[\[TOP\]](#)

[\[Chapter 7: Organization & Expression of Immunoglobulin Genes\]](#)

[\[Chapter Resources\]](#)

[\[Immunology Home Page\]](#)

[\[TOP\]](#)

[\[Chapter 7: Organization & Expression of Immunoglobulin Genes\]](#)

[\[Chapter Resources\]](#)

[\[Immunology Home Page\]](#)

Figure 7-11: Junctional Flexibility in V_K to J_K Rearrangements

Junctional flexibility is but one of several mechanisms responsible for generating the vast diversity of antibody sequences. Figure 7-11 illustrates the mechanism for junctional flexibility at the V_K -exon/ J_K -exon boundary resulting from slightly different antibody gene rearrangement events. Click here for [animations of the V-J exon rearrangements](#) that result in different V-J coding joints as shown in this figure.

Figure 7-11. Junctional flexibility in the joining of immunoglobulin gene segments is illustrated with V_K and J_K . In-phase joining (arrows 1, 2, and 3) generates a productive rearrangement, which can be translated into protein. Out-of-phase joining (arrows 4 and 5) leads to a nonproductive rearrangement, which contains stop codons and is not translated into protein.

© 1997, Duane W. Sears

[\[TOP\]](#)

[\[Chapter 7: Organization & Expression of Immunoglobulin Genes\]](#)

[\[Figure 7-13: Temporal Cascade of Antibody Gene Rearrangements\]](#)

[\[Chapter Resources\]](#)

[\[Immunology Home Page\]](#)

W. H. Freeman & Co. and Sumanas, Inc.
Immunology, January, 1997

Last Updated: August 4, 1997

Figure 7-13: Antibody Gene Rearrangements in the Maturing B Cell

In the progenitor B cell, antibody gene rearrangement is initiated by a D_H/J_H rearrangement of one of the two H chain gene alleles as illustrated in Figure 7-13 below. This is followed by a $V_H/(D_H-J_H)$ rearrangement of the same allele to create a fully rearranged μ heavy-chain gene. Antibody gene rearrangements are stochastic in nature, and some nonproductive rearrangements are possible (*i.e.*, the rearranged gene may not encode a full-length H chain polypeptide). If this first rearranged H chain gene is nonproductive, the rearrangement process switches to the second H chain allele. Once a productive μ heavy-chain gene has been successfully rearranged, a L chain gene rearrangement begins. In the mouse, rearrangement starts with one of the two κ chain alleles, whereas in humans, rearrangement commences with either a κ or λ allele. Again, a nonproductive L gene rearrangement is followed by rearrangement of the homologous L chain allele. If both L chain gene rearrangements fail, the rearrangement process switches to the genes encoding the other L chain isotype. As soon as one functional H chain gene and one functional L chain gene are created, the rearrangement process terminates. However, if temporal cascade of rearrangements ultimately fails to create either a functional H chain or L chain gene, the B cell is eliminated.

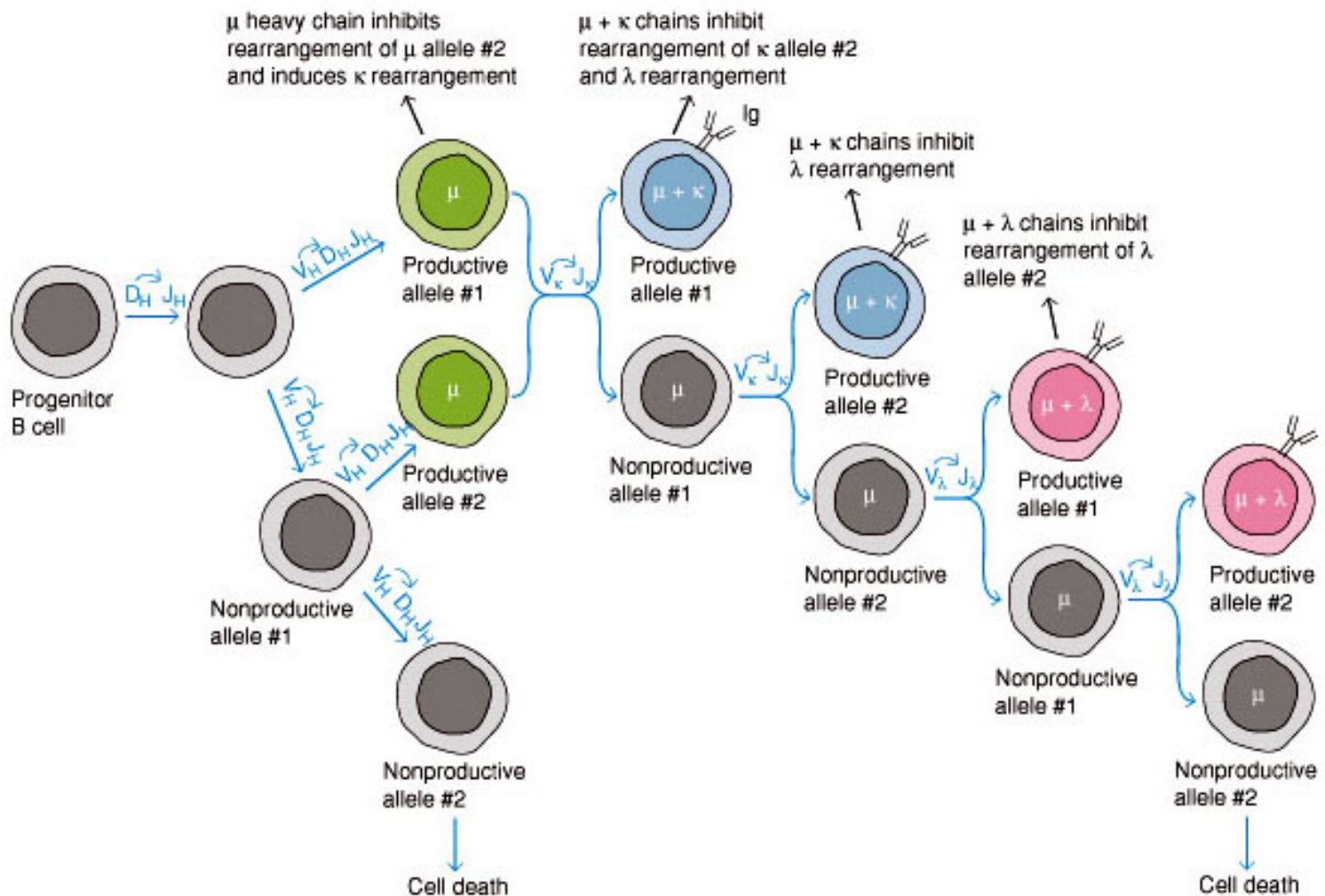


Figure 7-13. Model to account for allelic exclusion. Heavy-chain genes rearrange first, and once a productive heavy-chain gene rearrangement occurs, the μ protein product prevents rearrangement of the other heavy-chain allele and initiates light-chain gene rearrangement. In the mouse, rearrangement of the κ light-chain genes precedes rearrangement of the λ genes, as shown here. In humans, however, either κ or λ rearrangement can proceed once a productive heavy-chain rearrangement has occurred. Formation of a complete immunoglobulin inhibits further light-chain gene rearrangement. If a nonproductive rearrangement occurs for one allele, then the cell attempts rearrangement of the other allele. [Adapted from G. D. Yancopoulos and F. W. Alt, 1986, *Annu. Rev. Immunol.* 4:339]

On first consideration, the rearrangement process seems to be excessively wasteful given its randomness and inefficiency. Inefficient as it may be, this process still greatly enhances the B-cell repertoire by producing a multitude of different antibody combining sites for antigen. Even though additional biochemical controls could conceivably be imposed to reduce the net failure rate of the rearrangement process, more efficient regulation has not yet evolved, presumably because no harm is done to the organism by continuously discarding nonfunctional B-cells as they arise during B-cell maturation.

© 1997, Duane W. Sears

[\[TOP\]](#)

[\[Chapter 7: Organization & Expression of Immunoglobulin Genes\]](#)

[\[Figure 7-11: V-J Junctional Flexibility\]](#)

[\[Chapter Resources\]](#)

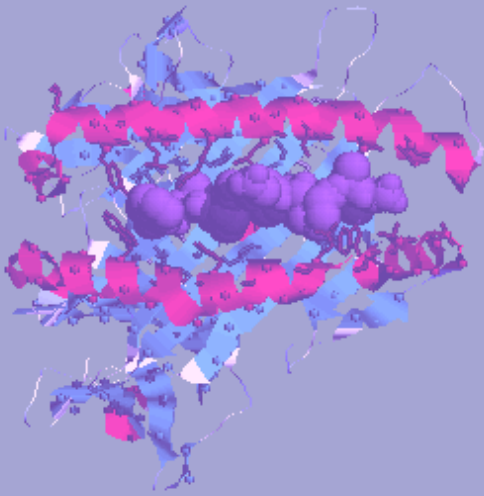
[\[Immunology Home Page\]](#)

W. H. Freeman & Co. and Sumanas, Inc.
Immunology, January, 1997

Last Updated: August 4, 1997

Chapter 9: Major Histocompatibility Complex

NOTE: This chapter requires Netscape Navigator v3.x or greater.



MHC Topic Bookmarks

Class I	Class II
T cytotoxic	T helper
CD8	CD4
overview	overview
animate	animate
endogenous	exogenous
antigenic oligopeptides	
$\alpha\beta$ subunit composition	
cell type expression	
MHC class I + 2 peptides	

The primary immunological function of Major Histocompatibility Complex (MHC) molecules is to bind and "present" antigenic peptides on the surfaces of cells for recognition (binding) by the antigen-specific T cell receptors (TCRs) of lymphocytes.

- Differential structural properties of MHC **class I** and **class II** molecules account for their respective roles in activating different populations of T lymphocytes.

Cytotoxic T_C lymphocytes bind antigenic peptides presented by **MHC class I molecules**.

Helper T_H lymphocytes bind antigenic peptides presented by **MHC class II molecules**.

- MHC **class I** and **class II** molecules differentially bind **CD8** and **CD4** cell adhesion molecules.

MHC class I molecules specifically bind **CD8 molecules** expressed on **cytotoxic T_C lymphocytes**.

MHC class II molecules specifically bind **CD4 molecules** expressed on **helper T_H lymphocytes** (see [Figure 4-9](#)).

- The sizes of the antigenic peptide-binding pockets of MHC class I and II molecules differ.

MHC class I molecules bind smaller antigenic peptides, 8-10 amino acid residues in length.

MHC class II molecules bind larger antigenic peptides, 13-18 amino acid residues in length.

- MHC class I and class II molecules are characterized by distinctive α and β polypeptide subunits that combine to form $\alpha\beta$ heterodimers characteristic of mature MHC molecules.

MHC class I α subunits are encoded by polymorphic MHC genes while the common β subunit is encoded by the conserved, non-MHC gene for β_2 -microglobulin (β_2m).

Both MHC class II α and β subunits are encoded by polymorphic MHC genes. Class II $\alpha\beta$ dimers also tend to associate as $(\alpha\beta)_2$ "dimers of dimers."

- As illustrated in [Figure 9-16](#), co-dominant class I and class II gene expression by diploid cells gives rise to heterologous as well as homologous $\alpha\beta$ dimer combinations on the cell surface since the subunits encoded by the maternally and paternally inherited genes mutually associate.

MHC class I molecules expressed on F_1 cells match maternal or paternal class I molecules since only the α subunit genes exhibit species-specific polymorphism. Thus, homologous and heterologous dimers are functionally equivalent.

MHC class II molecules expressed on F_1 cells include homologous and heterologous $\alpha\beta$ dimer mixtures since both α and β subunit genes exhibit species-specific polymorphism. Homologous dimers match class II molecules expressed on either parental cell type while heterologous dimers are unique to the F_1 genotype and are functionally non-equivalent to parental class II molecules.

[Figure 9-6 provides an overview](#) of the heterodimeric polypeptide structure of an **MHC class I molecule**, and [animated MHC class I images](#) further explore the folding and peptide binding sites of these proteins.



[Figure 9-8 provides an overview](#) of the heterodimeric polypeptide structure of an **MHC class II molecule**, and [animated MHC class II images](#) further explore the folding and peptide binding sites of these proteins.



- Differential cell-type expression of **class I** and **class II** molecules correlates with the specialized functions of different types of immune cells.

MHC class I expression is widespread on virtually **every cell** of the body. This is consistent with the protective function of **cytotoxic T_C lymphocytes** which continuously survey cell surfaces and kill cells harboring metabolically active microorganisms.

MHC class II expression is restricted to "**antigen presenting cells**." This is consistent with the functions of **helper T_H lymphocytes** which are locally activated wherever these cells encounter **macrophages, dendritic cells, or B cells** that have internalized and processed antigens produced by pathogenic organisms.



- Different antigen degradation and processing pathways produce MHC-peptide complexes where "**endogenous**" peptides associate with **class I** molecules and "**exogenous**" peptides associate with **class II** molecules.

MHC class I molecules bind peptide fragments derived from proteolytically degraded proteins **endogenously synthesized** by a cell. Small peptides are transported into the endoplasmic reticulum where they associate with nascent MHC class I molecules before being routed through the Golgi apparatus and displayed on the surface for recognition by **cytotoxic T_C lymphocytes**.

MHC class II molecules bind peptide fragments derived from proteolytically degraded proteins **exogenously internalized** by "antigen presenting cells," including **macrophages, dendritic cells, and B cells**. The resulting peptide fragments are compartmentalized in the endosome where they associate with MHC class II molecules before being routed to the cell surface for recognition by **helper T_H lymphocytes**.



The most intriguing feature of MHC proteins is their ability to form complexes with an assortment of different peptides many of which exhibit only marginal structural homology. The multi-peptide-binding property of MHC molecules represents the antithesis of most biological receptor/ligand systems where structural specificity in the binding reaction is usually much more narrowly defined. Although MHC molecules appear to be somewhat indiscriminant in their peptide-binding properties, this is not entirely the case. [Three-dimensional structural analysis of the same MHC molecule with different peptides](#) bound to the antigen-binding pocket provide insights into the mechanisms for this unusual and important feature of MHC molecules.



For additional overview of MHC structure, check out the [Gallery of Immune Molecule Structures](#) created by Eric Martz at the University of Massachusetts, Amherst, where the structures of these and other immune molecules are also featured.

- [Human MHC Class I HLA-A \(U. Mass.\)](#)
- [Mouse MHC Class I complex with Sendai \(U. Mass.\)](#)
- [Human MHC Class II HLA-DR complex with Influenza peptide \(U. Mass.\)](#)
- [MHC 3D Structure \(U. Mass.\)](#)

[\[Chapter 9: Overview of MHC class I three-dimensional structure\]](#)

[\[Chapter 9: Animation of MHC class I domain structure\]](#)

[\[Chapter 9: Animation of dual peptide binding by one MHC class I molecule\]](#)

[\[Chapter 9: Overview of MHC class II three-dimensional structure\]](#)

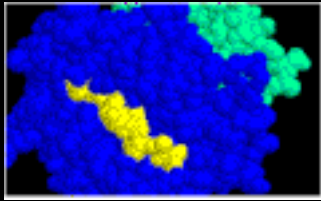
[\[Chapter 9: Animation of MHC class II domain structure\]](#)

[\[Chapter Resources\]](#)

[\[Immunology Home Page\]](#)

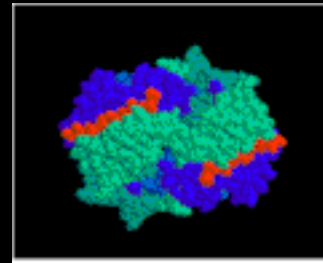
W. H. Freeman & Co. and Sumanas, Inc.
Immunology, January, 1997

Last Updated: June 26, 1997



$\alpha\beta_2m$ MHC class I molecule complexed to a peptide (multicolored) **9 amino acid residues** long.

$(\alpha\beta)_2$ MHC class II dimer complexed to two peptides (multicolored), each **13 amino acid residues** long.



© 1997, Duane W. Sears & Sean R. Christensen

[\[TOP\]](#)

[\[Chapter Resources\]](#)

[\[Immunology Home Page\]](#)

W. H. Freeman & Co. and Sumanas, Inc.
Immunology, January, 1997

Last Updated: June 26, 1997

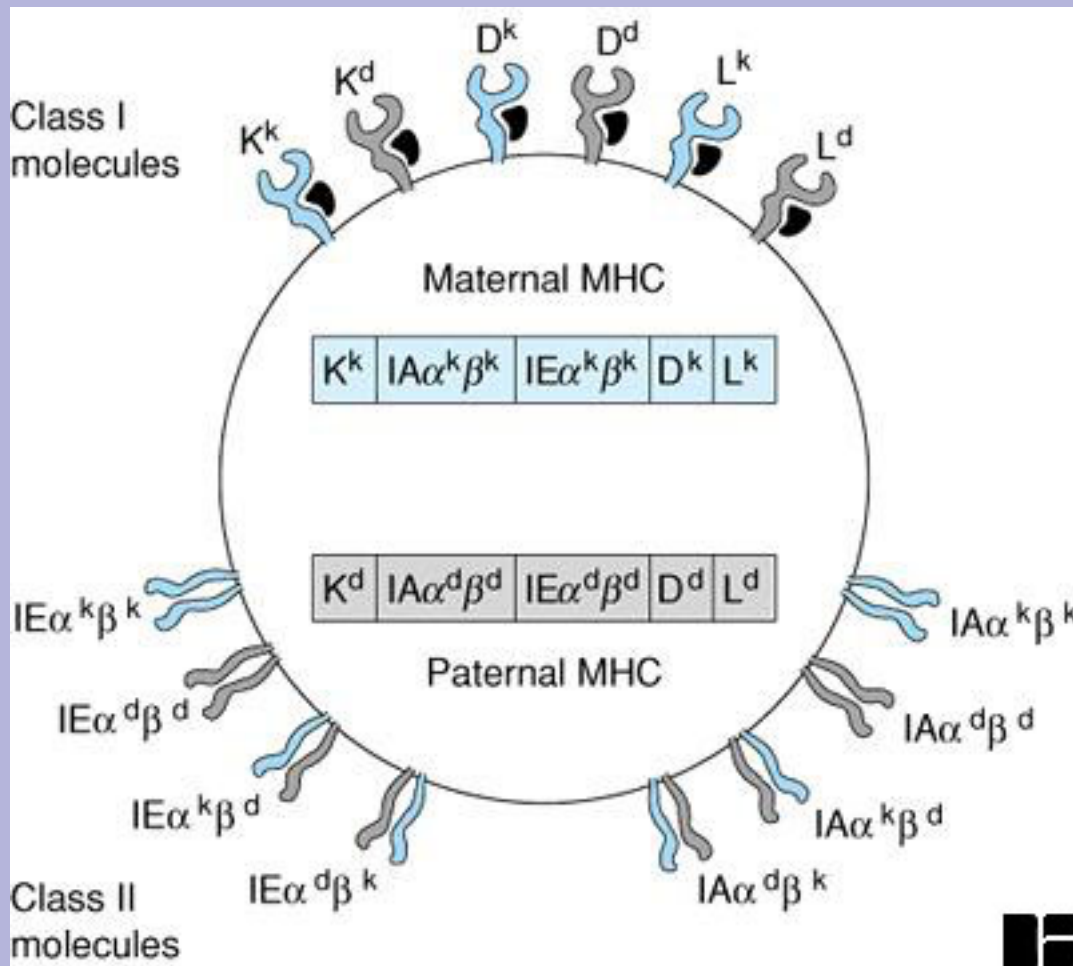
Figure 9-16: MHC Molecules Expressed on H-2^{k/d} Antigen Presenting Cells

Diagram illustrating various MHC molecules expressed on antigen-presenting cells of a heterozygous H-2^{k/d} mouse. Both the maternal and paternal MHC molecules are expressed. Because the class II molecules are heterodimers, heterologous molecules containing one maternal-derived chain and one paternal-derived chain are produced. The β₂-microglobulin component of class I molecules (black) is encoded by a gene on a separate chromosome and may be derived from either parent.

© 1997, Duane W. Sears

[\[TOP\]](#)

[\[Chapter Resources\]](#)

[\[Immunology Home Page\]](#)

W. H. Freeman & Co. and Sumanas, Inc.
Immunology, January, 1997

Last Updated: June 26, 1997

Chapter 13: Cytokines

NOTE: This chapter requires Netscape Navigator v3.x or greater.

Most cytokines are secreted proteins with wide-ranging immunoregulatory and hematopoietic activities as illustrated in [Figure 13-4](#). One cytokine featured at this web site is lymphotoxin- α (LT- α) or tumor necrosis factor- β (TNF- β) as it is sometimes called. This cytokine represents one of several members of the TNF family of cytokines which bind to one or more receptors of the [TNF receptor](#) family. Because TNF-like cytokines are homotrimeric in structure, they can bind one, two, or three TNF-like receptors. Receptor clustering appears to be a key initiating step for cytokine function. The three-dimensional structure of the [LT- \$\alpha\$ /\(TNF-R1\)₃](#) complex is examined on the linked web page.

The following web sites have additional interesting and useful information regarding cytokine and cytokine receptor structure and function:

- [Cytokine web site](#) created by [Cytokine Structures](#)
[Psynix Information Systems Ltd](#) [Cytokine Receptor Structures](#)

© 1997, Duane W. Sears

[\[TOP\]](#)

[\[Lymphotoxin- \$\alpha\$ Complex with Tumor Necrosis Factor Receptor 1\]](#)

[\[Chapter Resources\]](#)

[\[Immunology Home Page\]](#)

W. H. Freeman & Co. and Sumanas, Inc.
Immunology, January, 1997

Last Updated: June 30, 1998

Figure 13-4: Cytokines

Interaction of antigen (blue) with macrophages and the subsequent activation of resting T_H cells leads to release of numerous cytokines, generating a complex network of interacting cells in the immune response.

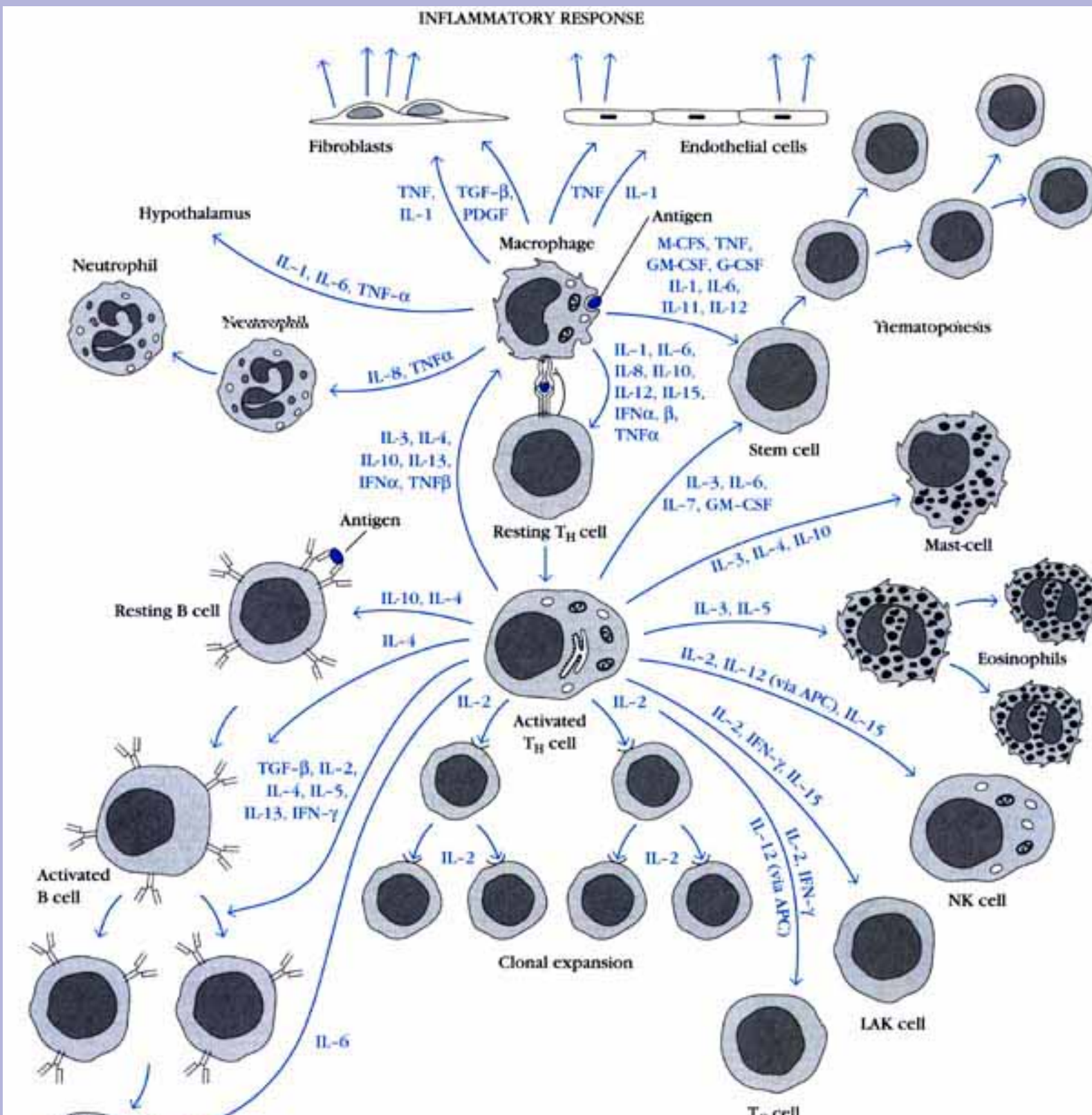
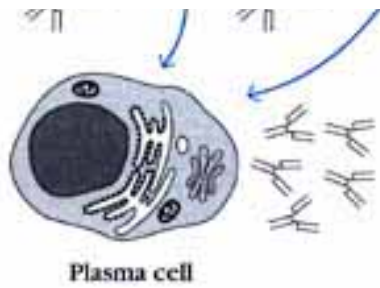


Figure 13-4



© 1997, Duane W. Sears

[\[TOP\]](#)

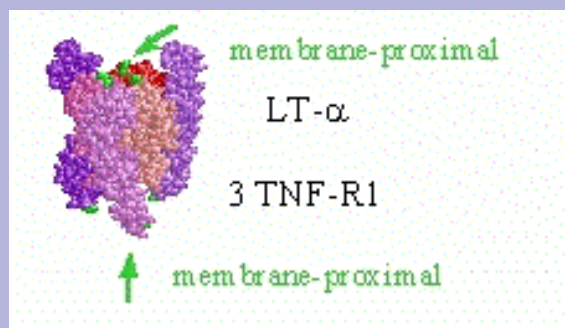
[\[Chapter Resources\]](#)

[\[Immunology Home Page\]](#)

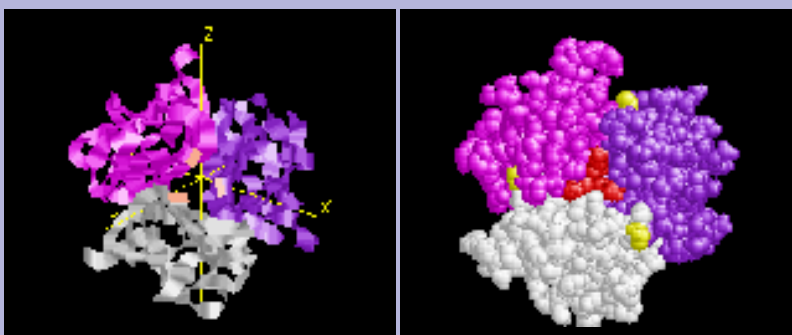
W. H. Freeman & Co. and Sumanas, Inc.
Immunology, January, 1997

Last Updated: June 24, 1997

Lymphotoxin-alpha/TNF-R1 Complex -- Overview



Binding of **LT- α** (center) to 1, 2, & 3 **TNF-R1** polypeptide chains (outside)



[Top view](#) of **LT- α** complexed to 3 **TNF-R1** receptor chains.

[Side view](#) of **LT- α** complexed to 3 **TNF-R1** receptor chains

ribbons	soluble : soluble LT-α : TNF-R_n	spacefilled
	◀ LT-α ▶	
	◀ + TNF-R1₁ ▶	
	◀ + TNF-R1₂ ▶	
	◀ + TNF-R1₃ ▶	

© 1997, Duane W. Sears

[\[TOP\]](#)

[\[Chapter Resources\]](#)

[\[Immunology Home Page\]](#)

NOTE: This page requires Netscape Navigator v3. x or greater.

Wire	Space	Ribbon	Trace	"Picking"	In	Out	
------	-------	--------	-------	------------------	----	-----	--

Lymphotoxin homotrimer (chains **a**, **b**, & **c**)
+ 3 **TNF-R1** receptors (chains **r**, **s**, & **t**)

Enter [RasMol Commands](#) below [1tnf.pdb](#)

PDB Info	Sequence	Labels off
Set Picking	None	Label
	Identify	Distance
	Monitor	Monitor off
	Center	

Move the mouse pointer over the center of an atom in the Chime window and click the mouse button (while holding down the **Control, **Option**, or **Command** key if using a **Macintosh**).**

© 1997, Duane W. Sears

[\[TOP\]](#)

[\[Chapter Resources\]](#)

[\[Immunology Home Page\]](#)

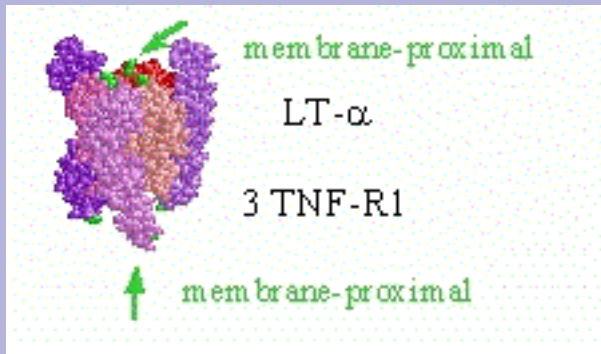
W. H. Freeman & Co. and Sumanas, Inc.
Immunology, January, 1997

Last Updated: June 25, 1997

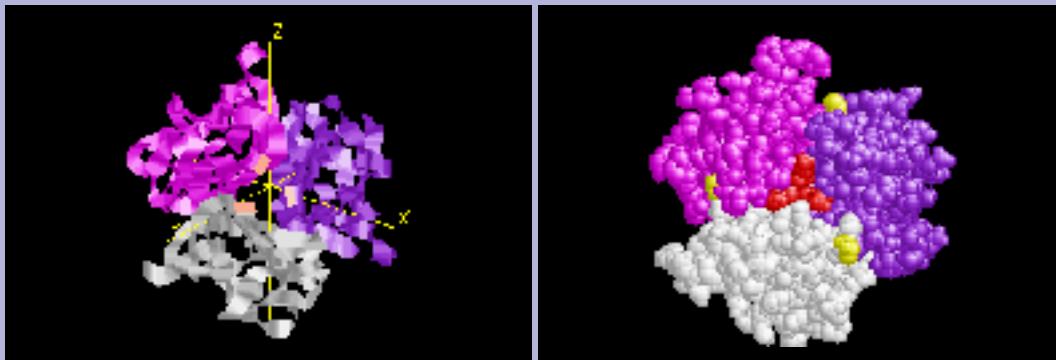
W. H. Freeman & Co. and Sumanas, Inc.
Immunology, January, 1997

Last Updated: June 25, 1997

Lymphotoxin-alpha/TNF-R1 Complex -- Overview



Binding of **LT- α** (center) to 1, 2, & 3 **TNF-R1** polypeptide chains (outside)



[Top view](#) of **LT- α** complexed to 3 **TNF-R1** receptor chains.

[Side view](#) of **LT- α** complexed to 3 **TNF-R1** receptor chains

ribbons	soluble : soluble LT- α : TNF-R _n	spacefilled
	◀ LT- α ▶	
	◀ + TNF-R1 ₁ ▶	
	◀ + TNF-R1 ₂ ▶	
	◀ + TNF-R1 ₃ ▶	

© 1997, Duane W. Sears

[\[TOP\]](#)

[\[Chapter Resources\]](#)

[\[Immunology Home Page\]](#)

W. H. Freeman & Co. and Sumanas, Inc.

Immunology, January, 1997

Last Updated: June 25, 1997

NOTE: This page requires Netscape Navigator v3.x or greater.

Wire	Space	Ribbon	Trace	" Picking "	In	Out	
------	-------	--------	-------	-----------------------------	----	-----	---

Lymphotoxin homotrimer (chains **a**, **b**, & **c**)

+ 3 **TNF-R1** receptors (chains **r**, **s**, & **t**)

Enter [RasMol Commands](#) below

[1tnf.pdb](#)

PDB Info	Sequence	Labels off
Set Picking	None	Label
	Identify	Distance
	Monitor	Monitor off
	Center	

Move the mouse pointer over the center of an atom in the Chime window and click the mouse button (while holding down the **Control, **Option**, or **Command** key if using a Macintosh).**

© 1997, Duane W. Sears

[\[TOP\]](#)

[\[Chapter Resources\]](#)

[\[Immunology Home Page\]](#)

W. H. Freeman & Co. and Sumanas, Inc.
Immunology, January, 1997

Last Updated: June 25, 1997

Chapter 22: The Immune System in AIDS

NOTE: This chapter requires Netscape Navigator v3.x or greater.

Human immunodeficiency virus, or HIV (see [Figure 22-2a](#)), causes the deadly disease known as [Acquired Immunodeficiency Syndrome](#) (AIDS) and presents an unprecedented paradox to the immunologist and the physician. HIV uniquely infects and destroys two types of cells -- [T_H helper cells](#) and [macrophages](#) -- that are essential for immunity against viruses. [HIV infection](#) is initiated by the HIV [gp120](#) envelope glycoprotein which binds to [CD4](#) molecules expressed on the surfaces of these two cell types. The virus then fuses to the host cell with another HIV envelope protein, [gp41](#), which binds to host-cell "[fusin](#)" as illustrated in [Figure 22-4](#) which has been identified as one of the [chemokine receptors](#). Once the virus infects the cell, a complex biology and pathology ensue.

From an understanding of the precise biology of the virus itself and the nature of the pathogen-host cell interaction, new strategies have emerged for treating AIDS. One approach is to block key viral enzymes in order to disrupt the viral life cycle. [HIV protease](#) is one such enzyme. [Inhibitors or HIV protease](#) administered in combination with other anti-HIV drugs make a particularly effective treatment for HIV-infected individuals helping to delay the onset of the more serious symptoms associated with AIDS.

© 1997, Duane W. Sears

[\[TOP\]](#)

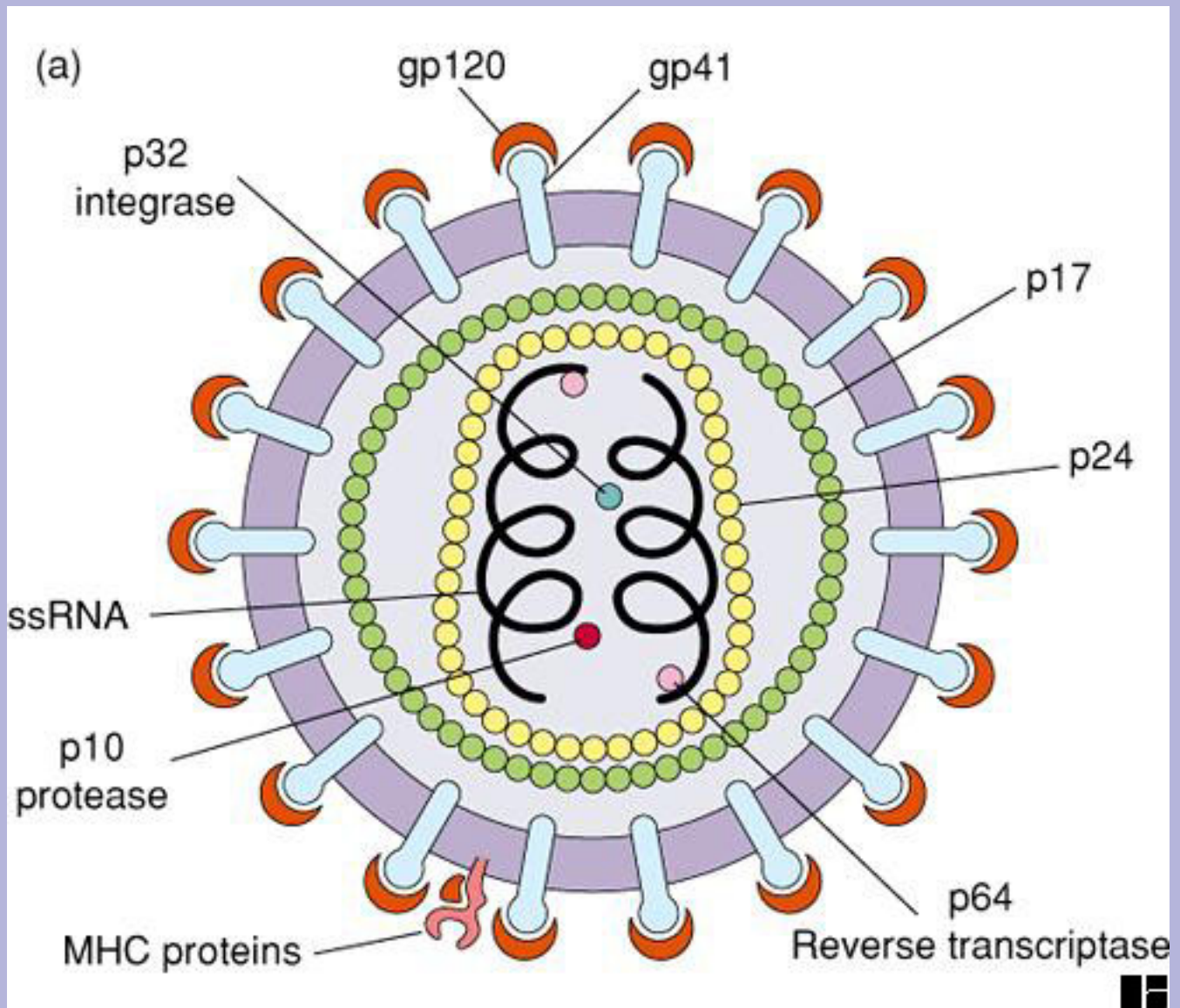
[\[Chapter 22: HIV Protease Complex with Pepstatin Inhibitor\]](#)

[\[Chapter Resources\]](#)

[\[Immunology Home Page\]](#)

W. H. Freeman & Co. and Sumanas, Inc.
Immunology, January, 1997

Last Updated: June 25, 1997

Figure 22-2(a): Structure of HIV

(a) Cross-sectional schematic diagram of HIV virion. Each virion expresses 72 glycoprotein projections composed of **gp120** (orange) and **gp41** (light blue). **Gp41** is a transmembrane molecule that crosses the lipid bilayer of the envelope. **Gp120** is noncovalently associated with **gp41** and serves as the viral receptor for **CD4** on host cells. The viral envelope also contains some host-cell membrane proteins such as class I and class II MHC molecules. Within the envelope is the viral core, or nucleocapsid, which includes a layer of a protein called **p17** (green) and an inner layer protein called **p24** (yellow). The HIV genome consists of two copies of **ssRNA**, which are associated with two molecules of reverse transcriptase **p64** (light red) and nucleoid proteins **p10**, a **protease** (red), and **p32**, an integrase (dark blue).

[\[TOP\]](#)

[\[Chapter Resources\]](#)

[\[Immunology Home Page\]](#)

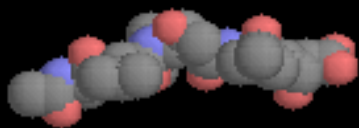
W. H. Freeman & Co. and Sumanas, Inc.
Immunology, January, 1997

Last Updated: June 25, 1997

Inhibition of Human Immunodeficiency Virus Protease

Animate the pepstatin inhibitor complex with HIV protease at the right. (approx. 30 sec.)

pepstatin:



reset

Acetylpepstatin Inhibitor of HIV Protease

Recent development of effective HIV protease inhibitors has revolutionized treatment of AIDS patients when these inhibitors are combined with other anti-viral drugs. HIV protease serves a critical role in the replication of virus in infected human cells. Specifically, it cleaves the single, long polypeptide encoded by the viral genome, thereby generating the essential proteins needed for viral maturation. The biochemical basis for inhibition of HIV protease is illustrated by the structure at the right where a **pepstatin inhibitor** is visibly bound to the active site formed by a pocket between the two identical subunits of this enzyme (shown in green and blue). The inhibitor specifically

NOTE: This page requires Netscape Navigator v3.x or greater.

Wire	Space	Ribbon	Trace	" Picking "	In	Out	
------	-------	--------	-------	-----------------------------	----	-----	--

HIV protease homodimer (chains **a** & **b**)
+ pepstatin (chain **c**)

Enter [RasMol Commands](#) below [hvp.pdb](#)

PDB Info	Sequence	Labels off
Set Picking	None	Label
	Identify	Distance
	Monitor	Monitor off
	Center	

Move the mouse pointer over the center of an atom in the Chime window and click the mouse button (while holding down the Control, Option, or Command key if using a Macintosh).

© 1997, Duane W. Sears

[\[TOP\]](#)

[\[Chapter Resources\]](#)

[\[Immunology Home Page\]](#)

W. H. Freeman & Co. and Sumanas, Inc.
Immunology, January, 1997

Last Updated: June 25, 1997

blocks substrate entry and the catalytic activities of two centrally located aspartate residues which are revealed through a series of *Chime* buttons below.

Squarely press these buttons in sequence and wait until the image stops moving before pressing the next button. If the *Chime* image fails to respond when the button is pushed, press the browser's "Reload" button.

reset

- 1) Visualize the active site "hole" in the middle of the enzyme with its two interior catalytic aspartate residues.**
- 2) Visualize the inhibitor, acetyl pepstatin, bound to HIV protease and note how this substrate analog completely occludes the active site.**
- 3) Zoom into the enzyme active site and note the interaction between pepstatin and the two catalytic aspartate residues, (Asp₂₅ & Asp₂₂₅).**
- 4) Scan the active site and view the relationship between the bound inhibitor and the catalytic aspartate residues. Note the close proximity of the side chain oxygen atom of Asp₂₅ and the hydroxyl oxygen of pepstatin indicating that these are hydrogen bonded through a hydrogen (not shown) donated**

**either by the hydroxyl group
itself or by the un-ionized
carboxylic acid side chain of
Asp₂₅.**

© 1997, Duane W. Sears

[\[TOP\]](#)

[\[Chapter Resources\]](#)

[\[Immunology Home Page\]](#)

W. H. Freeman & Co. and Sumanas, Inc.
Immunology, January, 1997

Last Updated: June 25, 1997

NOTE: This page requires Netscape Navigator v3.x or greater.

Wire	Space	Ribbon	Trace	"Picking"	In	Out	
------	-------	--------	-------	------------------	----	-----	---

HIV protease homodimer (chains **a** & **b**)
+ pepstatin (chain c)

Enter [RasMol Commands](#) below

hvp.pdb

PDB Info	Sequence	Labels off
<u>Set Picking</u>	<u>None</u>	<u>Label</u>
	<u>Identify</u>	<u>Distance</u>
	<u>Monitor</u>	<u>Monitor off</u>
	<u>Center</u>	

Move the mouse pointer over the center of an atom in the Chime window and click the mouse button (while holding down the **Control, **Option**, or **Command** key if using a Macintosh).**

© 1997, Duane W. Sears

[\[TOP\]](#)

[\[Chapter Resources\]](#)

[\[Immunology Home Page\]](#)

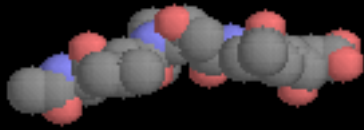
W. H. Freeman & Co. and Sumanas, Inc.
Immunology, January, 1997

Last Updated: June 25, 1997

Inhibition of Human Immunodeficiency Virus Protease

Animate the pepstatin inhibitor complex with HIV protease at the right.
(approx. 30 sec.)

pepstatin:



reset

Acetylpepstatin Inhibitor of HIV Protease

Recent development of effective HIV protease inhibitors has revolutionized treatment of AIDS patients when these inhibitors are combined with other anti-viral drugs. HIV protease serves a critical role in the replication of virus in infected human cells. Specifically, it cleaves the single, long polypeptide encoded by the viral genome, thereby generating the essential proteins needed for viral maturation. The biochemical basis for inhibition of HIV protease is illustrated by the structure at the right where a **pepstatin inhibitor** is visibly bound to the active site formed by a pocket between the two identical subunits of this enzyme (shown in green and blue). The inhibitor specifically blocks substrate entry and the catalytic activities of two centrally located **aspartate residues** which are revealed through a series of *Chime* buttons below.

Squarely press these buttons in sequence and wait until the image stops moving before pressing the next button. If the *Chime* image fails to respond when the button is pushed, press the browser's "Reload" button.

reset

- 1) **Visualize the active site "hole" in the middle of the enzyme with its two interior catalytic aspartate residues.**
- 2) **Visualize the inhibitor, acetyl pepstatin, bound to HIV protease and note how this substrate analog completely occludes the active site.**

- 3) **Zoom into the enzyme active site and note the interaction between pepstatin and the two catalytic aspartate residues, (Asp₂₅ & Asp₂₂₅).**
- 4) **Scan the active site and view the relationship between the bound inhibitor and the catalytic aspartate residues. Note the close proximity of the side chain oxygen atom of Asp₂₅ and the hydroxyl oxygen of pepstatin indicating that these are hydrogen bonded through a hydrogen (not shown) donated either by the hydroxyl group itself or by the un-ionized carboxylic acid side chain of Asp₂₅.**

© 1997, Duane W. Sears

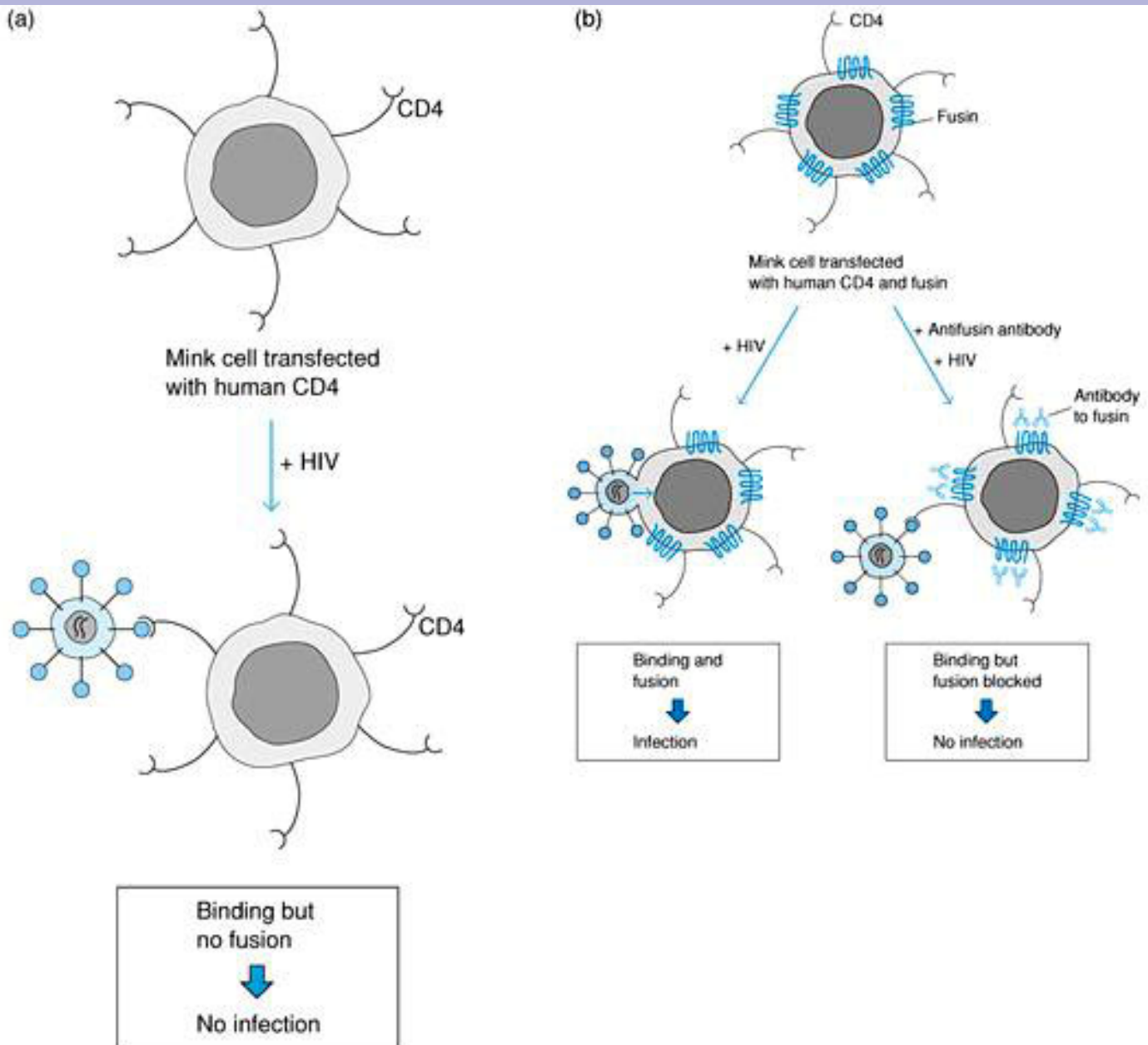
[\[TOP\]](#)

[\[Chapter Resources\]](#)

[\[Immunology Home Page\]](#)

W. H. Freeman & Co. and Sumanas, Inc.
Immunology, January, 1997

Last Updated: June 25, 1997

Figure 22-4: CD4 and Fusin Receptors Required for HIV Infection

Experimental demonstration of requirement for fusin in fusion of HIV with host cells. Mink cells normally do not express either CD4 or fusin and thus are resistant to HIV infection. (a) Cells transfected with human CD4 bind HIV but are not infected. (b) Cells cotransfected with both human CD4 and fusin become infected as both HIV binding and fusion occur. In the presence of antifusin antibody, cotransfected cells only bind HIV.